

DIGITAL DISASTERS: THE ECONOMIC COSTS OF SUBMARINE CABLE BREAKS*

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Abstract

I estimate the economic costs of submarine cable disruptions, the backbone of cross-border data traffic, using a large panel covering 2008-2020. Exploiting plausibly exogenous variation in accidental disruptions timing and repair duration, a dynamic difference-in-differences design identifies a persistent GDP per capita drop of 5–7% relative to trend within four to five years, with unaffected regional peers experiencing growth gain from activity reallocation. Disruptions erode trade flows and exporter participation, reduce private consumption, FDI and productivity, and tighten access to finance. The effects are concentrated in countries with infrastructures promoting redundancy, pointing to a low-cost policy lever for digital resilience.

Keywords: Submarine cables, connectivity disruptions, economic growth, trade, spillovers, digital infrastructure.

JEL Classification: O40 , L96 , F43 , O33 , H54.

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1 Introduction

Modern economies rely on cross-border data flows to coordinate trade, finance, and production. Submarine cables (SMCs) carry 99% of international data traffic ([Telegeography 2023](#)), forming the physical backbone that connects national economies to the global economy. A growing literature shows that the arrival of submarine cables fosters economic growth, trade integration, and financial development, particularly in developing economies ([Hjort and Poulsen 2019](#); [Simione and Li 2021](#); [D’Andrea and Limodio 2024](#)). Yet these gains come with a new source of vulnerability. Where international connectivity rests on only a handful of submarine routes and landing stations, cable failures can simultaneously disrupt payments, logistics, and digital services for days or even weeks, with consequences potentially cascading across sectors and amplifying across time ([ITU et al. 2026](#)). The policy importance of these risks has become increasingly evident, prompting the European Commission, among others, to elevate submarine cable resilience to a strategic priority¹, while its economic incidence may fall hardest elsewhere. Despite this growing attention, the economic costs of submarine cable disruptions have yet to be measured.

In 2024, cable breaks were the leading cause of global internet shutdowns by countries affected, and the second most frequent after government-ordered shutdowns.² The macroeconomic consequences of these failures remain poorly understood. The literature lacks causal evidence on SMC failures specifically: existing assessments of connectivity disruptions are largely accounting exercises and consider other disruption causes, such as government shutdowns. [Deloitte \(2016\)](#) estimates that connectivity disruptions can destroy hundreds of millions of dollars in GDP within days, while The Brookings Institution ([West 2016](#)) estimated global GDP losses from government-induced internet shutdowns at \$2.4 billion annually, with some country-level losses reaching nearly \$1 billion. Beyond identification concerns, these studies largely overlook submarine cable failures, arguably the most economically damaging source of connectivity disruption given their role as critical first-mile infrastructure in the world economy’s architecture, their capacity to affect multiple countries simultaneously, and the extended durations required for underwater repairs. Second, they implicitly treat disruptions as short-lived events, rather than as shocks that can generate cascading effects through persistent output losses across production and service networks, and reshape regional patterns of economic activity. For power outages, reduced-form estimates understate the output cost three- to fivefold once incumbent firms expand and new firms enter ([Fried and Lagakos 2023](#)). A joint ITU–UNDRR–Sciences Po report ([ITU et al. 2026](#)) underscores the same gap, noting that the economic impacts of digital disruptions remain poorly understood and that existing risk frameworks lack models adequate to the depth of modern societies’ digital dependence.

Connectivity disruptions can depress economic activity well beyond digitally-intensive sectors. When international connectivity is impaired, firms face immediate information and financial frictions. Orders cannot be verified, payments cannot be executed, and the data-intensive services embedded in modern trade—finance, logistics, transport, and ICT support—are delayed or disrupted. These frictions propagate through supply chains and trade networks as firms reliant on just-in-time logistics and cross-border coordination incur order, payment and delivery delays³, and are likely amplified today by the increasing digitalization of economies and “servicification” of production and merchandise trade ([Baldwin 2016](#)). Direct evidence that information frictions resulting from infrastructure

¹See, for instance, the European Commission Recommendation 2024/779 of 26 February 2024 on secure and resilient submarine cable infrastructure ([link](#)).

²<https://radar.cloudflare.com/year-in-review/2024>

³Early evidence shows that each additional day of delay in product delivery reduces bilateral product trade by more than 1 percent on average ([Djankov et al. 2010](#)), and can be compared to an ad-valorem tariff of 0.6 to 2.1 percent on imported goods ([Hummels and Schaur 2013](#)).

failures distort trade comes from [Steinwender \(2018\)](#), who shows that temporary technical failures of the transatlantic telegraph cable widened the US–UK cotton price gap by 23% per additional day of information lag, with the telegraph’s permanent elimination of those frictions raising trade volumes by 37%. Firm-level evidence confirms that internet access directly governs the scope of these trade relationships, with broadband expansion increasing firm imports by around 25%, primarily through the range of products sourced and countries involved ([Malgouyres et al. 2021](#)). Internet access also determines the geographic structure of trade relationships, with broadband adoption expanding firms’ effective choice sets and making bilateral flows more sensitive to distance and economic size, so that higher information frictions push sourcing and distribution toward geographically proximate and already-familiar partners ([Akerman et al. 2021](#)). Cable failures can thus also generate spillovers to nearby non-disrupted economies, by weakening competitiveness and triggering regional diversion of trade flows, foreign investment, and bandwidth-intensive activities toward proximate economies that offer more reliable connectivity.

The paper provides the first quasi-experimental estimates of the short- and medium-run economic costs of submarine cable failures. Using a new cross-country dataset spanning 2008–2020, I exploit plausibly exogenous variation in disruption timing and duration in a dynamic event-study difference-in-differences framework, separately identifying direct losses for disrupted countries and positive spillovers to proximate non-disrupted peers. An average disruption, requiring 10–11 cable repair days, is followed by a decline in GDP per capita that accumulates over the following years. Within four to five years, GDP per capita falls roughly 9% below the counterfactual trajectory implied by non-disrupted peers in the same region, narrowing to 5–7% once spillovers to unaffected regional peers are netted out. Once the atypical 2020 pandemic year is set aside, the loss attenuates to about –2.5% by the sixth year, leaving the medium-run cost anchored at the four-to-five-year horizon. These losses are confined to countries with sparse cable networks and concentrated cable ownership, and are fully eliminated where alternative routes, Internet exchange points, or a more diversified set of cable owners exist. Overall, even short-lived disruptions to international digital infrastructure can leave persistent scars on economic growth, that are largely avoidable through targeted investment in network redundancy.

How do cable disruptions translate into such GDP losses? Disruptions depress merchandise and services trade, particularly imports, reduce private consumption, and tighten financial conditions, with domestic credit and cross-border banking positions weakening over time. Mirroring the GDP responses, foreign direct investment, bandwidth-intensive service exports, domestic and foreign exporter participation, and total factor and labor productivity all fall in disrupted countries and rise in nearby non-disrupted countries, providing consistent evidence that cable failures reallocate trade, investment, and production across countries within the region. These findings suggest that submarine cable disruptions are systemic shocks whose economic consequences extend well beyond the directly affected countries.

Dynamic effects are estimated using the event-study framework of [de Chaisemartin and d’Haultfoeuille \(2024\)](#), which accommodates staggered, repeated, non-binary and non-absorbing treatments. Identification exploits the fact that cable failures are primarily triggered by maritime hazards and occur within a dense network architecture that makes disruption timing and location plausibly orthogonal to macroeconomic conditions.⁴ Exposure follows the physical geography of the network

⁴The International Cable Protection Committee (ICPC) argues that accurate charting of submarine cable routes on nautical charts is critical for maritime safety and cable protection, because most cable faults are accidental and largely caused by fishing activity and vessel anchoring (on the order of 70–80% of faults). The ICPC also notes that removing cables from charts would do little to deter deliberate damage, since actors intent on sabotage can locate cables regardless, while de-charting would materially increase accidental incidents ([link](#)).

(Imbs and Pauwels 2025). Faults concentrate at the maritime choke points, where many cable systems share the same shallow seabeds that also carry the densest shipping, and along the routes feeding peripheral economies that depend on a few landing cables. The choke points carry the highest risk of a fault, because anchoring and fishing in those crowded waters are the leading cause of breaks (Clare et al. 2023), and a single fault there severs many systems at once. The peripheral, low-redundancy economies are where faults do lasting damage, because they are the ones without the alternative routes or exchange points to contain it. Therefore, exposure is defined from the perspective of what each country experiences, since a single break can simultaneously impair connectivity among distant countries.

A first-stage analysis based on high-frequency connectivity data establishes that these coded disruptions correspond to a measured loss of international connectivity. Drawing on routed-prefix counts from the RIPE routing-information service, compiled independently of the disruption coding, I find that the share of a country’s networks reachable from abroad holds steady before a disruption and falls on its recorded onset. While it lasts the withdrawal is deep, with the worst daily shortfall reaching about 10 percent of a country’s routed connectivity in the disruption month. Routed connectivity then returns to its pre-event level within about three weeks. This window matches the average of 10 to 11 repair days that the dose treatment variable records per incident, linking the outage to the physical repair of the cable.

To address endogeneity concerns, I test for non-random exposure and apply the recentering approach of Borusyak and Hull (2023), I exclude incidents linked to natural disasters, sabotage, or government-ordered shutdowns which directly affect economic outcomes, and conduct sensitivity checks across alternative exposure definitions and sequential exclusions of countries, regions or income groups. I also document that the cross-country incidence of disruptions is uncorrelated with income and with pre-period growth once the 2009 financial-crisis year is set aside. Exposure loads instead on a country’s position in the cable network, so that which economies are hit is set by geography and topology rather than by their development level. Identification is further validated against an independent disruption dataset from the *Cloudflare Radar* platform covering 2022–2025.

The causal literature on cross-border connectivity has so far measured what its arrival delivers. Broadband and cable rollout raised growth, productivity, and the reallocation toward services across developing economies (Hjort and Poulsen 2019; Simione and Li 2021; D’Andrea and Limodio 2024), and access through shared cable systems fostered trade and widened firms’ participation in foreign markets (Herman and Oliver 2023; Imbruno et al. 2025). This paper addresses connectivity’s withdrawal. The output loss it identifies is larger than the mirror image of those arrival gains, because economies that organized trade, payments, and production around continuous connectivity hold no domestic substitute once a route fails; and the economies that cannot divert traffic to contain the shock are those with few alternative cable routes, no Internet exchange points, and cables concentrated among a small number of owners, a margin set by network redundancy rather than by income. These losses compound with disruption duration, since an unrerouted fault holds trade-time frictions in place for the days or weeks a deep-water repair requires (Djankov et al. 2010; Hummels and Schaur 2013). They also persist, since a cable break is a critical-infrastructure outage without a backstop. An electricity shortage can be partly buffered by on-site generation (Fried and Lagakos 2023), storage, or imports from a neighbouring grid. International bandwidth severed at sea has no such local substitute⁵, because the input is produced abroad and reaches the economy only through the physical route that has failed.

⁵Nor satellite alternatives, including low-orbit broadband, which carry too little capacity, at too high latency and cost, to replace a submarine route.

Therefore, submarine cable failures warrant treatment as *digital disasters*. They are sudden, exogenous shocks that disable critical infrastructure and propagate economy-wide with persistence, with estimated welfare losses in the same magnitude class as those documented for climate-attributable extreme weather events (Carleton and Hsiang 2016; Newman and Noy 2023), financial and socio-political shocks (Cerra and Saxena 2008), or inter-state wars (Federle et al. 2026). Unlike those shocks, their primary effects operate through disruptions to market access, production linkages, and cross-border financial flows rather than capital destruction, precluding the reconstruction-driven recovery that typically follows such events and consistent with the persistent GDP divergence estimated here. Effects concentrate in countries with limited rerouting capacity where the absence of redundancy amplifies transmission and persistence. Combining the spillover-inclusive declines with these economies’ average income yields per-capita output losses of roughly \$50 on impact, rising to \$240–400 in the medium run. Results also point to mitigation levers, since economies with diverse international routes, Internet exchange points, or diversified cable ownership absorb the same disruptions with losses close to zero.

The remainder of the paper is organized as follows. Section 2 reviews the institutional background, the transmission mechanisms, and the related literature. Section 3 sets out the event-study estimator, describes the data, presents a first stage linking coded disruptions to independently measured losses of routed connectivity. Section 4 reports baseline GDP growth and level effects, analyses regional spillovers, and documents heterogeneity by network redundancy. Section 5 checks the robustness of results, probing sample composition, testing for non-random exposure, and reporting a replication using external Cloudflare disruption data. Section 6 traces the transmission channels through trade, consumption, productivity, foreign investment, and finance. Section 7 concludes with policy implications.

2 Background

2.1 Cable disruptions as a critical and plausibly exogenous connectivity shock

As broadband access expands, exposure rises where international capacity hinges on a small number of undersea routes and landing points. European policymakers explicitly treat this layer as critical infrastructure: the EU’s NIS2 cybersecurity directive brings digital-infrastructure entities operating submarine cables into scope, while the Critical Entities Resilience (CER) directive sets a cross-EU framework to harden such assets.⁶ In 2025, the Commission went further with an EU Action Plan on Cable Security to coordinate protection and resilience measures across borders. This policy reflects the reality depicted in this paper: subsea cable networks are now critical and their failure can have cascading effects on the economy (Bueger et al. 2022).

Occurrence

SMC faults are common. The global network spans 1.8 million km of telecom cables; 100–200 damage events are recorded annually, concentrated in shallow water (<100 m) where accidental human interactions dominate, especially fishing (nearly half of documented faults) and anchoring. While suspicions of sabotaged cables make the headlines, those episodes remain marginal compared to accidental damages⁷, while the ICPC advocates for charting cables on nautical charts to prevent their

⁶ Available [here](#).

⁷ *Telegeography* (2025a) invokes *Hanlon’s razor* in a piece explaining the causes of undersea cable breaks: “*Never attribute to malice that which is adequately explained by stupidity.*” Talking about recent incidents in the Baltic Sea, a Washington Post

occurrence⁸. Put simply, the modal break is neither planned nor targeted, it is the by-product of routine seabed activity, and its timing is effectively random relative to a country’s macro conditions (Clare 2021; Telegeography 2025b). However, cable faults do not systematically lead to internet disruptions in connected countries.

Transmission

Because international connectivity follows shared trunk routes, where a fault occurs is not where it is materially felt. Figure 1 maps a December 2008 incident on the SEA-ME-WE and FLAG systems: the break lay off the Egyptian coast, yet the countries reported as experiencing multi-day outages or major slowdowns, and coded as treated, stretch across the Middle East and South Asia and reach as far as Zambia and the Maldives.⁹ What determines whether a country is hit is its position in the physical cable network, not its proximity to the break (Imbs and Pauwels 2025). Economies adjacent to the maritime choke points where many systems converge, and peripheral economies that rely on a handful of landing cables, are the most exposed, because a single fault on a shared route can remove a large share of their international capacity. This geographic basis of exposure, fixed network topology rather than a country’s macro conditions, is the main structural determinant of who is affected, and it is what allows cable disruptions to serve as a connectivity shock orthogonal to growth, examined directly in the first-stage and permutation checks of Sections 3.3 and 5.2.

Whether, where and to what extent a submarine-cable fault will map into internet disruptions is hence difficult to predict *ex ante* because a physical break does not translate mechanically into who loses connectivity. Depending on how networks are engineered (redundant links, capacity, and contractual circuits), the same fault may make some online destinations unreachable, or instead prompt traffic to be diverted onto alternative routes with heterogeneous effects across networks and locations.¹⁰ Evidence also indicates that impacts can extend beyond traffic that directly used the failed span. In fact, when traffic is shifted onto backups, congestion on those backups can degrade performance even for users whose traffic would not normally traverse the broken cable (Chan et al. 2011). Finally, as noted by the Regional Internet Registry for Europe, Middle East and Central Asia (RIPE), one cannot directly infer which end-to-end paths actually traversed the damaged cables and that comprehensive public data on “who uses which cable, when” are lacking, so the set of affected economies is largely revealed only *ex post*.¹¹

Duration

Around 99% of cable systems are privately owned and thus not “flagged” to any single state, making repair times primarily an issue of international coordination rather than national policy. Repairing a damaged system indeed follows a well-established sequence, independent from country’s policies and context.¹² Operators localize the fault, then a specialized repair ship mobilizes, grapnels and

article stressed that faults were not intentional but “accidents caused by inexperienced crews serving aboard poorly maintained vessels.” Nevertheless, as robustness check, I adjust the disruption exposure’s definition by dropping events associated with suspected or confirmed episodes of sabotage, but also to natural disasters that could directly affect growth through physical and human damages.

⁸See ICPC viewpoint: “Charting Submarine Cables Is Critical for Maritime Safety & Infrastructure Protection”, International Cable Protection Committee, Updated Wednesday 25 June 2025.

⁹Other episodes show the same pattern. In June 2017 a MainOne fault about 3,000 km off Portugal degraded service in Nigeria and along the West African seaboard until repair (report). In March 2024 an undersea landslide off West Africa cut WACS, ACE, SAT-3 and MainOne at once, degrading connectivity in 10–13 countries despite the offshore damage (report 1, report 2, report 3). Further SubTel Forum alerts document similar multi-day disruptions in Vietnam, Pakistan, and Southern Africa.

¹⁰See RIPE blog: “Mediterranean Fibre Cable Cut - a RIPE NCC Analysis”, Eds: Rene Wilhelm, Chris Buckridge, RIPE NCC Science Group.

¹¹See RIPE blog: “Does the Internet Route Around Damage? - Baltic Sea Cable Cuts”, Emile Aben, RIPE Labs, 20 Nov 2024.

¹²Cable repair ships are regular civilian vessels flying a national flag, but they operate under international law and large regional maintenance contracts that cover vast ocean areas.

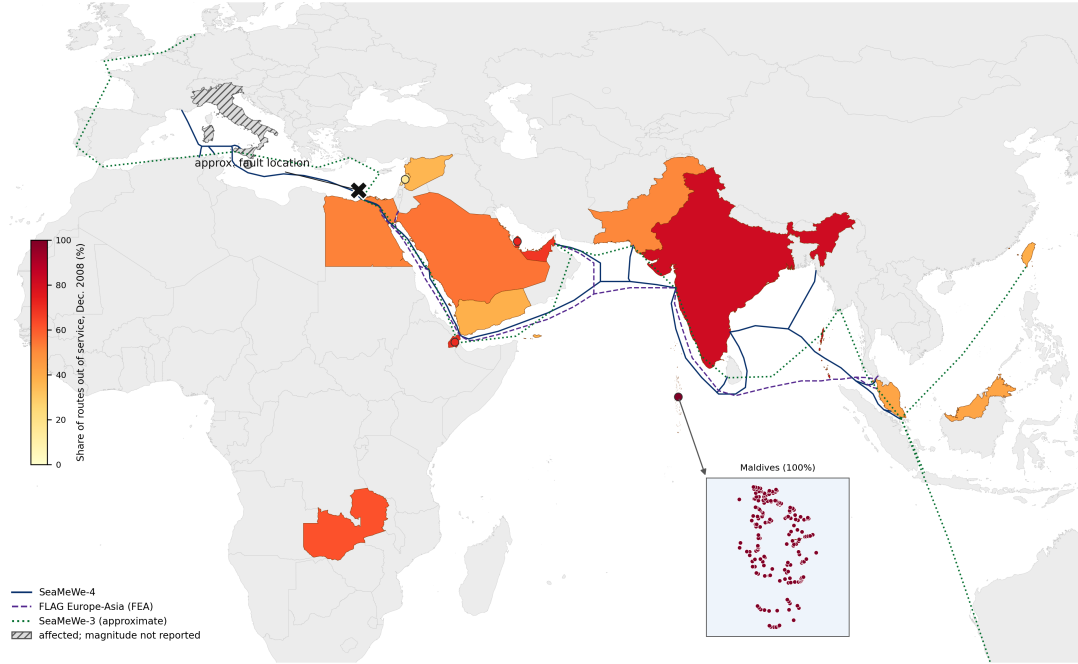


Fig. 1: Internet disruption events related to the SEA-ME-WE and FLAG systems failure, December 2008

Notes: Affected countries are those recorded in the *SubTel Forum*-based database for the December 2008 SEA-ME-WE and FLAG failures. Colour reports the share of routes out of service, as documented by [SubmarineNetworks](#). Malta and Italy, recorded as affected (see [Wikipedia page](#)) but without a documented outage magnitude, are shown hatched. Small countries (Qatar, Lebanon, Djibouti, Malta) are marked with a dot for legibility, and the Maldives, the most affected with all routes out of service, is enlarged in the inset. The cross marks the approximate location of the cable faults, in the Mediterranean off the Egyptian coast near the Alexandria landing stations.

recovers the cable, trims damaged sections, splices in new lengths, tests, and re-buries where required ([Telegeography 2025b](#)). Today, maintenance relies on regional contracts and zones (e.g., ACMA for the Atlantic, MECMA for the Mediterranean/Red Sea), with a handful of ships covering vast areas, highlighting capacity limits when multiple faults coincide: among the 77 cable ships, only 22 are designed for cable repair.¹³ Industry analyses warn that much of the global repair fleet is ageing, and that meeting demand and avoiding longer queues will require US\$3 billion for 15 replacement and 5 additional ships to sustain service levels as cable-km rise toward 2040.¹⁴ Therefore, determinants of cable repair duration are geophysical and logistical, not driven by national macroeconomic trends or policies, another reason repair duration is treated as a plausibly exogenous “dose” in the event-study and dose–response estimates.

2.2 Related Literature

Early work on the ICT–growth nexus established that telecommunications and broadband networks generate large growth effects, functioning as enabling technologies with network externalities that spread across sectors ([Röller and Waverman 2001](#); [Czernich et al. 2011](#)).

Building on these contributions, more recent work has turned specifically to submarine cables, the backbone of international connectivity, and extended the analysis to developing economies ([Hjort and Poulsen 2019](#)). First, recent studies have highlighted their contribution to Internet accessibility, especially in Africa ([Hjort and Poulsen 2019](#); [Cariolle 2021](#); [Cariolle et al. 2025](#)). Regarding their socio-economic impacts, several recent studies document trade effects of submarine cable deployments.

¹³See this animated [piece](#) from *The Verge*. See also the maintenance map in Appendix [OA.1.1](#).

¹⁴See [Telegeography’s blog](#): “It’s Going to Take \$3 Billion to Ensure Submarine Cable Repair Ships Can Keep the World Connected”, *Telegeography* staff, Jun 30, 2025.

Cariolle and da Piedade (2023) show that improved cable connectedness significantly promotes export upgrading in developing countries. Imbruno et al. (2025) find that bilateral connectivity through shared cable systems facilitates exporters’ access to foreign markets, particularly for high-productivity firms. Haltenhof (2019) reports that bilateral connectivity stimulates services trade flows, with elasticities highest in data-intensive sectors, while Herman and Oliver (2023) documents complementary effects in both goods and services trade.

A final strand of research extends the emphasis from trade to domestic and microeconomic activity. Eichengreen et al. (2023) show that the availability of international bandwidth facilitates cross-border financial transactions, lowering frictions in payments and settlements that are critical for trade and capital flows. Complementarily, D’Andrea and Limodio (2024) find that high-speed internet via cables promoted financial technology adoption, expanded interbank market activity, and increased firm access to credit and sales. These results underscore how cable connectivity affects not only cross-border flows but also the functioning of domestic financial systems, amplifying growth potential. In this regard, Simione and Li (2021) use the staggered arrival of submarine cables in Sub-Saharan Africa as a quasi-experiment and find that resulting higher internet penetration rates significantly boost per-capita GDP growth and productivity, while also shifting employment and output shares toward the services sector. This macro evidence is corroborated at the firm level by Hjort and Poulsen (2019), who provide the first causal evidence on how fast internet arrival in Africa, through gradual SMC laying over the last decades, boosts firm productivity, employment, and export performance.

A distinct strand of recent work characterizes the supply-side economics and network structure of the cable market itself. Caoui and Steck (2025) model the global internet backbone as a dynamic entry game and show that buyer demand for route diversification, which hedges against disruption risk, is a primary determinant of cable entry and long-run market structure. Distortions from operators’ inability to internalize the social value of diversity are as large as business-stealing distortions. Jeon and Rysman (2025) characterize cross-border data flows as international trade and show that cable capacity is a binding constraint on country-to-country data exchanges, with elasticities of data trade to capacity highest in low-provision settings. Imbs and Pauwels (2025) build a new dataset on global cable topology, redundancy, and capacity over 2010–2025 and show that network vulnerability is persistent, with fragility concentrated along the MENA–South Asia corridor and correlating with network centrality. None of these papers estimates the macroeconomic costs that countries bear when disruptions occur.

This body of evidence on trade and growth effects of SMC connectivity has a direct implication for disruptions: when submarine cables fail, the same mechanisms that stimulate and organize economic activity in normal times operate in reverse, depressing flows through higher transaction and information costs, unreliable transport and logistics, and impaired cross-border coordination. However, the effects of connectivity shocks may be asymmetric as the economic damage from disruptions likely exceeds the proportional benefits of connectivity access due to heightened digital dependence. Modern economies have slowly adjusted around continuous connectivity, with just-in-time supply chains, real-time financial systems, and cloud-dependent operations creating vulnerabilities that amplify disruption costs beyond simple connectivity gains in reverse.

The closest evidence to this study’s is provided by Steinwender (2018), who exploits the 1866 establishment of the transatlantic telegraph, and within the post-telegraph period its temporary technical failures, to identify a widening of the cross-market price gap and a contraction in trade in a single commodity market over a one-year window. That window captures the contemporaneous effect but cannot track a medium-run output path, and the contrast with the persistence documented here

is consistent with the broader role digital connectivity now plays across production, trade, payments, and finance and with the deeper integration of economies, so that a present-day break propagates further and unwinds more slowly than a telegraph-era information lag. This paper thus scales that evidence from a single traded product to macroeconomic aggregates across 146 economies, and frames it in the current digital era.

2.3 Mechanisms

International connectivity is an enabling intermediate input for domestic production, consumption, trade, and finance (Deloitte 2016; Hjort and Tian 2025). Submarine-cable outages restrain international and domestic connectivity access and therefore act as a supply and demand shock that simultaneously lowers operational efficiency and raises adjustment frictions. Notably, these shocks can generate (i) *direct* losses in the affected country due to higher coordination costs, financial and trade frictions; and (ii) *regional spillovers* as trade flows, contracts, and investment are diverted toward substitute locations.

2.3.1 Direct effects

Operational disruptions, payments frictions, and immediate productivity losses

Connectivity loss immediately impairs firms’ ability to operate: internal communication and monitoring, digital workflows, cloud-based tools, and remote coordination are disrupted, reducing effective labor input and measured productivity. In parallel, outages disrupt the settlement layer of the economy. Even when production and sales can continue, firms may be unable to issue invoices, receive electronic payments, reconcile accounts, or execute domestic and cross-border settlements. These disruptions delay cash receipts while wages, rent, and input bills remain due, lengthening firms’ cash-conversion cycles precisely when uncertainty rises. In working-capital frameworks, longer payment lags tighten liquidity and lead firms to cut employment and production (Barrot and Nanda 2020). This channel therefore predicts an immediate contraction in activity through operational downtime and cash-flow stress.

Supply-chain disorganization and trade frictions

Beyond within-firm disruptions, outages raise the costs of coordinating and executing trade activities. International shipments rely on continuous information exchange across firms, banks, insurers, freight forwarders, ports, and customs authorities. By slowing documentation, compliance, tracking, and release of goods, degraded connectivity acts like a time trade barrier: each additional day of delay reduces bilateral trade (Djankov et al. 2010) and is equivalent to a sizable ad-valorem cost (Hummels and Schaur 2013). Because trade is increasingly “servicified”—bundling finance, insurance, transport intermediation, and digital logistics (Baldwin 2016)—connectivity failures that impair these service inputs translate directly into weaker import and export performance. Steinwender (2018) provides quasi-experimental evidence of this mechanism, showing that temporary technical failures of the transatlantic telegraph cable raised the US–UK cotton price gap by 23% per additional day of information lag. Malgouyres et al. (2021) confirms this mechanism at the firm level, showing that broadband expansion in France increased firm imports by around 25%, primarily through the product and sourcing-country margins. When inputs are relationship-specific and difficult to substitute quickly, disruptions propagate along production networks as buyers cannot seamlessly switch suppliers (Barrot and Sauvagnat 2016), amplifying the initial shock beyond the directly exposed firms.

Amplification through financial frictions

Connectivity disruptions can amplify real effects through two complementary financial channels. First, banking and credit provision are information- and communication-intensive: improved connectivity can expand credit supply and enhance monitoring and competition in lending (D’Andrea and Limodio 2024; D’Andrea et al. 2025). The reverse implication is that disruptions can temporarily impair screening/monitoring, raise operating costs, and worsen credit terms or availability. Second, firms’ short-term liquidity needs rise during disruptions exactly when payment frictions and bank intermediation problems make liquidity scarcer. The interaction amplifies production and trade frictions, as constrained firms are less able to bridge temporary disruptions, which can turn transitory shocks into persistent losses of jobs and market shares (Barrot and Nanda 2020; Barrot and Sauvagnat 2016). Over longer horizons, if disruptions raise operating and financing wedges differentially across firms and sectors, they may depress investment and slow aggregate productivity growth through reallocation and pro-competitive channels, consistent with the mechanisms emphasized by Varela (2018).

2.3.2 Regional spillovers

Regional spillovers arise because higher trade, financial, information and production costs in the affected economy shift activity toward close substitutes. In spatial equilibrium and trade-based geography frameworks, changes in trade or interaction costs reallocate production, spending, and factor demands across locations, generating indirect effects beyond treated units (Allen and Arkolakis 2014; Ahlfeldt et al. 2015; Donaldson and Hornbeck 2016). In this context, a connectivity failure can induce trade and investment diversion, as contracts, customer support, outsourcing tasks, logistics intermediation, and other digitally enabled business services are rerouted toward nearby countries that remain reliably connected.

Trade diversion

Unlike many natural disasters, cable outages do not destroy physical capital (except for cable owners and operators). Persistent effects are therefore more naturally interpreted as arising from coordination and liquidity frictions, relationship disruption, and the reallocation of contracts and investment, rather than from a reduced capital stock.

Evidence from the natural-disaster literature confirms that large shocks generate trade diversion. Exporters cut sales to affected destinations by reallocating toward unaffected ones (London and Gigout 2023), while importers substitute toward alternative suppliers (Freund et al. 2022). This substitution is likely stronger for coordination-intensive digital services, where regional alternatives are more salient.

Competitiveness and investment reallocation

Investment responses can reinforce this mechanism. The disaster literature also provides evidence that localized shocks can reallocate mobile capital toward unaffected areas. Friedt and Toner-Rodgers (2022) document large and persistent reductions in FDI in disaster-hit regions within India and sizable positive spillovers into otherwise unaffected regions. They show that a substantial share of the lost investment reflects relocation within the country, and the effects persist for several years. In this setting, if connectivity disruptions impair the whole economy, they can lower expected returns to operating in the affected country by raising operational and financing costs, discouraging FDI and accelerating reallocation toward regional alternatives. Foreign investors may therefore postpone, scale down, or relocate projects toward proximate regional alternatives that can deliver services and coordinate cross-border activity more reliably.

Such shifts can also shape aggregate productivity. If disruptions disproportionately raise operating costs or financing constraints for outward-oriented and more productive firms (Imbruno et al. 2025)—those most reliant on cross-border coordination, payments, and digital logistics—these firms shrink, lose contracts, or delay expansion, and activity reallocates toward less exposed (and potentially less productive) producers. At the same time, weaker entry and competitive pressure can slow selection and reallocation toward the most efficient firms, with adverse implications for TFP growth, consistent with the competition-and-reallocation mechanisms emphasized by Varela (2018). As a result, exposed economies can experience persistent competitiveness losses, while nearby non-disrupted economies can attract investment and exhibit higher productivity growth as activity relocates within the region.

3 Empirical framework

In this paper, I estimate the dynamic treatment effects of Internet disruptions caused by submarine cable failures on economic growth and its components, using a country–year panel of 146 countries covering 2008–2020. I use the event-study estimator of de Chaisemartin and d’Haultfoeuille (hereafter CDH), designed to study staggered non-binary and non-absorbing treatment with possible heterogeneous dynamic effects on macro-economic outcomes (de Chaisemartin and d’Haultfoeuille 2024).

3.1 Econometric modelling

3.1.1 Baseline model

Standard two-way fixed effects event-study estimators are biased under heterogeneous treatment effects with staggered adoption (Goodman-Bacon 2021; Sun and Abraham 2021; Callaway and Sant’Anna 2021; de Chaisemartin and d’Haultfoeuille 2024). Because SMC disruptions are staggered, non-binary, and non-absorbing (countries may experience multiple disruptions of varying intensity, with no mechanical persistence across years), I rely on the estimator of de Chaisemartin and d’Haultfoeuille (2024) (hereafter DID_M), which accommodates this setting with heterogeneity-robust Difference-in-Difference (DiD) and nonnegative aggregation weights.

Let y_{it} denote the macroeconomic outcome (e.g. GDP per capita growth or log GDP per capita) for country i in year t , and let D_{it} denote treatment intensity in country i , year t (the number of SMC-induced internet disruptions or SMC repair days). Let F_i be the first year in which D_{it} departs from its initial baseline level D_{i1} (which is zero for almost all countries), and define event time as $k = t - F_i$. I estimate

$$y_{it} = \alpha_i + \lambda_t + \gamma_i t + \delta_{r(i),t} + \delta_{c(i),t} + \sum_{k=-K_0}^{K_1} \beta_k \text{Event}_{it}^k + \varepsilon_{it}, \quad (1)$$

where α_i and λ_t are country and year fixed effects, $\gamma_i t$ are country-specific linear trends, and $\delta_{r(i),t}$ and $\delta_{c(i),t}$ are region-by-year and income-group-by-year fixed effects, with $r(i)$ denoting country i ’s World Bank region and $c(i)$ its income group. The pre- and post-event horizons are set to $K_0 = 5$ and $K_1 = 7$, respectively. For never-treated units, $\text{Event}_{it}^k = 0$ by convention. The event-time-by-intensity regressor is

$$\text{Event}_{it}^k \equiv D_{it} \mathbf{1}\{t - F_i = k\}, \quad k = -K_0, \dots, K_1,$$

so Event_{it}^k equals the number of disruptions (or repair days) that country i experiences k years after its first departure from zero exposure, and is zero otherwise. The coefficients β_k trace out the effect

of a one-unit increase in treatment intensity at event time k .¹⁵ Identification relies on a conditional parallel-trends assumption for outcomes in the absence of changes in D_{it} .

3.1.2 Identification under regional spillovers

An important concern for the empirical analysis is that submarine-cable disruptions may generate interference: outcomes in one country can be affected not only by its own disruption exposure, but also by disruptions occurring elsewhere in the region through diversion and reallocation of trade, contracts, and investment. Since such interference may lead to a violation of the stable unit treatment value assumption (SUTVA), I therefore distinguish between direct effects in treated countries and spillover effects on untreated countries (Hudgens and Halloran 2008; Özler 2018). Huber and Steinmayr (2021) provide a general framework for this decomposition in a DiD setting, maintaining SUTVA at the aggregate (regional) level while allowing within-region spillovers to operate freely, an assumption I adopt at the country level.

I distinguish three groups: (i) *treated* countries T ; (ii) *spillover-only* countries S , untreated but located in a region with at least one disruption; and (iii) *pure controls* C , untreated in non-exposed regions. The baseline event-study with region-by-year fixed effects identifies the *direct effect net of spillovers* ($T - S$); a separate event-study on the non-treated sample identifies *spillover effects* ($S - C$). These satisfy the accounting identity $T - C = (T - S) + (S - C)$, from which I recover an *implied ITT* relative to pure controls. The formal potential-outcomes framework, identification assumptions, and mapping to empirical specifications are detailed in Online Appendix OA.4.

3.2 Data

The empirical analysis uses an annual country panel for 2008–2020 that combines macroeconomic variables from the World Bank’s Open Database, Global Productivity Database, Exporter Dynamics Database, the Bank for International Settlements Data Portal, with SMC infrastructure data from Telegeography and an original database on cable-related internet disruption episodes. The data coding protocol applied to build cable disruption data is explained in Online Appendix OA.1.1.

3.2.1 Connectivity disruptions and SMC failures

Definition

In this paper, the treatment is defined at the country–year level as an episode in which a submarine-cable failure generates a measurable loss of international connectivity for that country.¹⁶ Countries that are topologically connected to the faulted cable but for which traffic is rerouted without noticeable service degradation are coded as *not treated*. Therefore, cable failure attribution follows the principle of material impact rather than physical proximity. Under this approach, a single submarine cable failure affecting multiple countries is recorded as several country-specific events, with affected nations sometimes located far from the break site.

The disruption database was hand-compiled from *SubTel Forum* —the SMC industry’s main technical platform— which systematically documents national-level disruptions caused by cable failures.¹⁷

¹⁵By normalization, $\beta_{-1} = 0$. Because treatment is non-binary, I use the `normalized` option in `did_multipleregt_dyn` Stata command, which rescales each local effect by the size of the underlying treatment change before aggregation. The β_k can therefore be interpreted as the average effect, k years after first exposure, of a one-unit increase in treatment intensity (one additional disruption or one extra repair day), relative to a status-quo path in which the country remained at zero exposure throughout; see de Chaisemartin and d’Haultfoeuille (2024) for details.

¹⁶The unit of analysis is therefore “country experiences an SMC-induced internet disruption”, not “cable segment near the country is physically damaged”.

¹⁷Data are collected from the SubTel Forum’s [cable faults and maintenance desk](#), complemented with Akamai “State of Internet Connectivity” reports and cross-checked with targeted web searches.

For each event, I distinguish physical *cable breaks* from technical faults or planned maintenance, termed *cable outages*, and I record *repair duration* based on SubTel Forum updates. Unlike Internet disruptions driven by internal policy factors—such as government-ordered shutdowns—submarine cable failures constitute plausibly exogenous shocks that primarily impair cross-border data flows rather than domestic access alone, making them well suited for identifying the effects of international connectivity losses.¹⁸ In fact, most SMC failures stem from incidents caused by maritime activities (fishing, anchoring) (Carter 2009; Clare et al. 2023), but the dataset also records less frequent causes such as natural hazards (seismic events, landslides, storms, flooding), sabotage, and other technical causes (e.g. power failures).¹⁹ This taxonomy allows me to exclude failures plausibly linked to policy or to external shocks that might affect economic activity independently from connectivity.

To study the growth effects of SMC-related connectivity shocks, I use two key variables: (i) the annual number of Internet disruptions caused by cable failures, a *treatment-incidence* measure; and (ii) the number of SMC repair days, a *treatment-intensity* measure. In robustness analysis I separate SMC-related incidents into (i) *cable breaks* (confirmed physical damage requiring a repair ship), and (ii) *cable outages* (service-affecting faults without confirmed physical damage, such as repeater failures or power-feed issues). In robustness analysis, I also use alternative disruption data from the *Cloudflare Radar*²⁰—which provides since 2022 a systematic and documented reporting of connectivity disruptions across the world—to check the reliability of *SubTel Forum*’s data and the consistency of estimated relationships.

Timing and duration

The timing and duration of each event are read from the dates of the SubTel Forum posts that report a fault and its subsequent maintenance and restoration updates, with repair duration measured as the span between the first post flagging a fault and the post recording its resolution. The platform posts updates regularly, and for most events these posts date the onset of the disruption and the completion of the repair precisely, corroborated by operator notifications or press releases. Where such corroboration is unavailable, I date the fault and its resolution by the SubTel Forum posts themselves, which bound the true dates within the posting lag. For a few events the repair-completion date is never reported, leaving the repair duration missing.²¹

Descriptive statistics

Summary statistics of these variables are reported in Appendix Table A.1.1.2, and their time and geographic distributions in Figure 2. They reveal that submarine cable failures are relatively infrequent but highly variable events with sizable repair burdens. At the country-year level (Appendix Table A.1.1.2, Panel A), the average country experiences about 0.10 cable disruptions annually, with cable breaks (0.084) more common than outages (0.018); the high standard deviations (0.39 and 0.35) indicate substantial cross-country heterogeneity. On average, a typical disruption is associated with 11 repair days (Appendix A.1.1.4). Therefore, repair activities impose a substantial time cost, averaging almost one day per country-year (0.92) but reaching 80 days in extreme cases, with a standard

¹⁸Concerns about non-random exposure to these disruptions are addressed through pre-trend analysis and placebo-permutation tests in the spirit of Borusyak and Hull (2023).

¹⁹Natural hazards represent the second most common cause, accounting for less than 20% of failures (Clare et al. 2023), but often generate multiple deep-sea breaks where repairs are more complex. Clare et al. (2023) emphasize that climate change and sea-level rise will however intensify these risks.

²⁰See Cloudflare Radar webpage.

²¹To place every event on the same footing, I define the treatment at the country-year level for the main analysis and at the country-month level for the first-stage analysis with the full panel of Section 3.3. The daily event study in that section completes the SubTel Forum timestamps with independent contemporaneous operator and press reports for the subset of episodes whose onset day the SubTel posts alone do not pin down precisely.

deviation of 5.4. In global annual totals (Panel B), the sample averages 19 disruptions and 174 repair days per year over 2008–2020. Excluding 2020 (Panel C) reduces these to 14.4 disruptions and 118 repair days, confirming that 2020 was an unusually intense year.

Aggregating failures by region in Figure 2 (right graphs), Sub-Saharan Africa (SSA) accounts for the largest totals of both Internet disruptions and cable repair days, followed by East Asia & Pacific (EAP), the Middle East & North Africa (MENA), and South Asia (SA). Europe & Central Asia (ECA) and Latin America & the Caribbean (LAC) register much smaller tolls. The contrast between the number of disruptions and the stock of repair days is informative, as a few episodes are long-lived and queue repair capacity. So events’ duration, not just incidence, drives exposure. The annual series in Figure 2’s left graphs show lumpy time patterns. The count of disruptions rises from 2019, driven mainly by outages rather than breaks, with the larger shift occurring in 2020. Repair-day exposure exhibits much larger variance, with a pronounced spike in 2020. This reinforces the choice to use repair duration as the complementary “dose” variable in the empirical analysis.²²

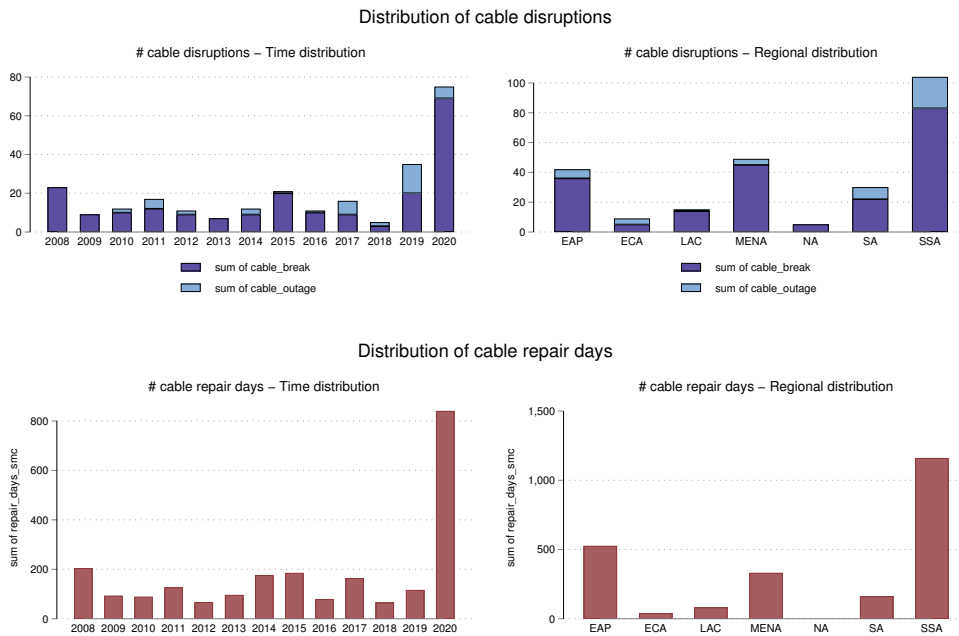


Fig. 2: Internet disruptions and repair days, regional and time distributions, 2008–2020

The geography of submarine cable failures

The country-level incidence of disruptions concentrates in locations defined by the physical layout of the cable network. Figure 3 maps the cumulative count of SMC-related disruptions per country over 2008–2019 against critical world maritime choke points. The highest counts fall on two types of location: economies on the periphery of the network that depend on a small number of landing cables, and countries adjacent to the maritime choke points where many systems converge — the Red Sea and Bab-el-Mandeb corridor feeding the Suez landings, the Strait of Malacca, the Mediterranean approaches off North Africa, and the West African coast. Isolated island economies of the Indian Ocean, the Pacific, and the Caribbean, shown in the insets, record disruptions out of proportion to

²²Results are robust to dropping the 2020 spike. See robustness Section 5.

their size for the same reason. This distribution follows the topology and vulnerability of the network rather than the economic characteristics of the countries it serves (Imbs and Pauwels 2025). Most faults originate in fishing and anchoring over shallow continental shelves (Clare et al. 2023), so a country's position relative to these shipping-dense seabeds, not its growth path, governs how often its international routes are severed. This geographic basis of exposure is what allows cable disruptions to serve as a connectivity shock. I test the assumption that exposure is unrelated to a country's growth trajectory directly in the robustness section.²³

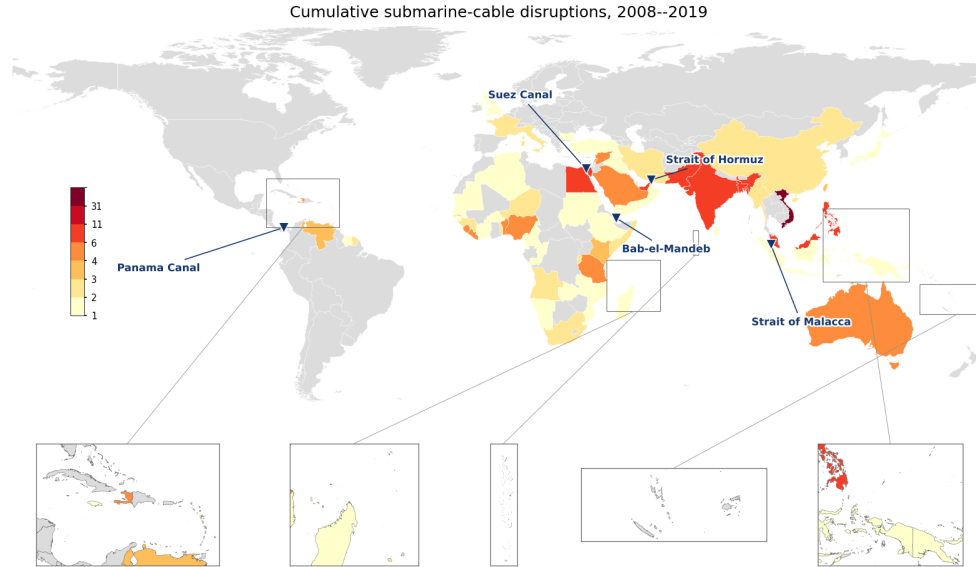


Fig. 3: Geographic distribution of SMC-related internet disruptions, cumulative 2008–2019.

Notes: Shading reports the cumulative count of submarine-cable-related internet disruptions recorded per country over 2008–2019 in the *SubTel Forum*-based database. Navy triangles mark the principal maritime choke points where many cable systems share crowded seabeds (Panama and Suez Canals, Bab-el-Mandeb, and the Straits of Hormuz and Malacca). Insets zoom on isolated island economies of the Indian Ocean, the Pacific, and the Caribbean, whose disruption counts are otherwise hard to read at world scale.

3.2.2 GDP growth and other economic outcomes

I study growth and macroeconomic aggregates that plausibly respond to connectivity shocks. For each following aggregate I focus on levels in constant 2015 USD (log) or their growth rates: GDP, household final consumption, government final consumption, fixed gross capital formation, exports and imports of goods and services, and Foreign Direct Investment (FDI).

Because submarine cables are the main digital infrastructure for cross-border data flows, I place particular emphasis on trade, productivity, and financial channels. Therefore, in addition to total trade in goods and services, I study disaggregated trade flows, such as merchandise trade and services trade, but also exchanges of commercial services, ICT services, financial services or transport services.²⁴ To investigate the mechanisms behind macroeconomic adjustments, I also study the productivity effects of cable disruptions, using the World Bank Global Productivity Database, and financial frictions, using Credit and Locational Banking Statistics from the Bank of International Settlements. Finally, I use the

²³Through the first-stage analysis of Section 3.3 and the permutation tests of Section 5.2

²⁴These services-trade series are available only in current USD; I obtain series in constant dollars using implicit deflators, which are the ratio of the exports (or imports) of goods and services expressed in current USD over exports (imports) in constant USD.

country-level exporter data from the World Bank’s Exporter Dynamics Database to connect empirical evidence from macroeconomic aggregates to those from observed import and export extensive margins.

3.3 First stage: do disruptions reduce measured connectivity?

Identification assumes that a *SubTel Forum*-coded disruption maps into a loss of international connectivity. I test this against routing data built independently of the disruption coding. The connectivity measure is the count of IPv4 address blocks (*routed prefixes*) that a country originates and that remain visible in the global routing table, from the Routing Information Service (RIS) of the Réseaux IP Européens (RIPE). A fall in this count means part of a country’s address space has become unreachable, which is the operational definition of a connectivity loss.

I use two designs, reported together in Figure 4. The first is a daily event study around the multi-country episodes whose onset date is fixed from the SubTel Forum timestamps and contemporaneous operator and press reports (17 episodes, 62 countries, 107 country-events; Online Appendix Table OA.2.1). Routed prefixes are flat before onset, drop on the event date, and recover within three weeks (panel a). Over the first two weeks the level holds 2.5% below baseline (panel b). The second design is a country-month panel covering 172 economies, estimated with the DID_M estimator (de Chaisemartin and d’Haultfoeuille 2024) on the full *SubTel Forum* event dataset. Online Appendix Table OA.2.2 reports the full panel, including two-way fixed-effects estimates and the outage-day, severity, and routed-level outcomes. In the month of the disruption the deepest within-month draw-down deepens by 10.0 percentage points (panel c) and the count of distinct outage episodes rises by 0.16 (panel d), both significant on impact with flat placebos, and both fade over the following months. Outage days and severity instead remain elevated through the second month after onset, with outage days up 0.85 at one month and 1.87 at two and severity up by roughly 53 at both (Online Appendix Table OA.2.3). The withdrawal’s depth registers on impact, while the time a country spends below baseline keeps accumulating into the second month before reverting. Therefore, daily event study and the monthly panel analysis both show routed prefixes falling at disruptions.

3.4 Estimation sample

Summary statistics of cable failures for the estimation sample are reported in Appendix Table A.1.1.3. Summary statistics of main macroeconomic outcomes are reported in Appendix Table A.1.1.5.

The estimation sample retained in baseline DID_M estimations consists of 1,898 observations covering 146 countries and 13 years (2008-2020). The estimation sample distribution by region and income group is given in Table 1. It includes 650 observations from high-income countries (HICs) (34%), 234 from low-income countries (LICs) (12%), and 1,014 observations from middle-income economies (MICs) (53%). Regional representation is likewise diverse: Europe and Central Asia (ECA) accounts for the greatest sample share (33%), followed closely by Sub-Saharan Africa (SSA) (27%), Latin America and the Caribbean (LAC) (23%), East Asia and Pacific (EAP) (12%), the Middle-East and North Africa (MENA) (4%) and South Asia (SA) (1%).²⁵ Among treated countries, SMC disruptions are more recorded among MICs, more located in SSA, while ECA is over-represented in the group of never-treated countries. Additional analysis shows that results are robust to successively removing regions and income-groups from the sample, including removing SSA (Section 5).

²⁵Due to identification concerns, North America (USA and Canada) is not represented in the sample since identified cable-related internet disruptions in the region are all located in the US, and caused by sabotage (in 2014 and 2015, in the San Francisco Bay) and natural hazards (in 2008). Results remain however robust to considering these events and region in the estimation sample and are available on request.

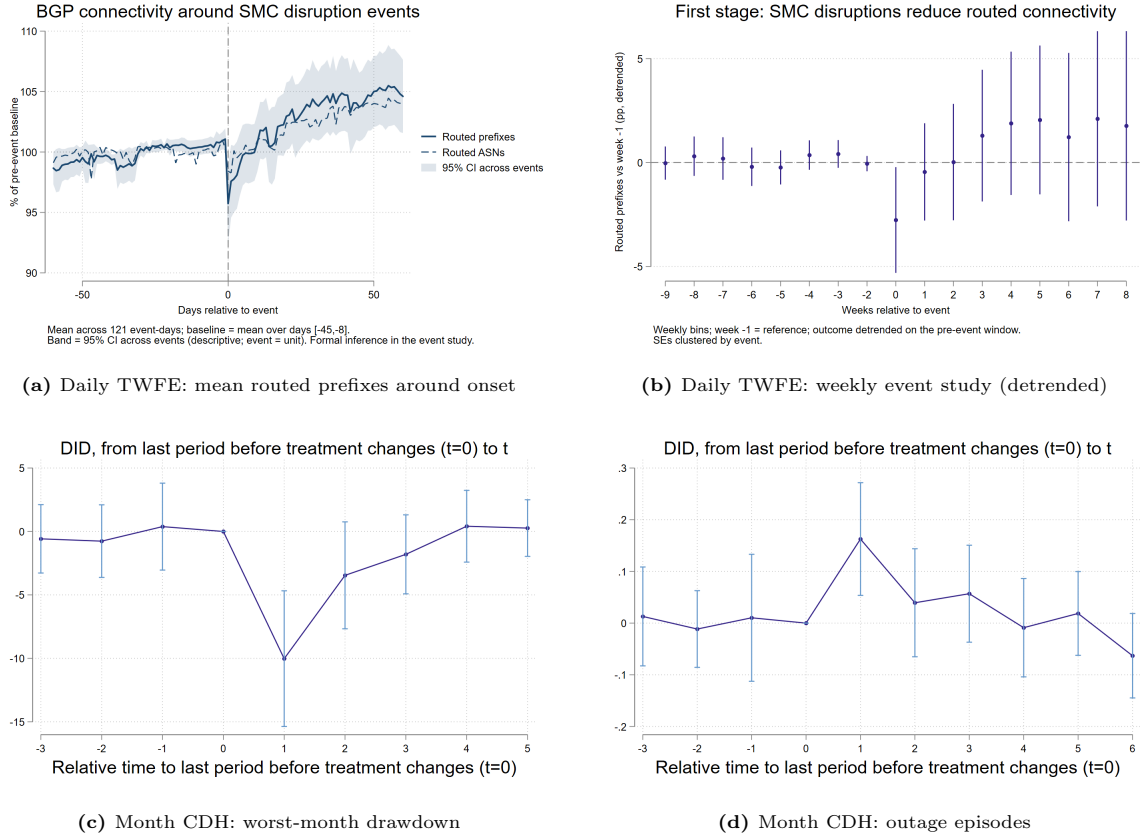


Fig. 4: First stage: submarine-cable disruptions reduce routed connectivity.

Notes: The outcome is each country's routed connectivity in the global BGP routing table (Routing Information Service, RIPE), measured daily. Routed prefixes = the number of distinct IPv4 address blocks the country originates that are visible in the global routing table; a fall means part of the address space is withdrawn and becomes unreachable. Routed ASNs = the number of the country's autonomous systems (independently administered networks — ISPs, content and hosting providers, enterprises) that appear as a route origin in the same table. (a) cross-event mean routed prefixes in event time, as a percentage of the pre-event baseline (mean over days [-45, -8]), 95% band across the 107 country-events (17 episodes) of Online Appendix Table OA.2.1. (b) event-study coefficients in weekly bins, outcome detrended by an episode-specific linear pre-trend, week -1 the reference, SE clustered by country-event. (c)–(d) dynamic effects from de Chaisemartin and d'Haultfoeuille (2024) (`did_multiplgt_dyn, normalized`, SE clustered by country) on the 172-country country-month panel; *x*-axis in months relative to a disruption, with placebo (pre-) estimates to the left of 0. *Worst-month drawdown* = deepest daily shortfall of routed prefixes relative to their trailing 60-day baseline within the month (percentage points, negative). *Outage episode* = a run of consecutive days with routed prefixes more than 10% below that baseline (gaps ≤ 3 days merged).

Baseline treatment–path diagnostics in Appendix Table A.1.2.1 show that both treatment variables generate a wide variety of paths that allow recovering six post-treatment effects. In fact, almost 40% of switchers (22 of 58) have a fully-observed window of 6 post-treatment periods in the baseline estimation sample. For the cable-disruption count, roughly 80-90% of groups follow four main paths in which treatment intensity switches on once and then remains stable. For the repair-days measure, treatment paths are more dispersed across groups, reflecting the fact that outage duration varies more flexibly year by year. Nonetheless, a sizeable share of groups still follows a small number of dominant paths, and the detected paths cover all switching groups, so dynamic effects are well identified in the full estimation sample. Repeating the exercise excluding 2020, which features an unusually high number of disruptions at the very end of the sample, yields an even more concentrated distribution of treatment paths (a few simple paths account for a larger share of treated groups).²⁶

²⁶Nevertheless, I retain 2020 in the baseline estimation framework to show that the paper's results are robust to the pandemic year inclusion, which acts as a statistical noise across estimations, as shown in Section 4. See also Appendix A.1.3.

Table 1: Estimation sample composition

	Observations	Percent
Panel A. Estimation sample		
<i>By income group</i>		
High income (HIC)	650	34.25
Middle income (MIC)	1,014	53.42
Low income (LIC)	234	12.33
<i>By region</i>		
Europe and Central Asia	624	32.88
Sub-Saharan Africa (SSA)	520	27.40
Latin America & Caribbean (LAC)	429	22.60
East Asia and Pacific	221	11.64
Middle East & North Africa (MENA)	78	4.11
South Asia	26	1.37
Total	1,898	100.0
Panel B. Switchers		
<i>By income group</i>		
High income (HIC)	104	13.56
Middle income (MIC)	481	62.71
Low income (LIC)	182	23.73
<i>By region</i>		
Sub-Saharan Africa (SSA)	403	52.54
Latin America & Caribbean (LAC)	117	13.8
South Asia	13	1.69
East Asia and Pacific	104	13.56
Europe and Central Asia	91	11.86
Middle East & North Africa (MENA)	39	5.08
Total	767	100.0

4 Main Results

This section presents baseline evidence on the growth effect of cable failures. The goal is twofold. First, to establish whether exposure to cable disruptions is followed by movements in GDP per-capita growth, and second, to assess whether the magnitude of these effects scales with the duration of disruptions. The analysis begins with baseline estimations assessing the average dynamic effect across exposed countries, before turning to heterogeneity tests that explore whether digital infrastructure characteristics condition the transmission of shocks to the whole economy.

4.1 Baseline estimations

The upper panels of Figure 5 report the response of GDP per capita to a cable disruption, relative to non-disrupted economies in the same region. Detailed estimates are reported in Appendix A.1.2. Placebo leads are flat, consistent with parallel pre-trends. The GDP response declines steadily, reaching a trough of about 9% at four to five years, having passed through roughly 4% by $t + 2$ and 6% by $t + 3$.²⁷ The loss then recedes rather than compounding.

Appendix Figure A.1.2.1 reports significant and consistent estimates from a baseline specification excluding income-group-by-year fixed effects. The parallel specification excluding region-by-year fixed effects is presented in Section 4.2 (Figure 6) as part of the regional spillover analysis.²⁸ Country-level leave-one-out re-estimations (Online Appendix Figures OA.3.1 and OA.3.2) further confirm that point estimates remain negative across all 41 single-switcher exclusions: at $t+6$, the leave-one-out (LOO) distribution spans roughly -8 to -10.5 pp for GDP per capita growth and -6 to -10.5% for log GDP per capita, with no single switcher moving the headline outside the full-sample 95% confidence band.

²⁷The medium-run estimates rest on 22 switchers and 150 contributing observations at $t + 6$; see Appendix A.1.2.2. Coefficients on log GDP per capita cumulate the effect, while the growth-rate panels report the annual effect.

²⁸Additional estimations reported in Appendix A.1.6.5 show that the GDP decline is not caused by the recurrence of those connectivity shocks.

Replacing the cable disruption number with the number of cable repair days confirms previous growth patterns and adds interpretability to the results (Figure 5, bottom panel). One additional cable repair day depresses per capita growth by 0.17 pp at short horizons, reaching over time roughly -0.48 pp (year $t + 3$), -0.61 pp (year $t + 4$), and stabilizing at -1 pp ($t + 5$ and $t + 6$). With an average duration of repair of around 10-11 days, this estimate is consistent with estimates using the disruption episode variable. Log-level specifications show a comparable dynamic effect of repair days, with an immediate drop of 0.14% in year t , amplifying thereafter and reaching roughly 0.7% by years $t + 5$ – $t + 6$.

Once the 2020 pandemic year is set aside, GDP per capita stands about 2.5% below its counterfactual by the sixth year, so the medium-run cost peaks at four to five years and survives dropping any single switcher (Appendix A.1.3). In magnitude, this 8-9% medium-run loss is of the order documented for power outages (Fried and Lagakos 2023), banking crises (Cerra and Saxena 2008), and inter-state wars (Federle et al. 2026).

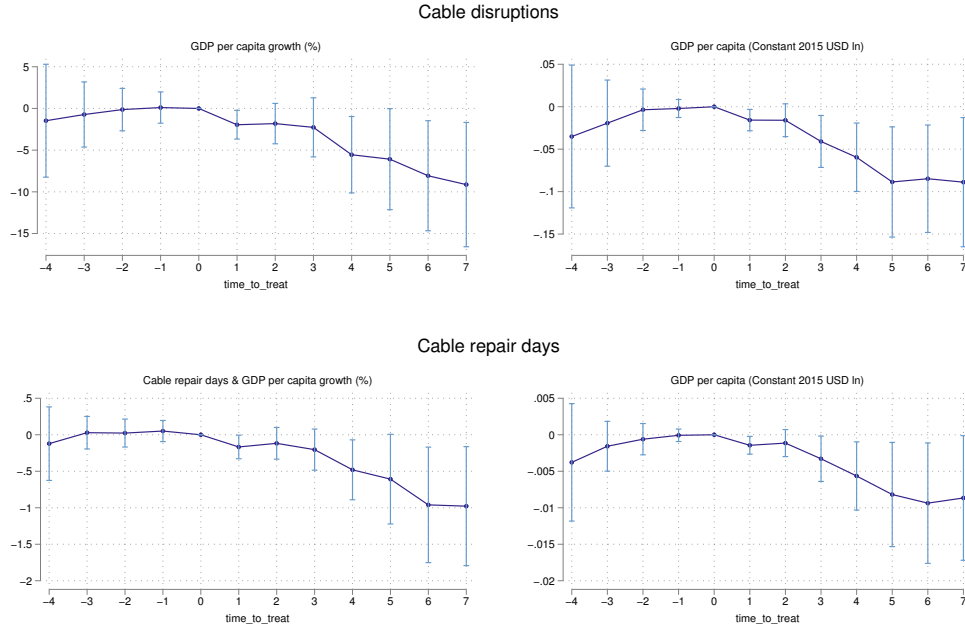


Fig. 5: Cable disruption: GDP per capita growth and level effects.

Notes: Left panel reports event-time coefficients for GDP per capita growth (%); right panel reports event-time coefficients for log GDP per capita (constant 2015 USD). Top panel reports event-time coefficients with the disruption count variable; bottom panel reports event-time coefficients with the repair days count variable. All panels show 95% confidence bands. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

4.2 Regional spillovers

In the presence of interference the baseline GDP effects should be interpreted as *net* losses reflecting a divergence process, since part of the relative decline in exposed economies may reflect contemporaneous gains in proximate, non-disrupted countries. This mechanism finds empirical support when broadening the counterfactual to non-disrupted countries outside the affected regions. Removing region-by-year fixed effects indeed yields a GDP per capita divergence of roughly 5–7% in the medium

run (Figure 6), against roughly 9% in the within-region baseline.²⁹ This attenuation is consistent with regional spillovers offsetting part of the within-region decline.

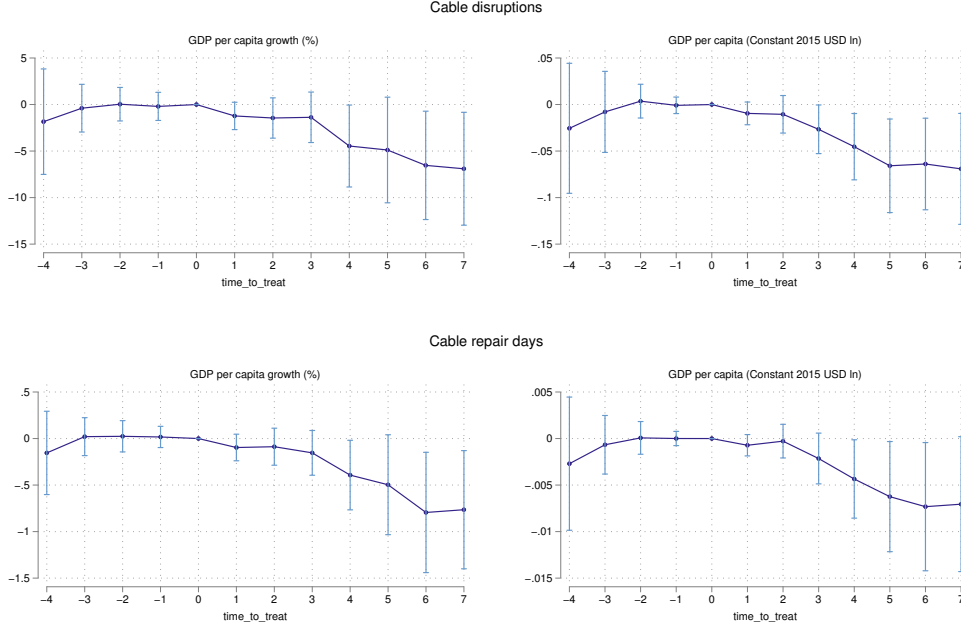


Fig. 6: Total effect of cable disruptions on GDP per capita: broader counterfactual.

Notes: Event-study estimates from the baseline specification re-estimated without region-by-year fixed effects, broadening the counterfactual to non-disrupted countries outside the affected regions. Top row shows effects of cable disruptions on overall GDP growth (%), log GDP per capita (center), and log GDP (right). Bottom row shows analogous effects for cable repair days. All panels report event-time coefficients with 95% confidence bands. Specifications include country fixed effects, year fixed effects, linear country trends, and income-by-year fixed effects, with standard errors clustered at the country level.

To probe the neighbour response directly, I construct two peer-disruption measures that aggregate, at the World Bank region level, the number of disrupted economies and the number of cable-disruption episodes occurring in a country's region, where one episode can affect several economies. I re-estimate the event study on the subsample of country-years with no domestic failure ($D_{it} = 0$), so that treatment varies only through exposure to disruptions elsewhere in the same region, including country fixed effects, year fixed effects, and a linear country trend. Figure 7 reports the neighbour response in both growth rates and log levels.

Placebo leads are flat and the post-treatment coefficients turn positive. Measured in growth rates, an unaffected neighbour gains about 2 percentage points per year four to six years after a regional disruption when exposure is the count of disrupted economies, and 4 to 5 percentage points when exposure is the count of disruption episodes. Measured in log levels, the cumulative gain is small, about 1 to 2 percent at $t+4$ – $t+5$. The two readings describe one process. Activity diverts to unaffected peers while the disruption is active, which raises their growth rate transitorily, and normalises once the link is repaired, so the accumulated level gain stays near the 2-percentage-point gap between the within-region baseline and the broader counterfactual. Because almost every non-disrupted country lies in a region that experiences some disruption over the sample, only two countries remain never-exposed controls. To enlarge this control pool, I also measure regional exposure among countries in

²⁹Country-level leave-one-out re-estimations (Online Appendix Figure OA.3.3) yield point estimates ranging from –4% to –8% at $t+6$, with all 41 LOO estimates remaining significantly negative.

the same region *and* income group, so that a country is treated only when a same-income regional peer is disrupted. The spillovers get sharper under this narrower measure (Appendix Figure A.1.4.4).

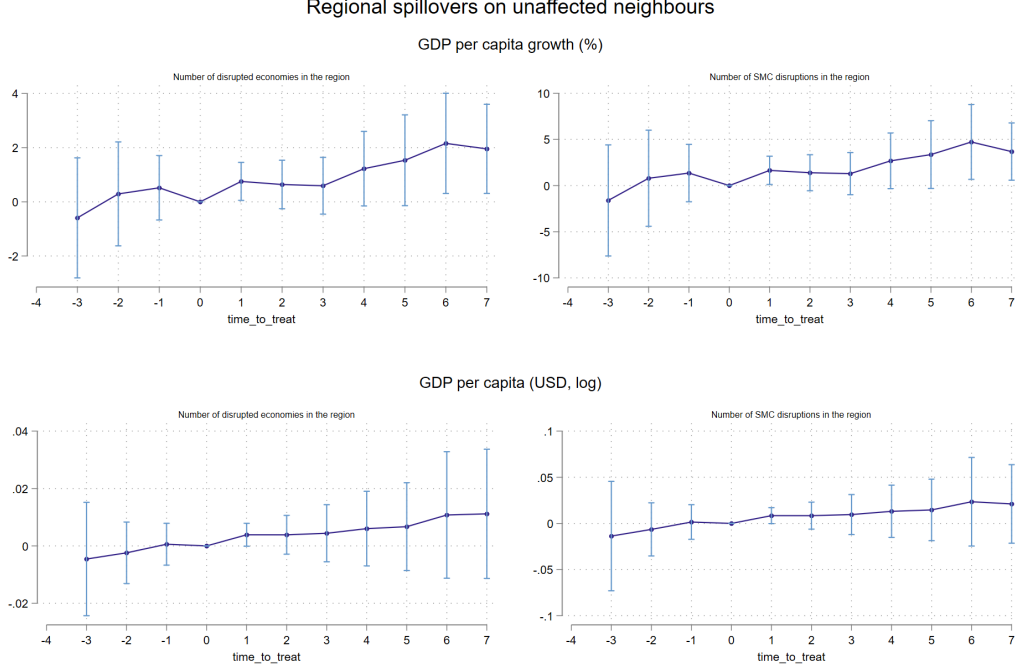


Fig. 7: redRegional spillovers of cable disruptions on unaffected neighbours: growth rates and log levels.

Notes: Event-study coefficients estimated on the subsample of country-years with no domestic cable failure ($D_{it} = 0$), where treatment varies through exposure to cable disruptions elsewhere in the same World Bank region. The top row reports GDP per capita growth (%). The bottom row reports log GDP per capita. Within each row, the left panel measures regional exposure by the number of disrupted economies and the right panel by the number of cable-disruption episodes. Specifications include country fixed effects, year fixed effects, and linear country trends, with standard errors clustered at the country level and 95% confidence bands. The joint test of post-treatment effects yields $p < 0.01$ in growth rates and $p = 0.11$ in log levels. The estimates are identified from 71–73 switching groups in growth rates and 64–66 in log levels among the non-disrupted countries, against only two never-exposed control countries. Year 7 corresponds to six years after the regional disruption.

Combining the within-region direct effect identified by the baseline ($T-S$) with these spillover estimates ($S-C$) recovers the total intention-to-treat effect through the identity $T-C = (T-S) + (S-C)$, developed in Appendix OA.4. Appendix Table OA.4.1 reports the decomposition with country-cluster bootstrap confidence intervals. Averaging over horizons $[t+3; t+6]$, the bootstrap-implied $T-C$ ranges from -5 to -7 percentage points of GDP per capita growth, in line with the broader-counterfactual estimate of Figure 6. Regional spillovers therefore offset part of the within-region decline, and the net loss is smaller than the within-region estimate.

4.3 Heterogeneity analysis

Should some affected countries weather SMC failures better than others? Four features of the international infrastructure stack may support an economy’s resilience to connectivity shocks. First, having more *SMC landings* can provide spare capacity. Second, *SMC connectedness*, reflected by a broader set of connected partners (distinct foreign landing countries/systems), expands the menu of available routes when one path fails. Third, a greater number of Internet Exchange Points (IXPs) improves networks’ efficiency and redundancy by keeping regional and domestic traffic locally rather than

hauling them across international links.³⁰ Fourth, *ownership diversity*, measured by the number of distinct commercial owners among the cables landed in a country, can limit the spread of a single disruption, since cables held by a common set of owners are more likely to be linked through joint infrastructure-sharing arrangements and common maintenance contracts, so that a fault affecting one owner’s operations can propagate across several of a country’s landed systems at once. These margins matter because the value of connectivity depends on how imperfectly correlated failure risks are across independent routes: where few independent paths exist, a single fault removes most substitution possibilities and the loss propagates at scale (Caoui and Steck 2025).

To identify heterogeneous effects of SMC failures on growth depending on network redundancy, I split countries at connectivity variables’ sample median, fix the sub-groups at their baseline values, and re-estimate the event studies (Figures 8, 9, and 10, and Appendix Figure A.1.5.1).

Route diversity across partners separates the groups more cleanly than the raw count of landed cables, suggesting it is the operative resilience margin. Countries with few connected partners experience GDP per capita growth losses reaching approximately 10 pp in the medium run (Figure 8, top panels), while high-connectedness countries show effects close to zero throughout. In log-level terms (bottom panels), the divergence compounds over time as connectivity shocks in poorly connected countries generate persistent income losses. Heterogeneity analysis based on the simple count of landed SMCs (Appendix Figure A.1.5.1) or Internet penetration (Appendix A.1.5.2) provides divergence patterns much less pronounced than for partner diversity.

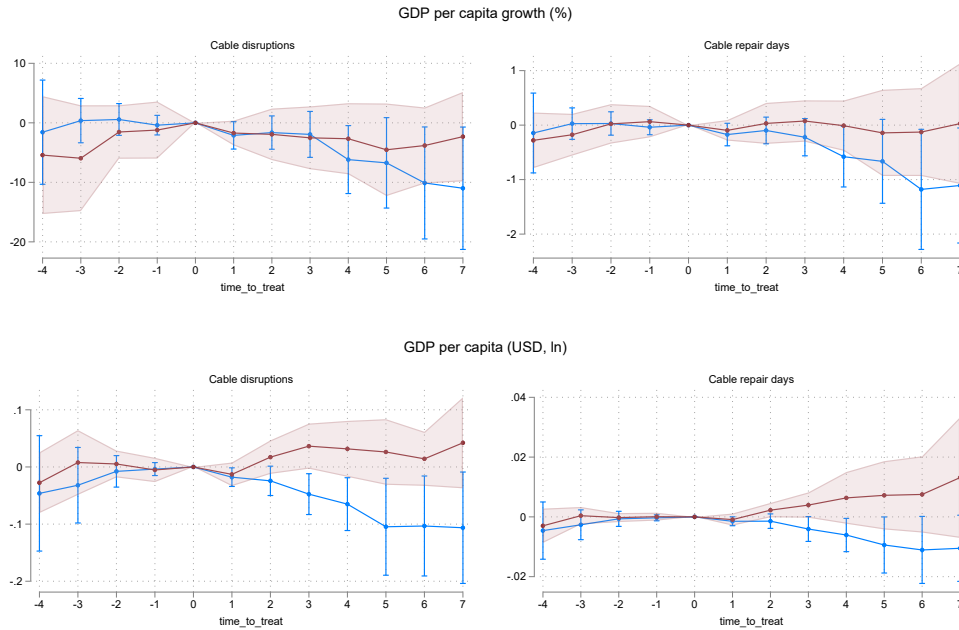


Fig. 8: Policy margin: SMC connectedness.

Notes: Data on SMC connectedness provided by the author, drawn from Telegeography’s [SMC map](#). Red hollow markers = above-median connectivity at baseline; blue filled markers = below-median. Group assignments are fixed at the start of the sample. Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

³⁰ An IXP is a facility where ISPs, CDNs and other autonomous systems interconnect to swap traffic directly without using international routes; dense IXP presence lowers latency, reduces transit dependence and offers more rerouting options.

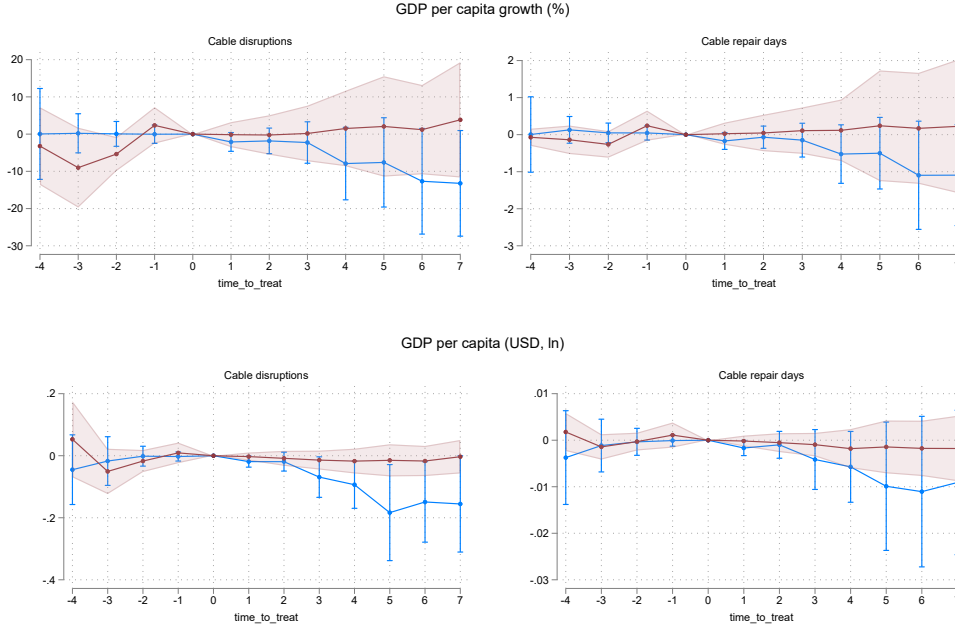


Fig. 9: Policy margin: IXPs.

Notes: Data on IXPs provided by the author, drawn from Telegeography’s [IXP map](#), Packet clearing House, and the Peering DB databases. Red hollow markers = above-median connectivity at baseline; blue filled markers = below-median. Group assignments are fixed at the start of the sample. Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

The role of IXPs produces a similar heterogeneity (Figure 9). Splitting the sample between countries with and without IXPs at the sample starting date reveals that countries lacking IXPs exhibit pronounced and persistent economic slowdowns in both growth rates and levels. Those with at least one IXP show statistically insignificant movements around zero across both metrics and throughout the event window. Since IXPs localize traffic and reduce dependence on long-haul international routes, this attenuation is consistent with the ability to maintain domestic traffic and to reroute regional and international traffic when international capacity is impaired.

Ownership diversity produces a comparable pattern (Figure 10). The measure counts the number of distinct commercial owners among the cables landed in a country, so a low count means that a small number of firms hold effective control over the country’s international capacity. Countries with few distinct cable owners experience GDP per capita growth and level losses reaching magnitudes comparable to the low-connectedness group in the medium run, while countries with a more diversified set of owners show effects statistically indistinguishable from zero throughout the event window.³¹

Across all four infrastructure characteristics, countries entering the period with more connected partners, more IXPs, or a more diversified set of cable owners do not experience measurable growth or level losses from SMC failures. More connected partners expand the feasible set of fail-over routes, IXPs confine a larger share of traffic to domestic or regional paths, and diversified ownership limits the share of a country’s capacity exposed to any single operator’s maintenance and repair schedule, together dampening transmission. Contrary to the assumptions in [Deloitte \(2016\)](#), low connectivity, rather than digitalization per se, drives economic losses. Neither is income per capita, since economies

³¹The connectedness and ownership-diversity results are robust to including the number of landed cables as a control. Results are available upon request.

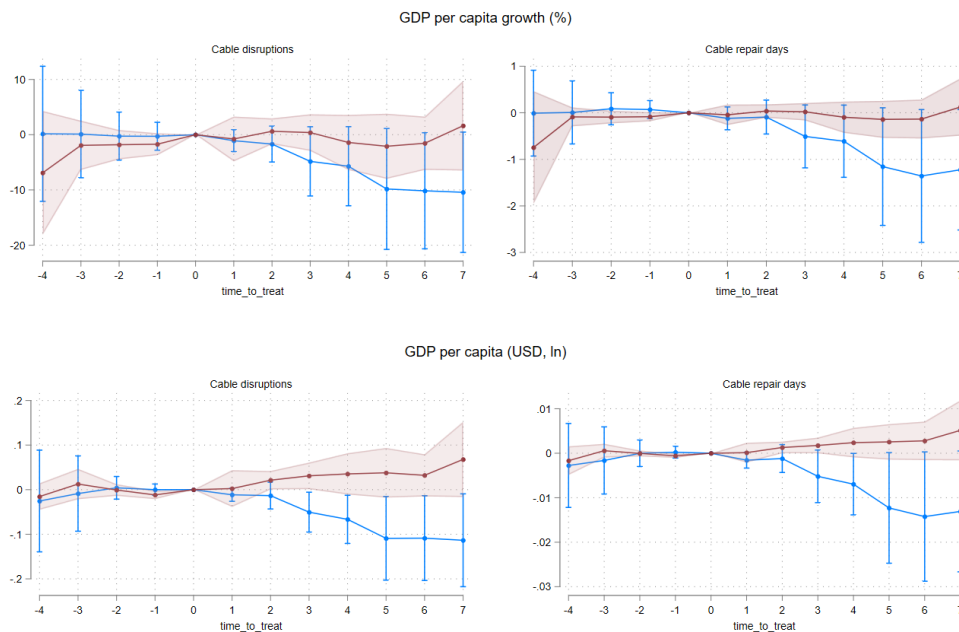


Fig. 10: Policy margin: cable owners diversity.

Notes: Cable owners diversity is measured by the number of distinct owners of cables landed in a given country. Data on cable owners provided by the author, drawn from Telegeography’s [IXP map](#), Packet clearing House, and the Peering DB databases. Red hollow markers = above-median connectivity at baseline; blue filled markers = below-median. Group assignments are fixed at the start of the sample. Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

in this low-connectivity group are not uniformly low-income.³² The same reading appears in the joint ITU–UNDRR–Sciences Po report ([ITU et al. 2026](#)), which observes that exposure to digital risk depends less on how digitalised a country is than on its network architecture, with some small coastal or island states relying on a single submarine cable.

5 Robustness

The baseline results show large and persistent growth losses after submarine-cable failures. This section probes their robustness against alternative explanations and identification threats. I first decompose the causes of failures, separating physical breaks from outages and from failures linked to natural hazards or sabotage, and I run sensitivity checks that sequentially exclude regions, income groups, and the outlier year 2020. I next address bias from non-random exposure to cable shocks through falsification tests in the spirit of [Borusyak and Hull \(2023\)](#). The concern is that countries with weak telecommunications policy accumulate both more disruptions and worse macro trends. A final subsection groups two further checks: estimates after controlling for time-varying confounders, and replication on an independent disruption dataset from *Cloudflare*.

³²The lists of low-connectedness and no-IXP countries (Online Appendix Tables [OA.5.1](#) and [OA.5.2](#)) span upper-middle-income economies such as Bulgaria, Botswana, Gabon, and Iraq alongside the poorest, which locates the exposure margin in network architecture, namely route diversity and IXP presence, rather than in per-capita income.

5.1 Sensitivity Analysis

5.1.1 Excluding regions and income groups from the sample

A natural concern is that the baseline dynamics might be driven by a single geographic block such as SSA, or a specific income group such as MICs, which concentrate the largest number of disruptions. To probe these potential sources of sample bias, I re-estimate the dynamic specification sequentially excluding each World Bank region and each income group from the estimation sample.

First, Appendix Figure A.1.6.2 reports the resulting event-time coefficients for GDP per-capita growth excluding regions one by one. Pre-trends remain flat in every exclusion, supporting the parallel-trends restriction independent of regional composition. The post-event contraction redremains consistent across sub-samples: coefficients move below zero within two-to-three years after first exposure and continue to decline thereafter, cumulating to medium-run losses of roughly 6–10 percentage points by horizons $[t + 5; t + 6]$, well within the baseline confidence intervals. The trajectory obtained after dropping EAP, LAC, MENA, SA, and more importantly SSA, is very close to the baseline. Dropping Europe and Central Asia (ECA), which represents about one third of the estimation sample and contains many never-treated countries that serve as controls (Figure 2 and Table 1), leaves the shape and magnitude of the dynamic effects essentially unchanged, but slightly widens confidence intervals.

Second, to ensure that effects are not driven by a specific income group—such as MICs which concentrate most cable failure events, or LICs which may be the most vulnerable to adverse connectivity shocks—I re-estimate the model sequentially excluding each income group from the estimation sample. Results are reported in Appendix Figure A.1.6.3 and show that estimated effects remain robust, suggesting that the significance, magnitude, and dynamics of the growth decline are not related to a specific income group category.³³

5.1.2 Composition of failures and exogeneity assumption

A first concern is that the “cable disruption” variable mixes events with different provenance, possibly related to policy factors, or directly affecting macroeconomic outcomes. First, physical breaks, typically due to anchors or trawling, are plausibly exogenous to macro conditions; whereas outages may reflect operator maintenance or soft faults and could be correlated with demand, national policies, or investment cycles. In addition, some disruptions coincide with natural hazards or sabotage, which may themselves be directly correlated to economic and policy factors. In Appendix Figure A.1.6.1, I therefore re-estimate the event studies (i) after excluding all events identified by the SubTel Forum as caused by natural hazards or sabotage, and (ii) separating physical breaks from lighter outages.

The top row of Appendix Figure A.1.6.1 shows that the growth effects remain unchanged when natural hazards and sabotage events are dropped. This indicates that the baseline results are not driven by concurrent macro shocks embedded in the treatment variables. The bottom row separates SMC breaks (left) from SMC outages (right). Break-only estimates display the baseline dynamic, that is, flat pre-trends and a persistent growth deterioration in the same magnitude as baseline estimates. Outage-only estimates are also negative from mid-horizon onward but are smaller in magnitude and non-significant. Taken together, the composition check suggests that baseline estimates are not a result of infrastructure maintenance capacity, natural hazards or acts of sabotage that could challenge the identification assumption.

³³The figure reports 10% confidence intervals due to the stringency of this sensitivity check.

5.2 Non-random exposure to connectivity shocks

Countries with weak telecom policy or shallow redundancy may accumulate more disruptions and grow more slowly for structural reasons unrelated to cable failures, biasing event-study coefficients downward. The network itself reinforces this concern, since [Caoui and Steck \(2025\)](#) model the global backbone as a dynamic entry game in which buyer demand for route diversification, hedging against disruption risk, shapes cable entry and long-run market structure, so a country’s exposure is in part an equilibrium outcome rather than purely exogenous. This non-random exposure problem, formalized by [Borusyak and Hull \(2023\)](#), is addressed with two complementary tests. First, following [Borusyak and Hull \(2023\)](#)’s test of non-random exposure to shocks, I randomly re-assign the timing of disruptions within countries and replicate the dynamic event-study specification 1,000 times over the 2008–2019 period, while enforcing the same estimation sample. Figure 11 presents the results for both GDP per capita growth (top row) and log GDP per capita level (bottom row). The left panel plots the real event-study coefficients against the 95% placebo band, and show that placebo effects remain centered around zero throughout the event window while the real coefficients diverge progressively into the lower tail. The right panel shows the permutation null distribution for the average post-treatment effect over $t-t+5$. The actual average effect (red dashed vertical line) lies clearly in the left tail, with permutation p-values of 0.068 for the growth rate and 0.010 for the log level. Randomly permuted shocks produce no significant effect on either GDP outcome, while the actual estimates lie clearly outside the null distribution, supporting the causal interpretation of the baseline results.

Second, I recenter the treatment variable following [Borusyak and Hull \(2023\)](#) by subtracting its expected value under uniform within-country timing, retaining only timing variation among countries with identical total disruption burdens. Appendix Figure A.1.6.4 reports the results. In the baseline specification with income-group-by-year fixed effects, the recentered estimates are negative and persistent but imprecisely estimated, reflecting strong sample attrition under high-dimensional fixed effects. Removing income-group-by-year fixed effects recovers estimates consistent with baseline dynamics in sign and magnitude, redjointly significant, with individually significant medium-run effects. The attenuation in the high-FE specification reflects loss of effective identifying variation under sample compression, not evidence of confounding.

As a cross-sectional complement, Appendix Table A.1.6.1 regresses each country’s total number of disruptions over 2008–2020 on its pre-period growth, income, and network position. Incidence is uncorrelated with income, and with pre-period GDP-per-capita growth once the 2009 financial-crisis year is excluded. It loads instead on cable connectedness, consistent with exposure being set by a country’s position in the network rather than by its growth path.

5.3 Additional robustness checks

Controlling for confounding factors.

The baseline dynamic is unchanged when I sequentially add controls for time-varying factors correlated with both disruptions and growth: disaster occurrence (EM-DAT), liner shipping connectivity, institutional quality, ICT regulatory quality, internet penetration, the number of submarine cables and cable path diversity, the real exchange rate, and the net barter terms of trade. Across all specifications the pre-treatment coefficients stay near zero and the post-disruption coefficients decline over the medium run, as in the baseline (Appendix Figure A.1.6.9).

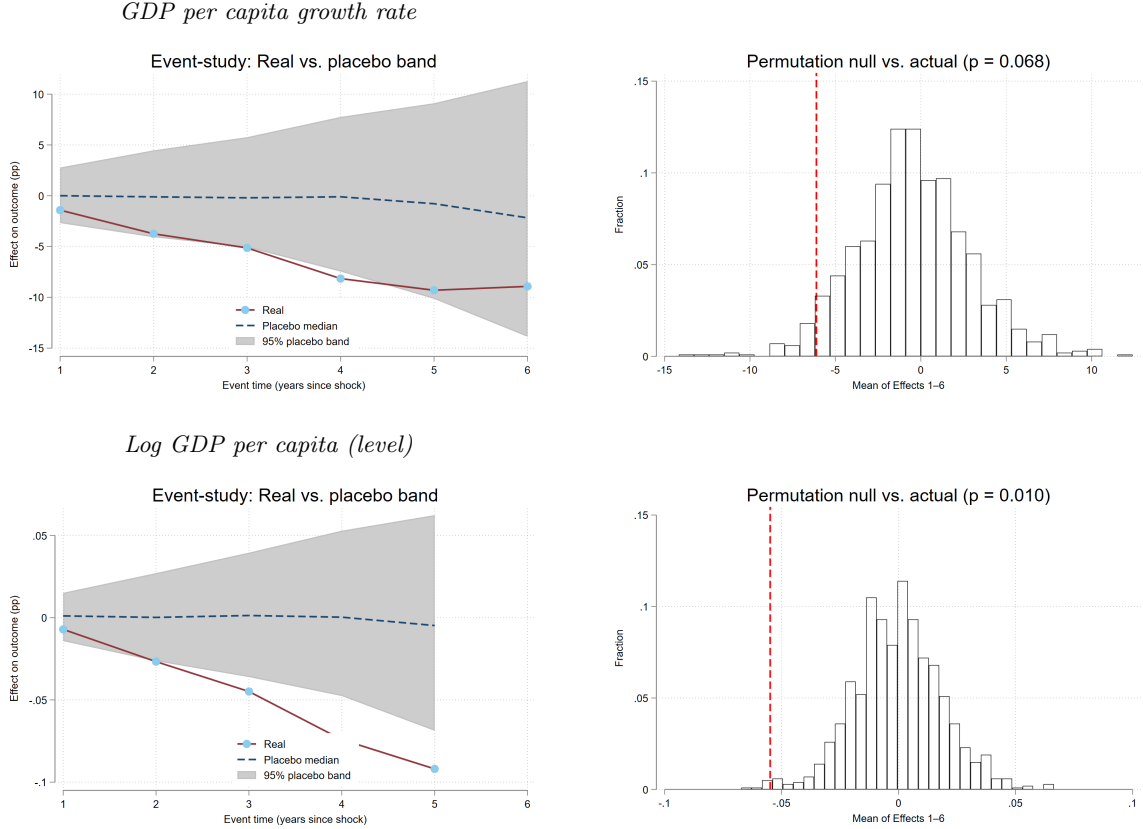


Fig. 11: Permutation test for non-random exposure: real vs. placebo (2008–2019).

Notes: All panels are based on 1,000 random re-assignments of cable disruption timing within countries, over the 2008–2019 sample. *Left panels:* Real event-study coefficients (dark red line with markers) plotted against the 95% placebo band (grey shaded area); the dashed blue line is the placebo median. *Right panels:* Histogram of the permutation null distribution for the post-treatment effect (average event-study coefficient over $t-t+5$); the red dashed vertical line marks the actual average effect. Top row: GDP per capita growth rate ($p = 0.068$). Bottom row: log GDP per capita level ($p = 0.010$). P-values are computed as the share of permutation statistics below the actual test statistic. Specifications include country fixed effects, year fixed effects, linear country trends, and region-by-year and income-group-by-year fixed effects, with standard errors clustered at the country level.

External validation with Cloudflare.

I replicate the analysis on Cloudflare Radar’s independent cable-cut records for 2022–2025, which are not used to build the 2008–2020 dataset.³⁴ Stacking SubTel disruptions (2008–2019/2020) with Cloudflare cable cuts (2022–2025) and estimating over 2008–2025 reproduces the baseline path. The first post-event coefficient is -1.28 pp (90% CI $[-2.54, -0.02]$; 52 switchers), and longer horizons stay negative and grow in absolute value (Appendix Figure A.1.6.10). Restricting treatment to Cloudflare events among countries that never switch in the SubTel data isolates a short-run effect of -4.86 pp (90% CI $[-7.61, -2.12]$), redjointly significant, with placebos not rejected. The small number of Cloudflare switchers leaves medium-run coefficients imprecise, so I read this as confirmation that independently coded cable cuts are followed by an immediate growth shortfall of the same sign as the baseline.

³⁴See Cloudflare Radar [webpage](#). Countries more exposed to SubTel disruptions over 2008–2019 record more Cloudflare nationwide cable-cut events in 2022–2025 (Appendix Figure A.1.7.1), which supports the reading that the disruption coding captures connectivity failures an independent monitor also detects.

6 Mechanisms

This section explores the mechanisms underlying the persistent growth effects of submarine cable failures.³⁵ I begin by examining the dynamics of the main GDP components. I then further explore the trade channel, documenting adjustments in exporter participation on both the export and import sides using customs transaction data, and showing that the estimated spillovers are consistent with service trade diversion toward nearby non-disrupted countries. Then, I highlight the central role of productivity and foreign direct investment: both total factor productivity (TFP) and labor productivity closely track the observed growth dynamics, are accompanied by a decline in FDI inflows, and exhibit similar regional spillover patterns. Finally, I explore the role of financial frictions as a potential shock amplification force.

6.1 Direct effects on trade, consumption, and investment

The most immediate adjustment is through trade. Goods-and-services imports and exports fall after a disruption and decline further over time (Appendix Figure A.1.6.6). Imports fall earliest and most persistently: about 5% on impact, reaching roughly -65% by $t+6$ (22 switchers), or about -43% once the single most-influential switcher (Sudan) is excluded.³⁶ All 41 leave-one-out estimates at $t+6$ stay negative and significant, spanning about -43% (Sudan excluded) to -76% (Sierra Leone excluded) (Online Appendix Figure OA.3.4). Exports also contract over the medium run, by less than imports and less precisely. Both merchandise and services trade respond, consistent with the “servicification” of goods trade (Baldwin 2016) and with the elasticity of cross-border data flows to bandwidth capacity (Malgouyres et al. 2021; Jeon and Rysman 2025). A breakdown by trade component is in Appendix Figures A.1.6.8 and A.1.6.7.

Household consumption per capita turns negative two to three years after the failure and declines monotonically, to about 10% below the pre-event level at $t+3$ and 17% at $t+6$, which points to a lasting demand contraction rather than postponed spending. Government final consumption instead rises from about two years out, moving counter-cyclically as private demand falls. Gross fixed capital formation shows only a small, statistically insignificant decline, consistent with multi-year project cycles insulating aggregate capital formation from temporary connectivity losses.

6.2 Import and export extensive margins

The aggregate trade response has a firm- and partner-level counterpart, which I trace with the World Bank Exporter Dynamics Database (EDD).³⁷ The import side uses the destination panel, recording foreign exporting firms and distinct origin countries per destination-year.³⁸ The export side uses the origin panel, recording entry, exit, incumbents, and survival of domestic exporters.³⁹

On the import side, disruptions thin the supplier base (Appendix Table A.1.8.1, Panel A1). The average number of foreign exporters per partner falls by 35.8 on impact (11% of the mean) and by 84.6 at $t+3$ (27%). In levels the total count falls by 15,642 at $t+3$ (1.5 times the mean), and a destination

³⁵Since cable breaks have been identified in Section 5 as the main drivers of growth responses, I use this cable disruption definition as the preferred (because cleaner) treatment variable across this section.

³⁶For comparison, Steinwender (2018) estimates that permanently eliminating information frictions via the transatlantic telegraph raised bilateral cotton trade volumes by 37%.

³⁷Unbalanced panel of about 30 exporting countries covering 2008–2014, the attrition resulting from the unequal time and geographic coverage in the dataset.

³⁸This yields three variables: total foreign exporters (summing bilateral cells across observed origins), foreign exporters per observed origin (normalising for partial bilateral coverage), and distinct exporter countries with at least one observed firm (bounded by the EDD maximum of 57).

³⁹For the spillover analysis, treatment is the number of disrupted economies or disruption episodes in the region, estimated on non-disrupted countries.

loses 4.4 distinct exporter countries (15% of the mean). The aggregate import decline thus reflects the withdrawal of foreign exporters and a narrowing supplier base, not merely lower volumes per existing partner, consistent with [Akerman et al. \(2021\)](#): as information frictions rise, firms fall back on geographically proximate, already-familiar partners. The disrupted country’s own exporter base contracts in parallel, with the exit rate rising 15 pp at $t+1$ and surviving entrants falling by 170 on impact (Panel A2; 4–5 switchers, directional).

Non-disrupted regional peers show the mirror adjustment (Panels B1–B2). Incumbents expand, exits decline, the domestic exporter pool rises by 366–733 at $t+2$, and peers gain 2–5 distinct exporter countries on impact. From the firm perspective, this corroborates the aggregate service trade diversion documented in the following section.

6.3 Service trade diversion

I next test whether the growth spillovers come with service trade diversion, asking whether cable breaks raise exports in unaffected neighbors (Appendix Figures [A.1.4.1](#) and [A.1.4.2](#)). Spillovers operate almost only through services. Merchandise exports of unaffected countries rise briefly and then turn insignificant, whereas total and commercial service exports rise and settle about 5% above their pre-event level after four years. The response concentrates in financial and insurance services, the most bandwidth-intensive category, which roughly double red over the medium run. Transport and ICT services rise by about +15% and +25% over the medium run. This concentration is consistent with the codifiability channel of [Juhász and Steinwender \(2018\)](#), under which services whose specifications travel in words rather than physical samples are the most sensitive to information frictions.

Together with the large losses in directly exposed countries, this pattern is consistent with regional reallocation of digitally intensive activity, with unaffected neighbors capturing part of the displaced demand for high-bandwidth services. The next subsections turn to competitiveness and financial frictions as the mechanisms behind both sides of this reallocation.

6.4 Productivity and FDI: direct and spillover effects

Appendix Figure [A.1.4.3](#) traces FDI inflows, total factor productivity (TFP), and labour productivity for treated countries and their neighbors after a break. Treated countries show sharp, persistent declines in FDI inflows per capita, TFP growth, and labour productivity. Neighbors show the mirror gains, consistent with diversion of investment and activity toward more reliable nearby locations.⁴⁰

The reallocation channel shows in the labour-productivity decomposition (right-most panels). In treated countries the contribution of labor reallocation turns negative, so productivity-enhancing shifts of labor toward higher-productivity activities slow or reverse, consistent with impaired coordination, market access, and cross-border contracting. In neighbors the contribution turns positive and grows, as diverted demand is absorbed by higher-productivity activities.

The productivity responses track the growth effects of Section 4 in timing and magnitude, consistent with the medium-run cost operating through productive efficiency rather than factor accumulation, since TFP and labour productivity fall while fixed capital formation does not. The neighbor TFP gains parallel the positive growth spillovers.

⁴⁰Variables are drawn from the World Bank Global Productivity Database ([Dieppe 2021](#)). I report 90% confidence intervals given the imprecision of factor-productivity and decomposition-based measures, which may not fully capture informal activity in lower-income economies.

6.5 Financial channels

The persistent declines in trade and productivity point to financial frictions as a further channel, alongside the short-run information and production frictions. Digital infrastructure underpins payment systems and information flows within and across borders (D’Andrea and Limodio 2024; D’Andrea et al. 2025), so connectivity losses that tighten domestic and cross-border finance depress competitiveness and productivity through the channels discussed in Section 2 (Varela 2018).

On the domestic side, credit to non-financial agents falls after a disruption and the contraction deepens over the following years rather than reverting (Online Appendix Figure OA.6.1; BIS data, within-country z-scores winsorized at the 90th percentile).⁴¹ The decline is driven by credit to private non-financial actors, while credit to general government follows a smoother path, so it is not mechanically driven by public borrowing. Pre-trends are flat, and leave-one-out exclusions show that no single economy drives the result (Online Appendix Figure OA.6.2).

On the external side, cross-border banking positions decline gradually after a disruption, with wider confidence intervals at longer horizons (Online Appendix Figure OA.6.3; BIS Locational Banking Statistics, non-bank loans and deposits per capita). The two results together indicate that financial frictions operate through both domestic and cross-border intermediation.

7 Conclusion and policy implications

Submarine-cable disruptions impose large and persistent output losses. Within four to five years, GDP per capita falls to roughly 9% below the level implied by a country’s own pre-disruption growth path relative to non-disrupted economies in the same region, narrowing to 5–7% once spillovers to unaffected regional peers are netted out, with non-disrupted neighbours gaining from the diversion of trade, investment, and bandwidth-intensive services. The loss troughs at four to five years and, once the atypical 2020 year is set aside, reverts to about 2.5% by the sixth year. The apparent full-sample persistence at that horizon reflects the pandemic cohort, so the medium-run cost is anchored at four to five years. The losses are concentrated in economies with low network centrality, no Internet exchange points, and cables concentrated among few owners, while well-connected and diversely-owned economies absorb the same shock with effects close to zero. Translating the spillover-inclusive declines into per-capita terms, and taking the low-connectivity subsamples’ average GDP per capita of \$4,800–5,700, a back-of-the-envelope calculation implies per-capita losses of roughly \$50 on impact and \$240–400 at the four-to-five-year trough, declining to about \$90–100 by the sixth year on the sample that excludes 2020.⁴² These magnitudes are disaster-scale, and the concentration of effects in low-connectivity countries indicates that resilience margins, not experience alone, drive the cross-country distribution of losses.

The persistence and magnitude of these losses point to a set of reinforcing mechanisms that extend well beyond the immediate interruption of connectivity. The channel that moves first and farthest is the capacity to import goods and services, which falls on impact and declines further over the medium run through the withdrawal of foreign suppliers rather than smaller volumes per existing relationship, consistent with impaired import access as a proximate channel from connectivity loss to output. Second, cable disruptions generate a sustained loss in competitiveness, reflected in persistent declines in productivity and foreign investment inflows. These supply-side effects weaken firms’ ability to

⁴¹I use z-scores rather than logs because credit stocks are far higher in the six switchers than in non-switchers (Online Appendix Figure OA.6.4); the variable distributions are in Online Appendix Figure OA.6.5. Given the smaller sample, treatment is cable disruptions rather than cable breaks, which retains more switchers.

⁴²Appendix Table A.1.9.1 reports the larger within-region relative costs, net of spillovers.

compete in international markets and are accompanied by trade and FDI diversion toward unaffected neighboring countries. Third, shocks are amplified by financial channels. I also document a persistent and abnormal tightening of domestic credit to non-financial actors, including in relatively large and financially developed economies, as well as a gradual weakening of cross-border banking relationships.

Taken together, these results support a policy agenda centered on reducing exposure to SMC failures and accelerating recovery when they occur. First, building *redundancy* through more diverse landing sites, partner routes, and cable ownership, and avoidance of shared choke points, reduces the probability that a single break isolates a country. Second, densifying *IXPs* and domestic peering localizes traffic and facilitates rapid fail-over when international links are impaired. Third, shortening *repair queues* through adequate maintenance agreements and fleet renewal directly lowers outage duration and the depth of output losses. Finally, simulation-based stress tests can help governments treat cable infrastructure as critical cross-border infrastructure on par with energy or transport networks. Network redundancy is itself an equilibrium outcome of demand and past investment (Caoui and Steck 2025), but over the sample period that capacity was deployed by multiple stakeholders and consortia serving many markets, so a country’s route diversity reflects the geography of the network and converging deployment decisions more than its own conditions. Because the losses concentrate in low- and middle-income economies with thin route diversity, few IXP, and concentrated cable ownership, rather than in the poorest alone, the return to redundancy investment is a question of development finance as much as one of national and global security. Deploying IXP and diversifying international routes falls within the remit of the UN, the World Bank, Regional Economic Communities, and national regulators, which can weigh those outlays against the per-event output losses estimated above. Consistent with the ownership-diversity result documented above, as new capacity is increasingly built as single-owner cables by a few large content providers, and as economies turn to single-owned satellite networks such as Starlink for a backup route, the resilience margin concentrates in private actors optimizing their own traffic, who need not internalize the value of route diversity for the economies they serve. Left to private incentives, redundancy is underprovided relative to the social optimum, a gap that public coordination can correct and that future research should investigate.

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Appendix

A.1 Sample statistics and additional estimations

A.1.1 Sample statistics

Table A.1.1.1: Estimation sample countries by World Bank region

East Asia and Pacific					
AUS	FJI	IDN	JPN	KHM	KIR
MMR	MNG	NZL	PLW	PNG	SLB
THA	TON	TUV	VUT	WSM	
Europe and Central Asia					
ALB	AND	ARM	AUT	AZE	BEL
BGR	BLR	CHE	CYP	CZE	DEU
DNK	ESP	EST	FIN	FRA	GBR
GEO	GRC	GRL	HRV	HUN	IRL
ISL	ITA	KAZ	KGZ	LTU	LUX
LVA	MCO	MDA	MKD	NLD	NOR
POL	PRT	RUS	SMR	SVK	SVN
SWE	TJK	TKM	TUR	UKR	UZB
Latin America and Caribbean					
ABW	ARG	ATG	BHS	BLZ	BRA
BRB	CHL	COL	CRI	CUB	CYM
DMA	ECU	GRD	GTM	GUY	HND
JAM	KNA	LCA	MEX	NIC	PAN
PER	PRI	PRY	SLV	SUR	TTO
URY	VCT				
Middle East and North Africa					
DZA	IRQ	JOR	LBY	MAR	TUN
South Asia					
BTN	MDV				
Sub-Saharan Africa					
AGO	BDI	BFA	BWA	CAF	CIV
CMR	COG	COM	CPV	GAB	GHA
GIN	GMB	GNB	GNQ	KEN	LSO
MDG	MLI	MOZ	MRT	MUS	MWI
NAM	RWA	SDN	SEN	SLE	SOM
STP	SWZ	SYC	TCO	TZA	UGA
ZAF	ZWE				

Notes: Regions follow World Bank's region classification. Sample corresponds to estimates reported in Figure 5.

Table A.1.1.2: Summary statistics of cable failure variables

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Panel A: Overall sample statistics (177 countries)						
Cable disruptions	0.101	0.392	0.357	0	5	2,478
Cable breaks	0.084	0.352	0.324	0	5	2,478
Cable outages	0.018	0.144	0.135	0	2	2,478
Repair days	0.938	5.437	5.132	0	80	2,468
Panel B: Treated countries, sample statistics (86 countries)						
Cable disruptions	1.328	0.625	0.419	1	5	189
Cable breaks	1.095	0.723	0.502	0	5	189
Cable outages	0.232	0.471	0.345	0	2	189
Repair days	12.938	15.926	10.976	1	80	179
Panel C: Annual sums, 2008–2020						
Cable disruptions	19.08	18.54	–	5	75	13
Cable breaks	15.77	16.89	–	3	69	13
Cable outages	3.38	4.17	–	0	15	13
Repair days	176.92	204.62	–	65	840	13
Panel D: Annual sums, 2008–2019						
Cable disruptions	14.42	8.20	–	5	35	12
Cable breaks	11.33	5.66	–	3	23	12
Cable outages	3.17	4.28	–	0	15	12
Repair days	121.66	48.76	–	65	204	12

Table A.1.1.3: Summary statistics of cable failure variables, estimation sample.

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
Panel A: Overall panel statistics (country–year)						
Cable disruptions	0.066	0.324	0.304	0	5	1,898
Cable breaks	0.053	0.283	0.268	0	5	1,898
Cable outages	0.015	0.133	0.126	0	2	1,898
Repair days	0.647	4.815	4.606	0	80	1,893
<i>Panel size: Countries (n) = 146; Avg. years per country ($\bar{T}$) $\approx$ 13.0.</i>						
Panel B: Annual sums, 2008–2020						
Cable disruptions	10.50	15.39	–	0	56	12
Cable breaks	8.25	14.15	–	1	52	12
Cable outages	2.33	3.09	–	0	11	12
Repair days	102.08	166.48	–	0	617	12
Panel C: Annual sums, 2008–2019						
Cable disruptions	6.36	5.90	–	0	22	11
Cable breaks	4.27	3.38	–	1	11	11
Cable outages	2.18	3.19	–	0	11	11
Repair days	55.27	39.55	–	0	122	11

Table A.1.1.4: Repair days by incidence of cable disruptions

Cable disruptions	Mean	Median	Std. Dev.	Freq.
All countries				
1	10.60	4.5	14.49	124
2	17.22	16	17.30	36
3	28.30	26.5	18.00	10
5	50	0	501	
Total	13.26	7	16.19	171
Estimation sample				
1	11.22	2	17.26	63
2	17.5	16	16.10	20
3	38	44	13.08	3
5	50	0	1	
Total	14.03	5	17.88	87

Table A.1.1.5: Summary statistics of macroeconomic variables, estimation sample.

	Mean	Std. Dev.	Within SD	Min	Max	Obs.
GDP per capita growth (%)	1.181	5.6686	5.379	-49.13	91.78	1,869
GDP per capita (USD, const 2015)	15,135	22,956	2,059	271	181,497	1,636
GDP (USD, const 2015)	2.38e11	6.22e11	3.91e10	3.06e7	4.58e12	1,863
Private consumption (USD pc)	7,494	8,995	607.7	208	44,823	1,349
Government consumption (USD pc)	2,835	4,158	219.6	20.0	21,775	1,350
Gross fixed capital formation (USD pc)	3,245	4,592	1,197.5	22.6	39,166	1,327
Exports of goods and services (USD pc)	8,172	21,353	2,937.5	17.6	214,858	1,353
Imports of goods and services (USD pc)	7,642	17,741	2,733.4	42.0	179,347	1,353
<i>Panel size: Countries (n) = 146; Avg. years per country ($\bar{T}$) $\approx$ 12.8.</i>						

A.1.2 Baseline estimations

Table A.1.2.1: Treatment paths at estimation time-window $[t-1; t+6]$

	#Groups	%Groups	year 0	year 1	year 2	year 3	year 4	year 5 – 6 – 7
Panel A. Cable disruptions								
Path 1	10	45.5	0	1	0	0	0	0
Path 2	3	13.6	0	2	0	0	0	0
Path 3	2	9.1	0	1	0	0	0	3
Path 4	2	9.1	0	1	0	0	0	4
Paths 5-9	5	22.7	–	–	–	–	–	–
Panel A'. Cable disruptions - dropping 2020								
Path 1	9	56.25	0	1	0	0	0	0
Path 2	3	18.75	0	2	0	0	0	0
Path 3	1	6.25	0	1	0	0	0	3
Path 4	1	6.25	0	1	0	0	1	1
Path 5	1	6.25	0	1	2	0	0	1
Path 6	1	6.25	0	1	2	1	0	1
Panel B. Cable repair days								
Path 1	4	20.8	0	1	0	0	0	0
Path 2	2	8.3	0	10	0	0	0	0
Path 3	2	8.3	0	16	0	0	0	0
Path 4	1	4.2	0	0	0	0	0	34
Paths 5-18	13	58.4	–	–	–	–	–	–
Panel B'. Cable repair days - dropping 2020								
Path 1	4	25	0	1	0	0	0	0
Path 2	2	12.5	0	10	0	0	0	0
Path 3	2	12.5	0	16	0	0	0	0
Path 4	1	6.25	0	7	0	0	4	1
Paths 5-11	7	44.75	–	–	–	–	–	–

Notes: Year 0 corresponds to the year before the disruption occurred. In panels A and B, 22 of 58 switchers have a fully-observed window of 6 post-treatment periods in the estimation sample. Interpretation of path 1, panel A: 45.5% of treated countries have a “one-off” break in the first treated year and then no additional breaks over the next 6 years.

A.1.3 Excluding the 2020 pandemic year.

The 2020 pandemic year is atypical, recording an exceptionally large number of cable failures and repair days and coinciding with the COVID-19 shock, whose surge in data traffic strained routing capacity. Re-estimating the baseline on the 2008–2019 sample isolates the cable-failure effect from the pandemic. The growth-rate response is cleaner and earlier, and in log levels GDP per capita stands 9.2% below counterfactual at $t+4$ and 8.4% at $t+5$ (Figure A.1.3.1), with the post-treatment effects

**Table A.1.2.2: Cable failures and GDP per cap. growth (%):
Event-study estimates and placebo tests, 2008-2020.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-1.950698	0.8796778	-3.674835	-0.2265612	392	58
Effect_2	-1.805730	1.2350040	-4.226293	0.6148332	364	58
Effect_3	-2.254004	1.8088430	-5.799271	1.2912630	289	42
Effect_4	-5.531820	2.3386700	-10.11553	-0.9481100	213	29
Effect_5	-6.063789	3.0920850	-12.12416	-0.0034144	198	27
Effect_6	-8.051168	3.3679690	-14.65227	-1.4500700	174	23
Effect_7	-9.112940	3.8034310	-16.56753	-1.6583530	150	22
<i>Test of joint nullity of the effects: p-value = 0.02957928</i>						
<i>Placebo leads</i>						
Placebo_1	0.0886296	0.9583748	-1.789751	1.967010	360	55
Placebo_2	-0.1293523	1.2975970	-2.672597	2.413892	282	49
Placebo_3	-0.7202078	1.9900810	-4.620696	3.180280	196	31
Placebo_4	-1.4702760	3.4537230	-8.239448	5.298897	70	13
Placebo_5	2.6136940	3.6681070	-4.575664	9.803052	21	6
<i>Test of joint nullity of the placebos: p-value = 0.86895659</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.166887	0.0826055	-0.3287907	-0.0049833	378	54
Effect_2	-0.1163032	0.1107374	-0.3333444	0.1007381	353	54
Effect_3	-0.2023648	0.1438693	-0.4843435	0.0796138	277	38
Effect_4	-0.4785969	0.209545	-0.8892974	-0.0678963	211	28
Effect_5	-0.6057429	0.3134311	-1.220057	0.0085707	199	26
Effect_6	-0.9583649	0.4037035	-1.749609	-0.1671205	175	22
Effect_7	-0.9759625	0.4158666	-1.791046	-0.160879	155	22
<i>Test of joint nullity of the effects: p-value = 0.01991066</i>						
<i>Placebo leads</i>						
Placebo_1	0.0491271	0.0739607	-0.0958332	0.1940875	346	51
Placebo_2	0.0235962	0.0976396	-0.1677739	0.2149664	271	45
Placebo_3	0.121817	0.1138709	-0.1940716	0.2522943	184	27
Placebo_4	-0.3021379	0.2568059	-0.6251473	0.3815134	67	12
Placebo_5	0.3021379	0.4874179	-0.6531836	1.257459	19	5
<i>Test of joint nullity of the placebos: p-value = 0.66211585</i>						

jointly significant in growth rates and marginally so in levels. The repair-days dose specification gives the same path. Detailed event-study and placebo estimates for both the growth-rate and log-level responses are reported in Appendix Tables A.1.3.1 and A.1.3.2. At $t + 6$ the effect stabilises at about 2.5% rather than persisting near 9%. The deeper, persistent decline at that horizon in the full sample is an artifact of the pandemic, which depresses the cohort whose sixth year falls in 2020.

The leave-one-out exercises confirm this reading. With 2020 included (Figure A.1.3.2) the $t + 6$ estimate is held near -9% , while the $t + 4 - t + 5$ trough is stable across exclusions. Once 2020 is excluded (Figure A.1.3.3) the $t + 6$ estimate stabilises at about -2.5% across every single-switcher exclusion, and the $t + 4 - t + 5$ trough is unchanged. The medium-run cost is therefore anchored at the four-to-five-year horizon.

**Table A.1.2.3: Cable failures and GDP per. cap. (const USD, ln):
Event-study estimates and placebo tests, 2008-2020.**

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-0.0160841	0.0062813	-0.0283953	-0.0037729	391	58
Effect_2	-0.0256202	0.0094812	-0.0438430	-0.0066775	363	58
Effect_3	-0.0228331	0.0116031	-0.0451871	-0.0009575	289	42
Effect_4	-0.0553423	0.0192004	-0.0929745	-0.0177102	213	29
Effect_5	-0.0730016	0.0265307	-0.1250008	-0.0210024	198	27
Effect_6	-0.0833617	0.0318762	-0.1458380	-0.0208854	174	23
Effect_7	-0.0938445	0.0383056	-0.1689220	-0.0187669	150	22
<i>Test of joint nullity of the effects: p-value = 0.00464478</i>						
<i>Placebo leads</i>						
Placebo_1	-0.0024135	0.0053333	-0.0128665	0.0080395	359	55
Placebo_2	-0.0035723	0.0110163	-0.0251637	0.0180192	281	49
Placebo_3	0.0017157	0.0154682	-0.0286014	0.0320327	196	31
Placebo_4	0.0285196	0.0361300	-0.0398089	0.0429297	70	13
Placebo_5	0.0121075	0.0593467	-0.1042099	0.1284249	21	6
<i>Test of joint nullity of the placebos: p-value = 0.71364384</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.0014357	0.0006021	-0.0026157	-0.0002557	382	56
Effect_2	-0.0018819	0.0008908	-0.0032678	-0.0003160	357	56
Effect_3	-0.0016615	0.0012616	-0.0040891	0.0008561	282	40
Effect_4	-0.0038174	0.0019629	-0.0076466	0.0000298	216	30
Effect_5	-0.0054685	0.0030426	-0.0114319	0.0004948	203	28
Effect_6	-0.0069178	0.0042630	-0.0152732	0.0014376	179	24
Effect_7	-0.0070662	0.0045473	-0.0159788	0.0018465	159	24
<i>Test of joint nullity of the effects: p-value = 0.03589684</i>						
<i>Placebo leads</i>						
Placebo_1	-0.0001192	0.0004355	-0.0009727	0.0007343	347	52
Placebo_2	-0.0005187	0.0009488	-0.0023783	0.0013410	272	46
Placebo_3	-0.0005669	0.0013133	-0.0031409	0.0020070	186	28
Placebo_4	0.0032949	0.0029541	-0.0090848	0.0024949	69	13
Placebo_5	-0.0005277	0.0076471	-0.0144602	0.0155157	19	5
<i>Test of joint nullity of the placebos: p-value = 0.58428955</i>						

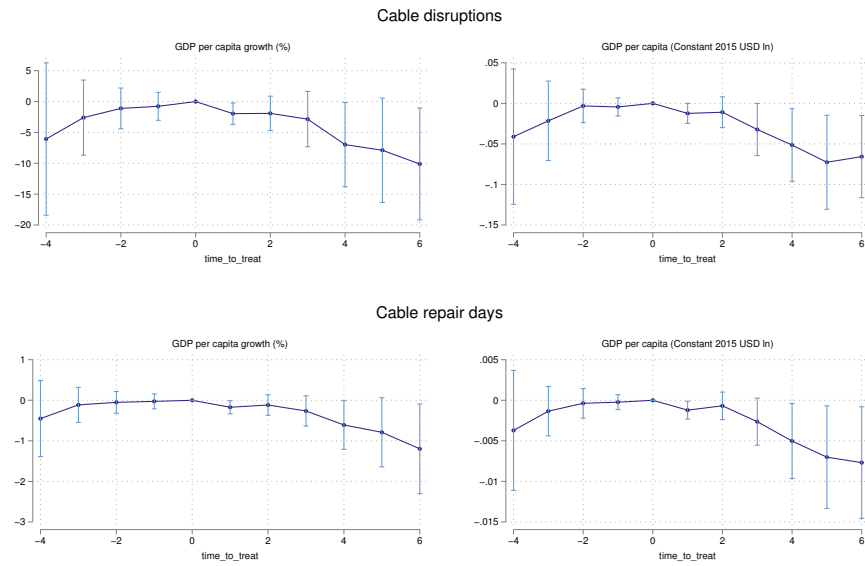


Fig. A.1.2.1: The growth costs of SMC failures: baseline specifications excluding income-by-year FEs.

Notes: Top row shows effects of cable disruptions on overall GDP growth (%), left) and log GDP per capita (center), and log GDP (right). Bottom row shows analogous effects for cable repair days. All panels report event-time coefficients with 95% confidence bands. Specifications include country fixed effects, year fixed effects, linear country trends, and region-by-year FEs, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

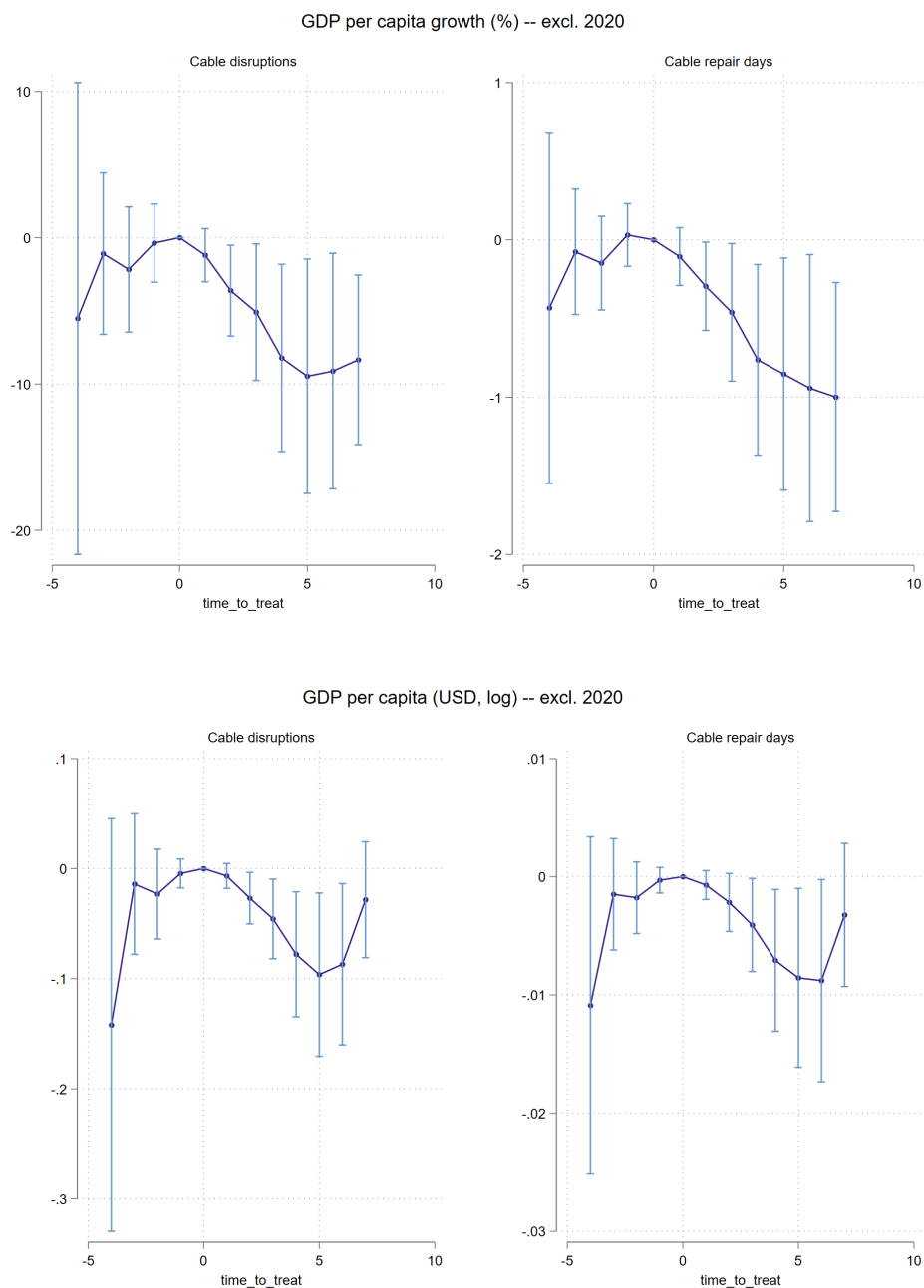


Fig. A.1.3.1: Baseline estimations excluding the 2020 pandemic year (2008–2019).

Notes: Event-study coefficients on the 2008–2019 sample (2020 excluded). Top row: GDP per capita growth (%) for cable disruptions (left) and repair days (right). Bottom row: log GDP per capita for cable disruptions (left) and repair days (right). Specifications include country and year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. 95% confidence bands. Year 7 corresponds to six years after the failure; the medium-run magnitude is anchored at $t + 4 - t + 5$.

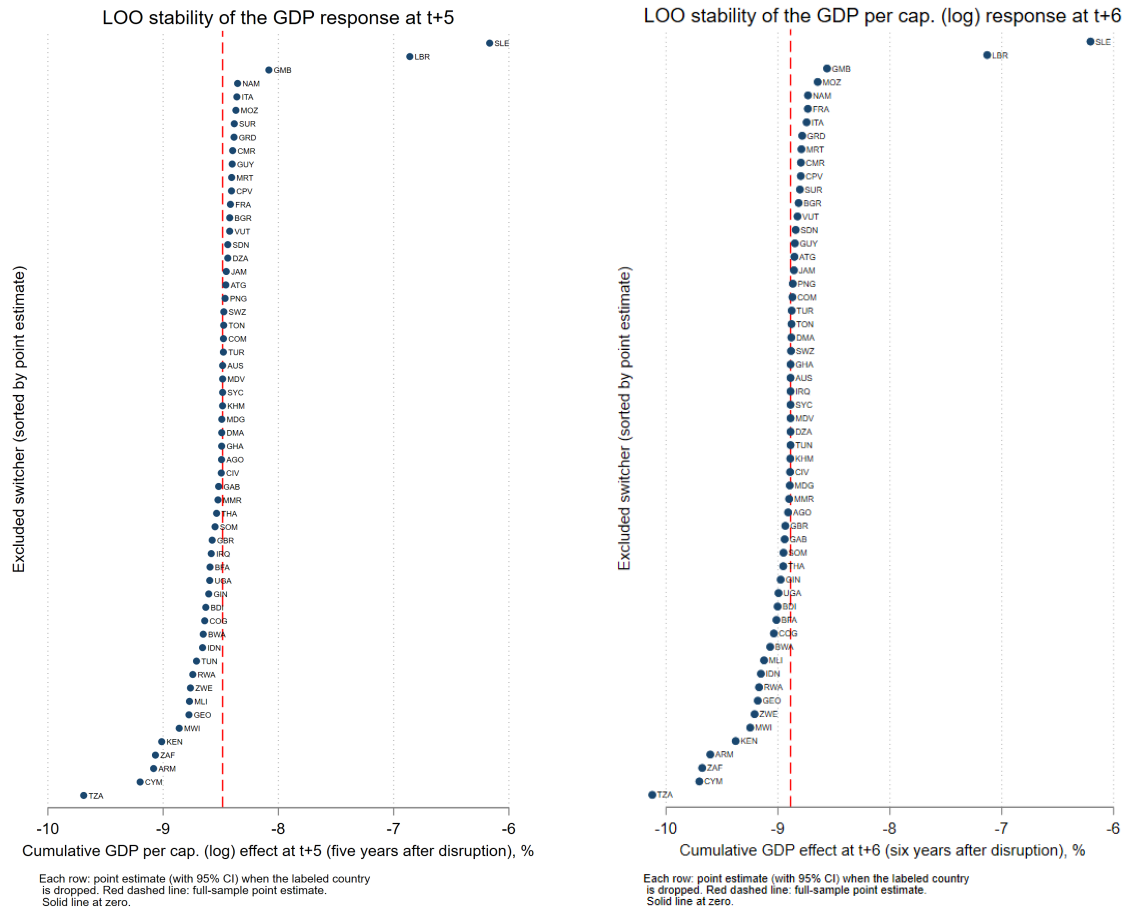


Fig. A.1.3.2: Leave-one-out stability, log GDP per capita, full sample (with 2020).

Notes: Each row drops one switcher and re-estimates the within-region log-level baseline. Left: effect at $t + 5$. Right: effect at $t + 6$. Red dashed line: full-sample point estimate; horizontal bars: 95% confidence intervals.

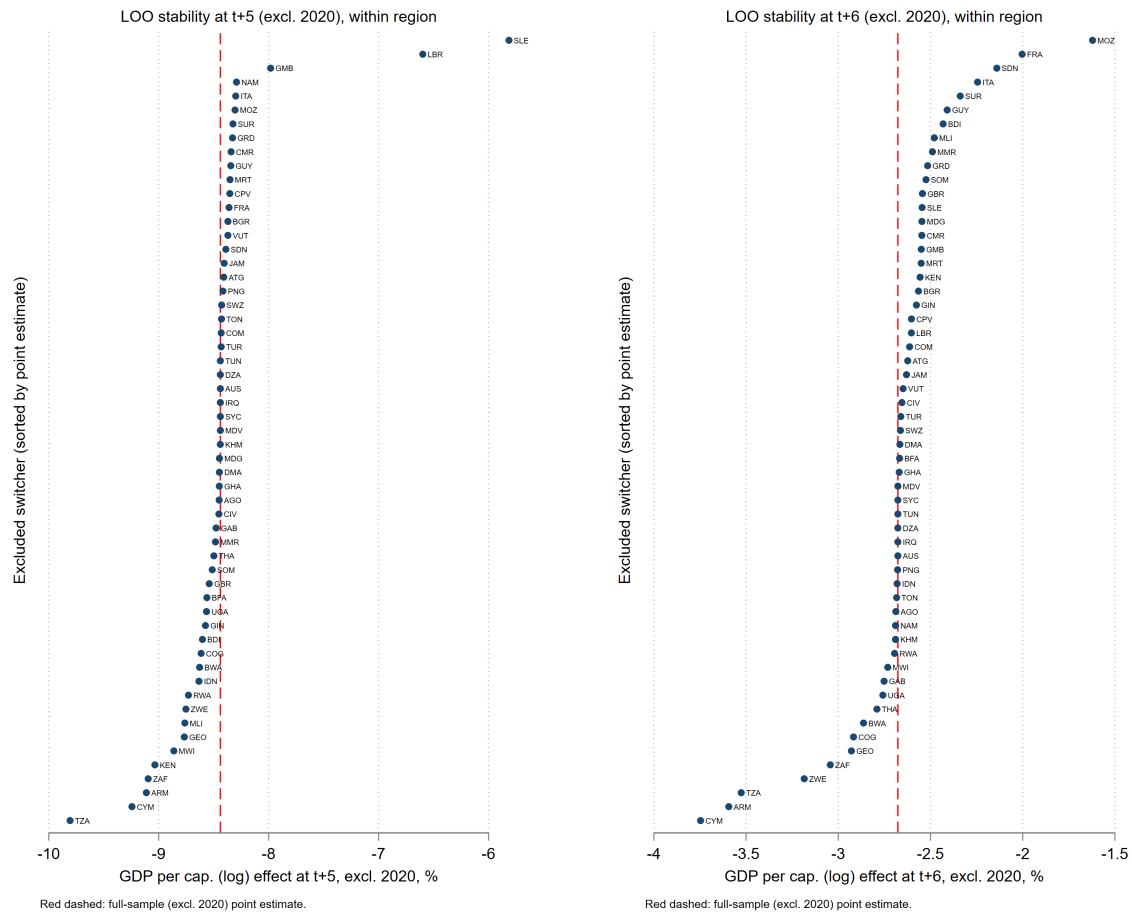


Fig. A.1.3.3: Leave-one-out stability, log GDP per capita, excluding 2020 (2008–2019).

Notes: As in Figure A.1.3.2, on the 2008–2019 sample. At $t+6$ the estimate stabilises near -2.5% across every single-switcher exclusion; the $t+4$ – $t+5$ trough is unchanged.

Table A.1.3.1: Cable failures and GDP per cap. growth (%), excluding 2020: event-study estimates and placebo tests, 2008–2019.

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-1.4167	0.9460	-3.2707	0.4374	334	42
Effect_2	-3.7454	1.5352	-6.7544	-0.7364	250	29
Effect_3	-5.1360	2.2501	-9.5461	-0.7259	223	27
Effect_4	-8.1545	3.0215	-14.0765	-2.2325	192	23
Effect_5	-9.3023	3.7720	-16.6954	-1.9093	182	22
Effect_6	-8.9206	3.8344	-16.4359	-1.4052	166	21
Effect_7	-7.9687	2.9623	-13.7747	-2.1627	128	16
<i>Test of joint nullity of the effects: p-value = 0.0057</i>						
<i>Placebo leads</i>						
Placebo_1	-0.4161	1.4133	-3.1862	2.3539	302	39
Placebo_2	-2.1550	2.0157	-6.1057	1.7957	168	20
Placebo_3	-1.3956	2.5478	-6.3892	3.5981	130	16
Placebo_4	-5.6134	7.3114	-19.9435	8.7167	49	7
<i>Test of joint nullity of the placebos: p-value = 0.7667</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.1198	0.0924	-0.3009	0.0614	318	38
Effect_2	-0.3114	0.1421	-0.5900	-0.0328	243	28
Effect_3	-0.4752	0.2176	-0.9016	-0.0488	217	26
Effect_4	-0.7535	0.2911	-1.3240	-0.1829	189	22
Effect_5	-0.8394	0.3540	-1.5331	-0.1456	185	22
Effect_6	-0.9229	0.4111	-1.7287	-0.1171	170	21
Effect_7	-0.9560	0.3692	-1.6797	-0.2324	131	16
<i>Test of joint nullity of the effects: p-value = 0.0021</i>						
<i>Placebo leads</i>						
Placebo_1	0.0285	0.1014	-0.1703	0.2273	286	35
Placebo_2	-0.1542	0.1450	-0.4383	0.1299	161	19
Placebo_3	-0.1027	0.1916	-0.4782	0.2728	124	15
Placebo_4	-0.4353	0.5154	-1.4454	0.5748	45	6
<i>Test of joint nullity of the placebos: p-value = 0.6959</i>						
<i>Notes:</i> Event-study estimates of cable disruptions (Panel A) and cable repair days (Panel B) on GDP per capita growth, estimated on the 2008–2019 sample (2020 excluded). Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. LB CI and UB CI are the 95% confidence-interval bounds. Effect_ k corresponds to year $t + (k - 1)$.						

Table A.1.3.2: Cable failures and GDP per cap. (const USD, ln), excluding 2020: event-study estimates and placebo tests, 2008–2019

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-0.0071	0.0058	-0.0184	0.0042	323	42
Effect_2	-0.0267	0.0114	-0.0491	-0.0044	243	29
Effect_3	-0.0449	0.0171	-0.0784	-0.0114	216	27
Effect_4	-0.0741	0.0262	-0.1255	-0.0228	185	23
Effect_5	-0.0920	0.0342	-0.1590	-0.0249	175	22
Effect_6	-0.0844	0.0341	-0.1513	-0.0175	159	21
Effect_7	-0.0268	0.0249	-0.0755	0.0220	121	16
<i>Test of joint nullity of the effects: p-value = 0.0565</i>						
<i>Placebo leads</i>						
Placebo_1	-0.0048	0.0068	-0.0181	0.0084	292	39
Placebo_2	-0.0243	0.0194	-0.0623	0.0138	164	20
Placebo_3	-0.0216	0.0306	-0.0816	0.0385	126	16
Placebo_4	-0.1265	0.0835	-0.2902	0.0373	49	7
<i>Test of joint nullity of the placebos: p-value = 0.5437</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.0007	0.0006	-0.0019	0.0005	307	38
Effect_2	-0.0022	0.0012	-0.0046	0.0002	236	28
Effect_3	-0.0041	0.0020	-0.0079	-0.0002	210	26
Effect_4	-0.0067	0.0029	-0.0123	-0.0010	182	22
Effect_5	-0.0082	0.0036	-0.0153	-0.0010	178	22
Effect_6	-0.0085	0.0042	-0.0167	-0.0004	163	21
Effect_7	-0.0030	0.0030	-0.0089	0.0028	124	16
<i>Test of joint nullity of the effects: p-value = 0.1997</i>						
<i>Placebo leads</i>						
Placebo_1	-0.0003	0.0005	-0.0013	0.0007	276	35
Placebo_2	-0.0019	0.0015	-0.0048	0.0010	157	19
Placebo_3	-0.0020	0.0024	-0.0066	0.0026	120	15
Placebo_4	-0.0097	0.0065	-0.0224	0.0030	45	6
<i>Test of joint nullity of the placebos: p-value = 0.5157</i>						

Notes: Event-study estimates of cable disruptions (Panel A) and cable repair days (Panel B) on log GDP per capita, estimated on the 2008–2019 sample (2020 excluded). Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. LB CI and UB CI are the 95% confidence-interval bounds. Effect_ k corresponds to year $t + (k - 1)$; coefficients on log GDP per capita are cumulative.

Table A.1.3.3: Cable failures and GDP per. cap. (const USD, ln): Average effects

	Coef.	Std. Err.	z	p-value	95% CI
Panel A: Cable disruptions					
avg_1.7	-4.967164	2.139255	-2.32	0.020	[-9.160027, -0.774301]
avg_2.7	-5.469909	2.397316	-2.28	0.023	[-10.16856, -0.771256]
avg_3.7	-6.202744	2.714713	-2.28	0.022	[-11.52348, -0.882004]
avg_4.7	-7.189929	3.041804	-2.36	0.018	[-13.15176, -1.228103]
avg_5.7	-7.742632	3.319909	-2.33	0.020	[-14.24795, -1.237317]
Panel B: Cable repair days					
avg_1.7	-0.491728	0.219293	-2.24	0.025	[-0.921536, -0.061920]
avg_2.7	-0.545136	0.246465	-2.21	0.027	[-1.028199, -0.062072]
avg_3.7	-0.629804	0.279791	-2.25	0.024	[-1.178184, -0.081425]
avg_4.7	-0.735498	0.317037	-2.32	0.020	[-1.356879, -0.114117]
avg_5.7	-0.825086	0.356996	-2.31	0.021	[-1.524578, -0.125360]

Table A.1.3.4: Cable failures and GDP per. cap. growth (%), excluding region-by-year FEs: Event-study estimates and placebo tests

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-1.224634	0.7492979	-2.693231	0.2439628	934	60
Effect_2	-1.448237	1.1043490	-3.612720	0.7162467	896	60
Effect_3	-1.374346	1.3838870	-4.086715	1.3380240	752	44
Effect_4	-4.457434	2.2467900	-8.861062	-0.0538070	660	30
Effect_5	-4.885011	2.8954640	-10.56001	0.7899936	554	28
Effect_6	-6.537678	2.9701560	-12.35908	-0.7162794	429	24
Effect_7	-6.900807	3.0939490	-12.96484	-0.8367783	390	23
<i>Test of joint nullity of the effects: p-value = 0.12817</i>						
<i>Placebo leads</i>						
Placebo_1	-0.2002856	0.7701497	-1.709751	1.309180	882	57
Placebo_2	0.0348312	0.9183815	-1.765163	1.834826	749	51
Placebo_3	-0.3929639	1.3068330	-2.954309	2.168382	594	33
Placebo_4	-1.8428420	2.8909710	-7.509042	3.823357	378	14
<i>Test of joint nullity of the placebos: p-value = 0.94817</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.0962363	0.0728768	-0.2390722	0.0465996	925	56
Effect_2	-0.0876711	0.1014472	-0.2865039	0.1111618	891	56
Effect_3	-0.1538398	0.1226799	-0.3942881	0.0866085	747	40
Effect_4	-0.3926938	0.1907398	-0.7665370	-0.0188506	660	29
Effect_5	-0.4964478	0.2739166	-1.0333140	0.0404189	556	27
Effect_6	-0.7943650	0.3297906	-1.4407430	-0.1479874	431	23
Effect_7	-0.7648841	0.3239352	-1.3997850	-0.1299828	400	23
<i>Test of joint nullity of the effects: p-value = 0.14102</i>						
<i>Placebo leads</i>						
Placebo_1	0.0170435	0.0581447	-0.0969181	0.1310050	873	53
Placebo_2	0.0240881	0.0856952	-0.1438715	0.1920476	744	47
Placebo_3	0.0202904	0.1040184	-0.1835819	0.2241626	589	29
Placebo_4	-0.1545479	0.2283079	-0.6020231	0.2929274	377	13
<i>Test of joint nullity of the placebos: p-value = 0.72843</i>						

Notes: Estimates from the baseline event-study specification re-estimated without region-by-year fixed effects (income-group-by-year fixed effects retained). All specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. F-tests are computed under `effects(7)`.

Table A.1.3.5: Cable failures and GDP per. cap. (const USD, ln), excluding region-by-year FEs: Event-study estimates and placebo tests

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
<i>Event-study effects</i>						
Effect_1	-0.0095209	0.0062256	-0.0217228	0.0026809	884	59
Effect_2	-0.0105136	0.0102868	-0.0306754	0.0096482	748	44
Effect_3	-0.0266178	0.0133292	-0.0527425	-0.0004932	654	30
Effect_4	-0.0452971	0.0181853	-0.0809396	-0.0096547	543	28
Effect_5	-0.0658371	0.0256377	-0.1160861	-0.0155881	436	24
Effect_6	-0.0638901	0.0251056	-0.1130962	-0.0146840	398	23
Effect_7	-0.0691478	0.0303987	-0.1287282	-0.0095674	328	22
<i>Test of joint nullity of the effects (effects(6)): p-value = 0.11908</i>						
<i>Placebo leads</i>						
Placebo_1	-0.0008711	0.0045311	-0.0097518	0.0080097	835	56
Placebo_2	0.0036298	0.0092456	-0.0144913	0.0217508	608	35
Placebo_3	-0.0079164	0.0222230	-0.0514726	0.0356398	503	19
Placebo_4	-0.0255676	0.0356369	-0.0954146	0.0442794	275	12
<i>Test of joint nullity of the placebos: p-value = 0.71588</i>						
Panel B: Cable repair days						
<i>Event-study effects</i>						
Effect_1	-0.0007224	0.0005846	-0.0018682	0.0004234	875	55
Effect_2	-0.0002797	0.0009233	-0.0020893	0.0015299	740	40
Effect_3	-0.0021465	0.0013917	-0.0048742	0.0005812	649	29
Effect_4	-0.0043449	0.0021470	-0.0085530	-0.0001368	541	27
Effect_5	-0.0062435	0.0030187	-0.0121600	-0.0003270	435	23
Effect_6	-0.0073223	0.0035149	-0.0142113	-0.0004333	406	23
Effect_7	-0.0070526	0.0036976	-0.0142998	0.0001945	336	22
<i>Test of joint nullity of the effects (effects(6)): p-value = 0.01381</i>						
<i>Placebo leads</i>						
Placebo_1	0.0000049	0.0003905	-0.0007605	0.0007702	826	52
Placebo_2	0.0000692	0.0008966	-0.0016881	0.0018264	600	31
Placebo_3	-0.0006697	0.0016061	-0.0038176	0.0024781	498	18
Placebo_4	-0.0027016	0.0036506	-0.0098566	0.0044534	272	11
<i>Test of joint nullity of the placebos: p-value = 0.79870</i>						

Notes: Estimates from the baseline event-study specification re-estimated without region-by-year fixed effects (income-group-by-year fixed effects retained). All specifications include country fixed effects, year fixed effects, linear country trends, and income-group-by-year fixed effects, with standard errors clustered at the country level. F-tests are reported from the **effects(6)** specification, as the **effects(7)** F-test for estimations using GDP per capita in level is flagged unreliable (near-singular variance matrix).

A.1.4 Regional spillovers

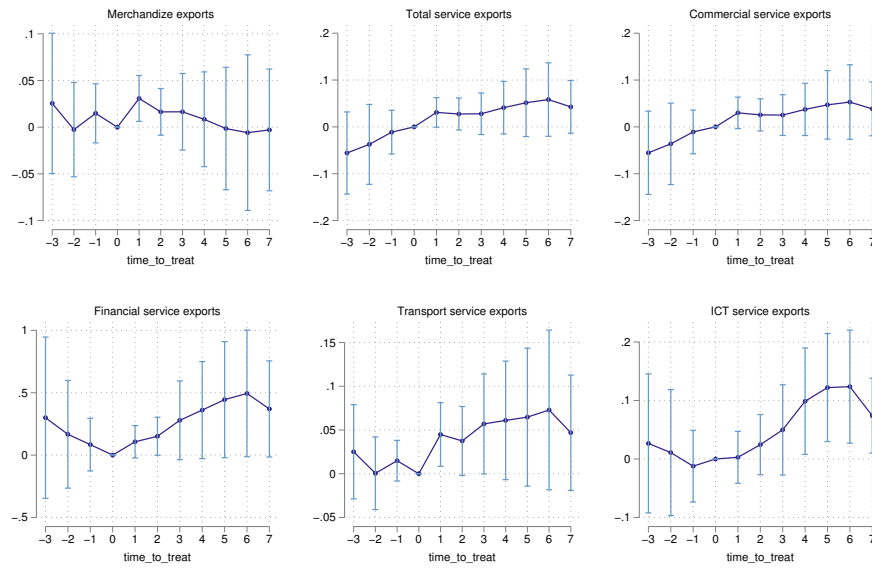


Fig. A.1.4.1: Regional spillovers on exports (USD, ln) - regional number of affected countries.

Notes: Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, and linear country trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence. The treatment is the number of affected economies recorded in the region of the non-affected country.

Table A.1.4.1: Regional peer disruptions: Event-study estimates and placebo tests

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Regional number of disrupted economies - with income-group×year FEs						
<i>Event-study effects</i>						
Effect_1	1.3014430	0.4078092	0.5021521	2.1007350	76	56
Effect_2	0.9219763	0.4834319	-0.0255328	1.8694850	74	54
Effect_3	0.9616457	0.6312889	-0.2756578	2.1989490	75	54
Effect_4	0.1886195	0.9474062	-1.6682620	2.0455020	44	39
Effect_5	0.7666671	1.0758510	-1.3419620	2.8752960	44	39
Effect_6	1.1972540	1.1752930	-1.1062770	3.5007860	45	39
Effect_7	1.0734810	1.0934610	-1.0696630	3.2166260	44	39
<i>Test of joint nullity of the effects: p-value = 1.067×10^{-6}</i>						
<i>Placebo leads</i>						
Placebo_1	0.1722157	0.7412890	-1.2806840	1.6251150	15	11
Placebo_2	-0.2504074	1.1070470	-2.4201810	1.9193660	15	11
Placebo_3	-1.0963900	1.2885600	-3.6219220	1.4291420	16	11
<i>Test of joint nullity of the placebos: p-value = 0.1120273</i>						
Panel B: Regional number of disrupted economies - no income-group×year FEs						
<i>Event-study effects</i>						
Effect_1	0.7381727	0.3508080	0.0506017	1.4257440	117	78
Effect_2	0.6197269	0.4434851	-0.2494880	1.4889420	115	76
Effect_3	0.5749558	0.5184753	-0.4412370	1.5911490	118	76
Effect_4	1.2246190	0.7023941	-0.1520478	2.6012860	80	73
Effect_5	1.5344730	0.8553296	-0.1419421	3.2108880	80	73
Effect_6	2.1583460	0.9441810	0.3077856	4.0089070	83	73
Effect_7	1.7404470	0.7475975	0.2751825	3.2057110	78	71
<i>Test of joint nullity of the effects: p-value = 0.00001842</i>						
<i>Placebo leads</i>						
Placebo_1	0.4900590	0.5729660	-0.6329336	1.6130520	37	33
Placebo_2	0.2651264	0.8856412	-1.4706980	2.0009510	37	33
Placebo_3	-0.5364274	1.0230560	-2.5415810	1.4687260	40	33
<i>Test of joint nullity of the placebos: p-value = 0.10079289</i>						
Panel C: Regional number of disruption episodes - with income-group×year FEs						
<i>Event-study effects</i>						
Effect_1	2.7292400	0.8552116	1.0530570	4.4054240	76	56
Effect_2	1.9342160	1.0141930	-0.0535653	3.9219970	74	54
Effect_3	2.0174380	1.3243820	-0.5783033	4.6131800	75	54
Effect_4	0.3966839	1.9924830	-3.5085110	4.3018790	44	39
Effect_5	1.6123710	2.2626140	-2.8222710	6.0470140	44	39
Effect_6	2.5179360	2.4717500	-2.3266050	7.3624760	45	39
Effect_7	2.2314890	2.2730220	-2.2235530	6.6865300	44	39
<i>Test of joint nullity of the effects: p-value = 1.067×10^{-6}</i>						
<i>Placebo leads</i>						
Placebo_1	0.4767450	2.0526811	-3.5465100	4.5000000	15	11
Placebo_2	-0.7802710	3.3062607	-7.2605420	5.7000000	15	11
Placebo_3	-3.3328800	3.8427959	-10.8657600	4.2000000	16	11
<i>Test of joint nullity of the placebos: p-value = 0.11202698</i>						
Panel D: Regional number of disruption episodes - no income-group×year FEs						
<i>Event-study effects</i>						
Effect_1	1.6480130	0.7831992	0.1129710	3.1800000	117	78
Effect_2	1.3916680	0.9958964	-0.5602536	3.3435896	115	76
Effect_3	1.2911290	1.1642950	-0.9908477	3.5731057	118	76
Effect_4	2.6802980	1.5373150	-0.3327847	5.6933807	80	73
Effect_5	3.3584680	1.8720420	-0.3106665	7.0276025	80	73
Effect_6	4.7239270	2.0665090	0.6736429	8.7742111	83	73
Effect_7	3.6831030	1.5820530	0.5823363	6.7838697	78	71
<i>Test of joint nullity of the effects: p-value = 0.00001843</i>						
<i>Placebo leads</i>						
Placebo_1	-3.6763695	4.1205967	-11.7527390	4.4000000	37	33
Placebo_2	0.7939525	2.6561467	-4.4120950	6.0000000	37	33
Placebo_3	-1.6123715	3.0675365	-7.6247430	4.4000000	40	33
<i>Test of joint nullity of the placebos: p-value = 0.10079309</i>						

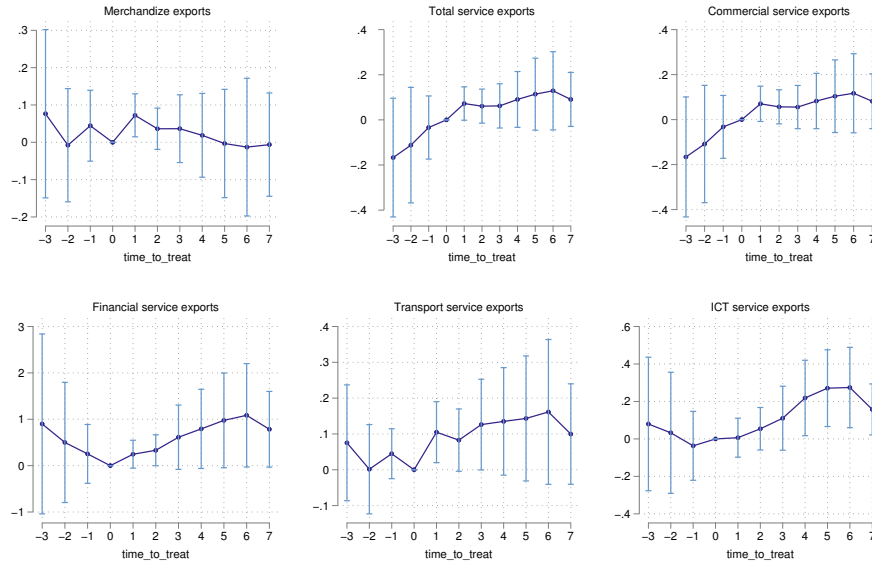


Fig. A.1.4.2: Regional spillovers on exports (USD, ln) - regional number of disruption episodes.

Notes: Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, and linear country trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence. The treatment is the number of cable disruptions recorded in the region of the non-affected country.

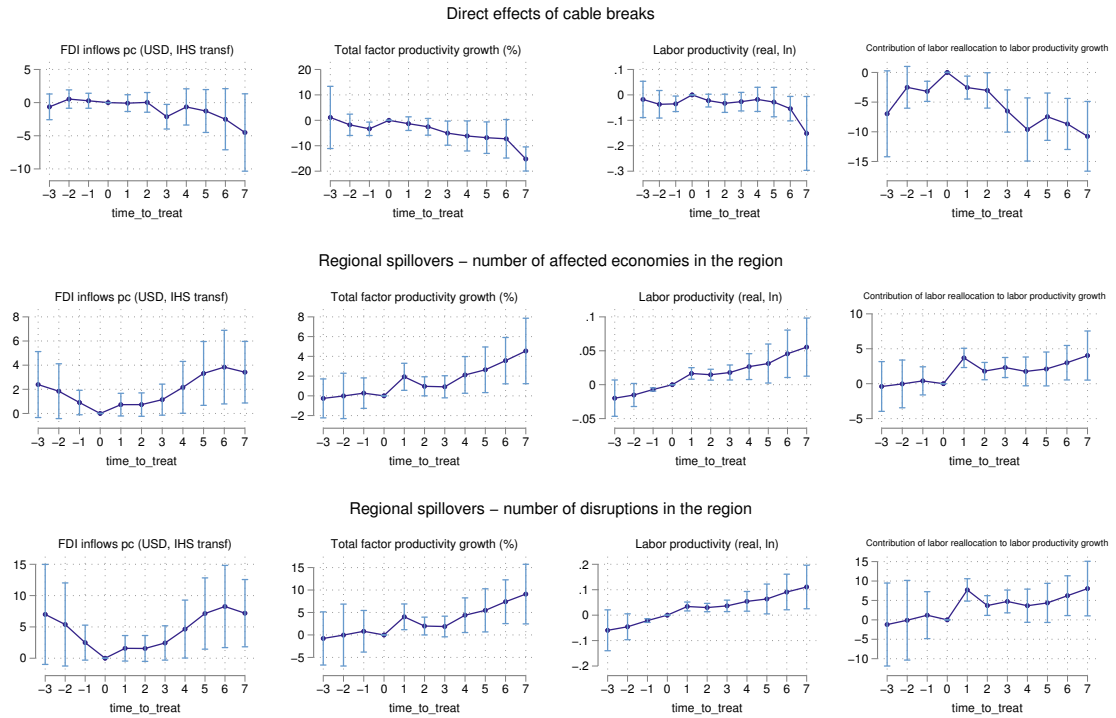


Fig. A.1.4.3: Cable breaks and competitiveness, direct effects and regional spillovers.

Notes: Figures report event-time coefficients with 90% confidence bands. All specifications include at least country fixed effects, year, and *income group* $\times$ *year* fixed effects, a linear country trends, with standard errors clustered at the country level. Top-row graphs include *region* $\times$ *years* fixed effects. FDI inflows per capita are transformed using the inverse hyperbolic sine. Labor productivity levels are measured as real GDP in US dollars measured in 2010 prices and exchange rates, divided by employment. TFP is a Human capital-adjusted TFP growth metric, calculated as a residual of labor productivity growth by subtracting the contribution of human capital and capital deepening to labor productivity growth. Year 7 corresponds to a period of six years after the cable failure occurrence. The regional treatment is either the number of affected economies or the number of disruptions—since one disruption can affect multiple economies—recorded in the region of the non-affected country.

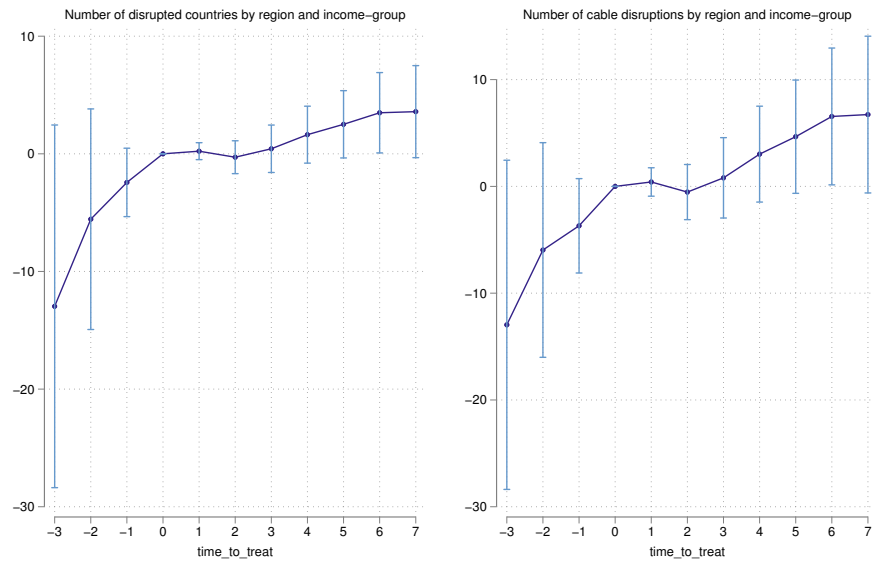


Fig. A.1.4.4: Regional–income group spillovers on per capita growth rates.

Notes: Figures report event–time coefficients with 90% confidence bands; all specifications include at least country fixed effects, year fixed effects, $income - group \times year$ fixed effects, a linear country trend, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence. The regional treatment is either the number of affected economies or the number of disruptions –since one disruption can affect multiple economies– recorded in the region–income–group of the non-affected country. When using the regional-by-income-group number of affected economies, the sample consist of 199 never-switchers (34 countries) and 971 (84) switcher-in countries. When using the regional-by-income-group number of cable disruption episodes, the sample consist of 206 never-switchers (41 countries) and 971 (84) switcher-in countries.

A.1.5 Exposure factors: additional estimations

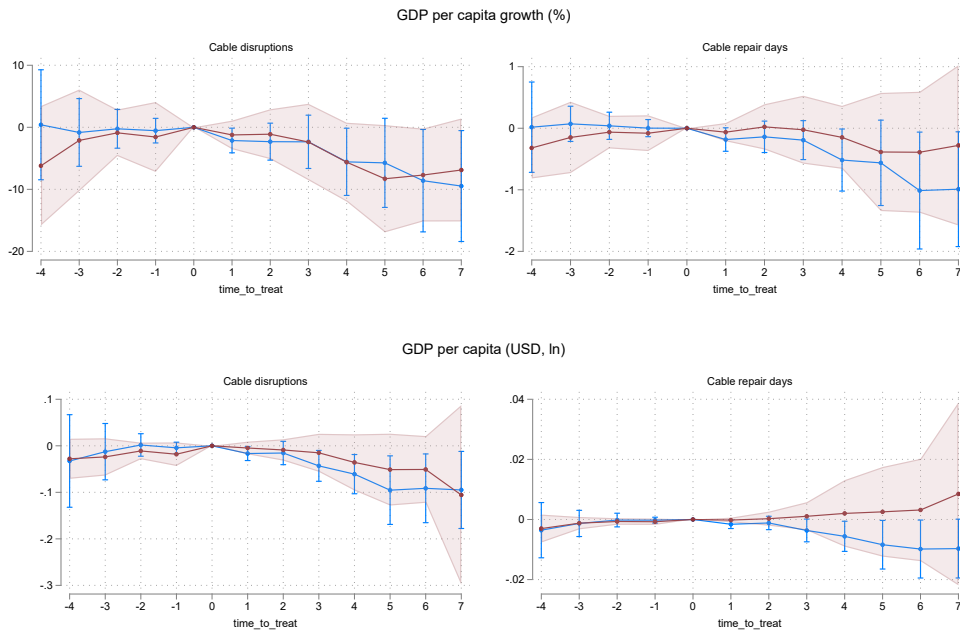


Fig. A.1.5.1: Policy margin: SMC density.

SMC density provided by the author, drawn from Telegeography's [SMC map](#). Red hollow markers = above-median connectivity at baseline; blue filled markers = below-median. Group assignments are fixed at the start of the sample. Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

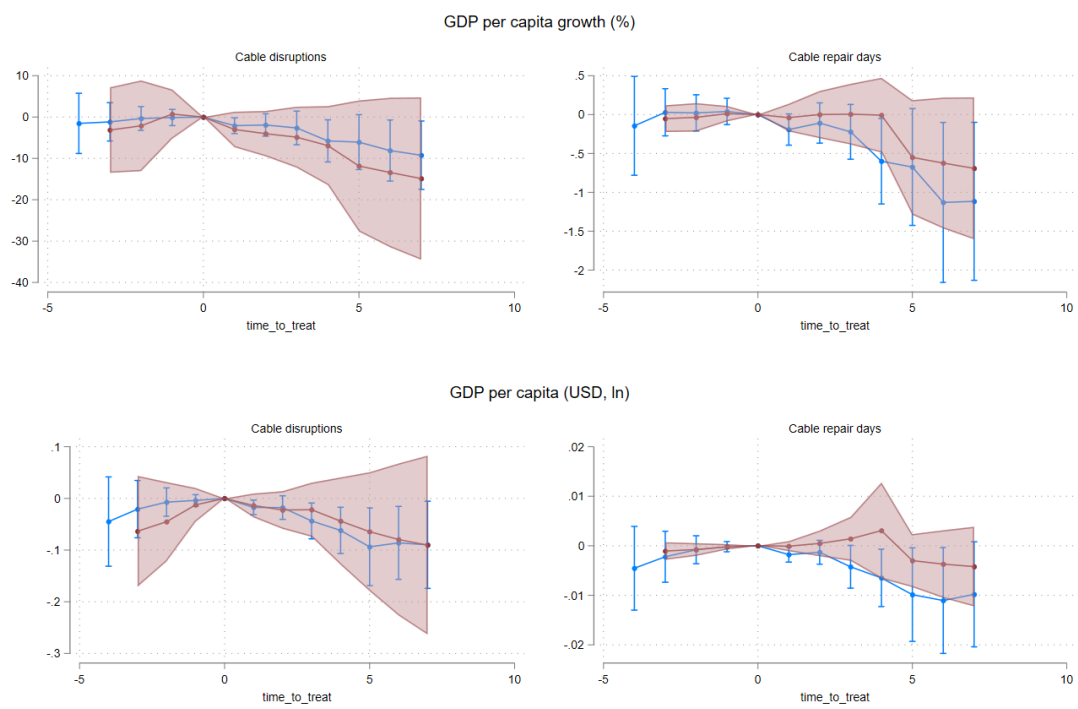


Fig. A.1.5.2: Policy margin: Internet penetration

A.1.6 Additional estimations

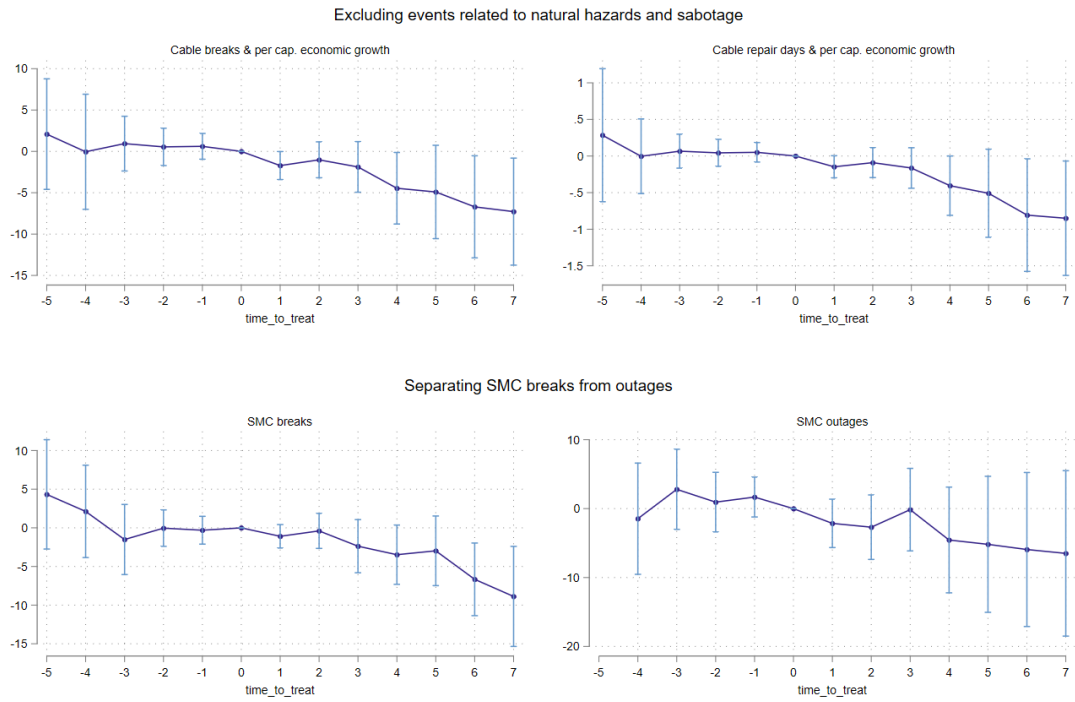


Fig. A.1.6.1: Composition of failures and exogeneity assumption.

Notes: Figures report event-time coefficients with 95% confidence bands; specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

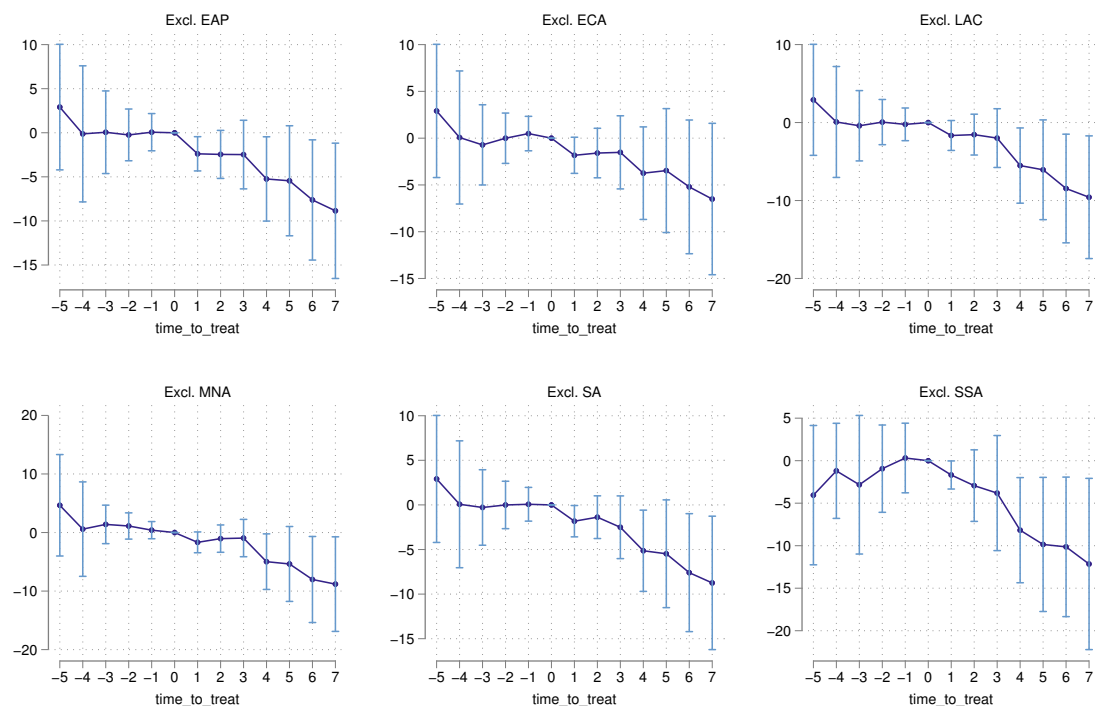


Fig. A.1.6.2: Cable disruptions and GDP per cap. growth (%) – Estimations sequentially removing regions.

Notes: Event-study coefficients for GDP per-capita growth after sequentially dropping the indicated region. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

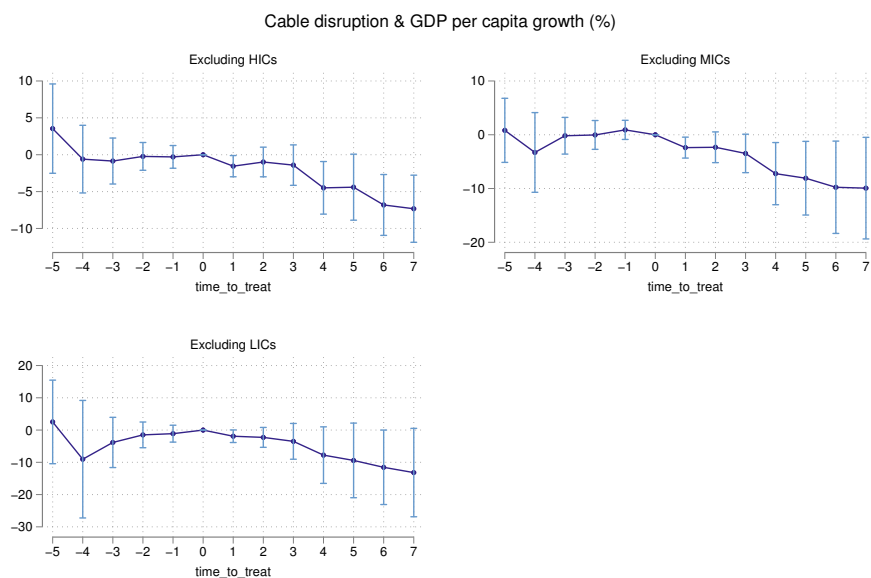


Fig. A.1.6.3: Estimations sequentially removing income groups.

Notes: Event-study coefficients for GDP per-capita growth after sequentially dropping the indicated income group. Markers show point estimates; bars denote 90% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

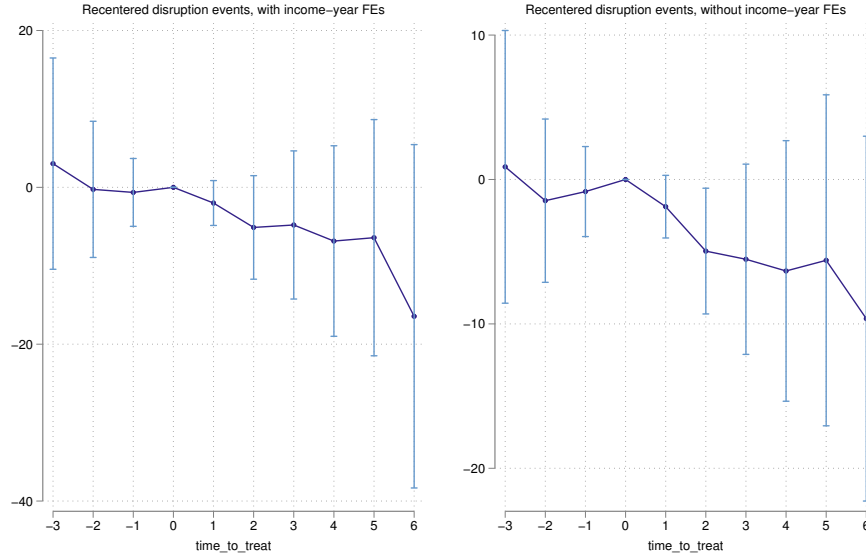


Fig. A.1.6.4: Cable disruptions and GDP per capita growth (%) – Recentering the treatment following [Borusyak and Hull \(2023\)](#).

Notes: Event-study coefficients for GDP per capita growth after recentering the treatment variable by subtracting each country's expected disruption count under a uniform within-country timing assumption: $\bar{D}_{it} = D_{it} - \bar{m}_i/T_i$, where $\bar{m}_i$ is the country's total disruptions over the sample and T_i its number of observed periods. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and region-by-year fixed effects, with standard errors clustered at the country level. Left panel includes income-group-by-year fixed effects (48 countries, 492 obs., 25 switchers); right panel excludes them (53 countries, 559 obs., 31 switchers).

Table A.1.6.1: Determinants of disruption exposure: cross-sectional balance.

Pre-period growth window	(1) 2008–2010	(2) excl. 2009	(3) pre-treatment
Pre-period GDP per capita growth	−0.394* (0.223)	−0.196 (0.243)	−0.141 (0.251)
Log GDP per capita	−0.181 (0.306)	−0.084 (0.300)	−0.002 (0.214)
Log population	0.390* (0.200)	0.303 (0.187)	0.231 (0.144)
Cable connectedness (relations)	0.823*** (0.304)	0.838*** (0.315)	0.252 (0.192)
Number of landed cables	−0.133 (0.163)	−0.124 (0.164)	0.003 (0.098)
Internet exchange points	−0.327 (0.209)	−0.314 (0.206)	−0.136 (0.155)
Region fixed effects	Yes	Yes	Yes
Countries	164	164	149
R^2	0.36	0.35	0.30

Notes: The dependent variable is the total number of SMC-induced cable disruptions per country over 2008–2020 (83 of 164 countries are ever disrupted; 241 events in total). All regressors are standardized to mean zero and unit variance and measured at baseline, so each coefficient is the change in the country's disruption count associated with a one-standard-deviation increase in the regressor. Estimation is OLS with region fixed effects and heteroskedasticity-robust (HC1) standard errors in parentheses. The three columns differ only in how pre-period growth is measured: column (1) averages GDP-per-capita growth over 2008–2010; column (2), the preferred specification, excludes the 2009 global financial-crisis year; column (3) uses each country's average growth before its first disruption, and the full-sample average for never-disrupted countries. Cable connectedness is the number of distinct cable relations. A Poisson count model yields the same pattern (column 2: connectedness incidence-rate ratio 1.49, $p = 0.002$; pre-period growth $p = 0.25$; income $p = 0.60$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

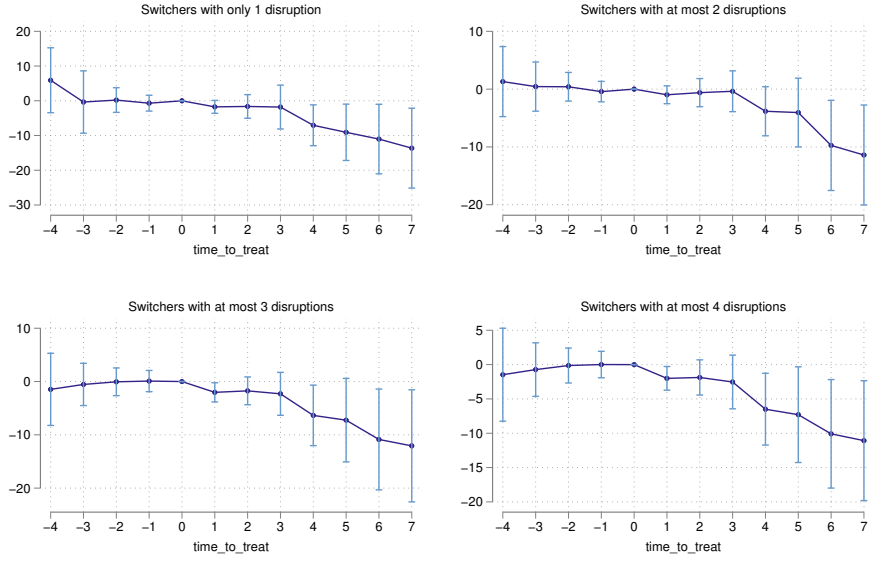


Fig. A.1.6.5: Recurrence of cable disruptions and GDP per cap. growth (%)

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

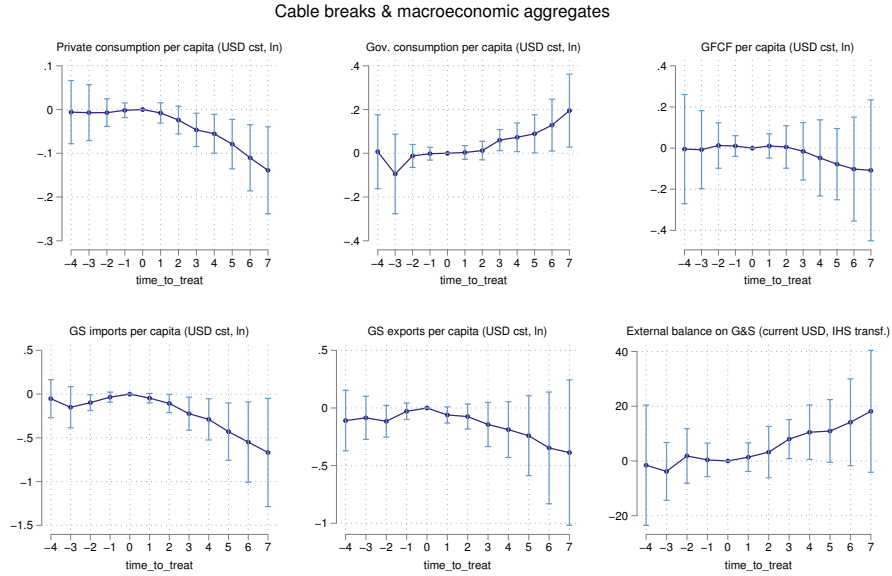


Fig. A.1.6.6: Macroeconomic adjustments following cable breaks.

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

Cable breaks & exports – by category (2015 USD, ln)

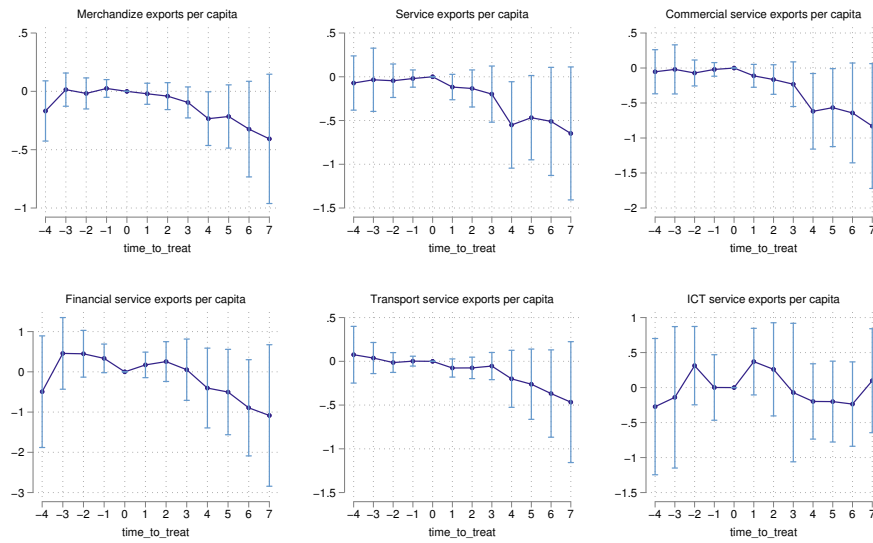


Fig. A.1.6.7: Cable breaks and exports (USD, ln), by category.

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and $region \times year$ and $income - group \times year$ fixed effects, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

Cable breaks & imports – by category (2015 USD, ln)

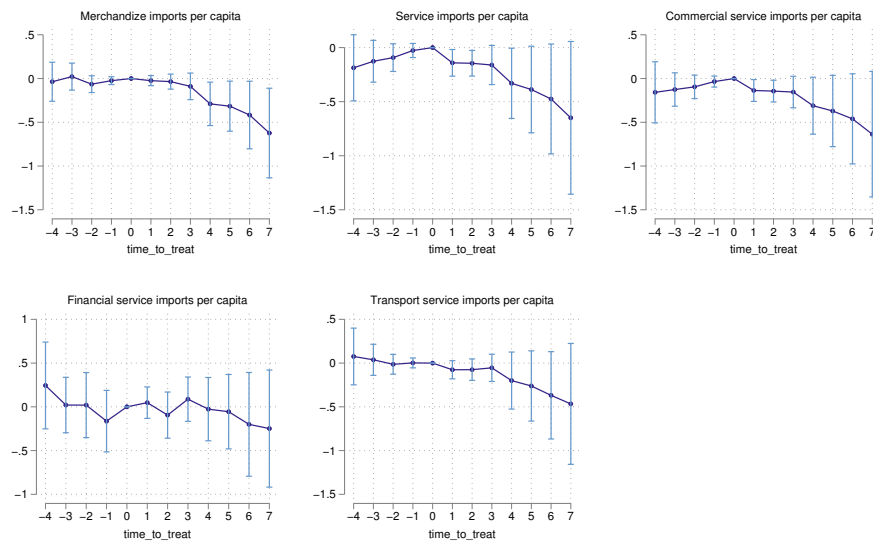


Fig. A.1.6.8: Cable breaks and imports (USD, ln), by category.

Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence.

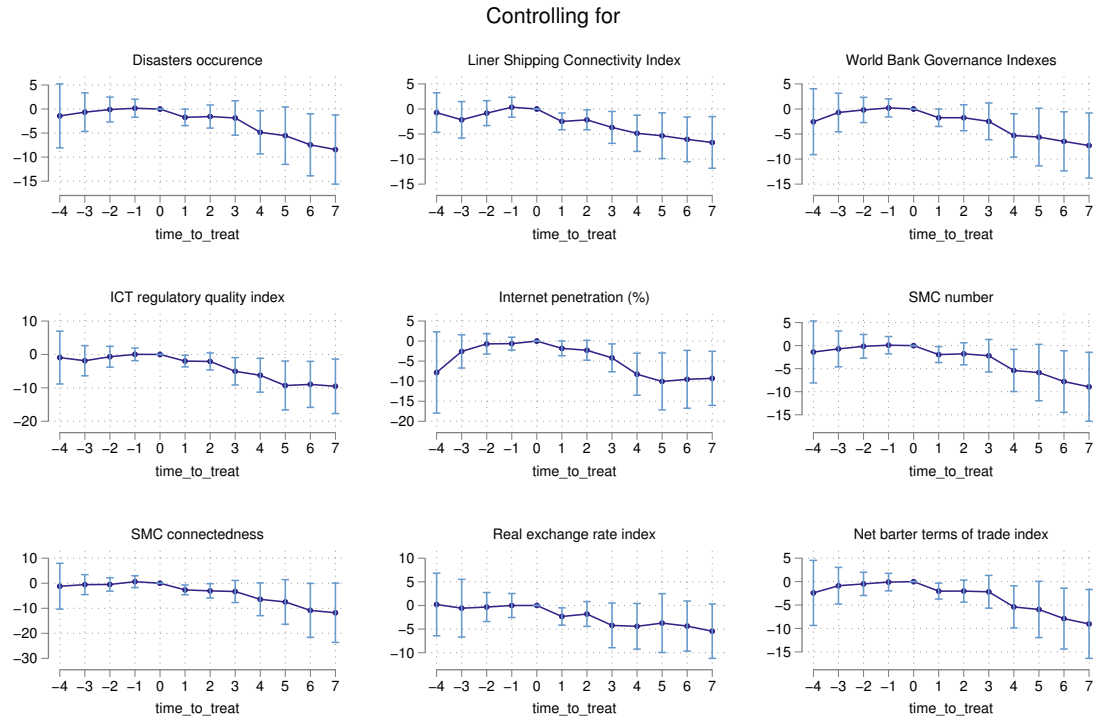


Fig. A.1.6.9: Cable disruptions and GDP per cap. growth (%), controlling for key confounding factors.

Notes: Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group non-parametric trends, with standard errors clustered at the country level. Year 7 corresponds to a period of six years after the cable failure occurrence. *Disaster occurrence* controls include the annual number of biological, climatological, geophysical, hydrological, and meteorological disaster events, as reported in the EM-DAT database. To avoid sample attrition, missing value are replaced by 0. Results remain robust to replacing missing values by region×income group×year averages or maximums, and can be provided on request. The *Liner Shipping Connectivity Index*, drawn from the UNCTAD, measures the level of integration of a country or port into the global maritime transport network, by assessing factors such as the number of ships, their capacity, the size of the ships, the number of services and the number of shipping companies. The *ICT Regulatory quality index* is measured by International Telecommunications Union's ICT Regulatory tracker. *World Bank Governance Indexes* include Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption indexes. *SMC connectedness* is the number of distinct cable-connected partners.

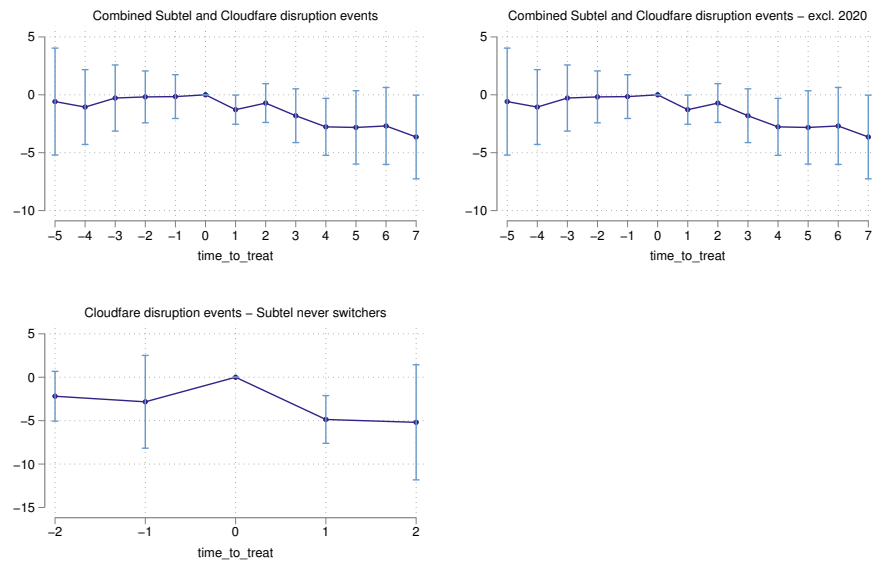


Fig. A.1.6.10: Cable disruptions and GDP per cap. growth (2008-2025) - Combining Subtel and Cloudflare events.

Notes: Figures report event-time coefficients with 90% confidence bands. All specifications include at least country fixed effects, year, and $region \times year$ fixed effects, a linear country trends, with standard errors clustered at the country level. Top-row graphs include $income\ group \times years$ fixed effects. The *Cloudflare* variable is the annual sum of cable disruption events detected by the [Cloudflare Radar](#) over 2022-2025. Top-row graphs combine (baseline) Subtelforum disruption events (over 2008-2019 or 2020) with Cloudflare disruption events (over 2022-2025). Bottom-row graph displays estimation's result using Cloudflare-only disruptions as treatment variable and a sample restricted to non-switchers according to Subtelforum data. Top-row (left) graph's sample: 102 countries (57 switchers), 1,449 observations (852 switcher-in). Bottom-row graph's sample (Subtel never switchers): 53 countries (9 switchers), 632 observations (96 switcher-in).

A.1.7 SubTel Forum *versus* Cloudflare data

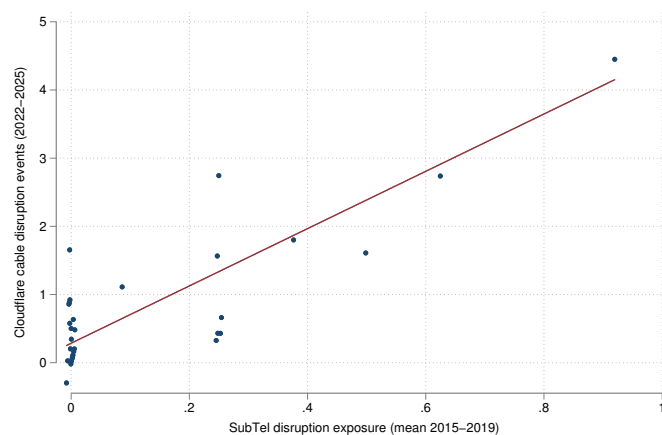


Fig. A.1.7.1: Correlation between SubTel disruption exposure and Cloudflare nationwide cable-cut events, binned scatter plot.

Notes: The Cloudflare variable is the cumulated sum of cable disruption events detected by the [Cloudflare Radar](#) from 2022 to 2025. The Subtel variable is the average number of cable disruption events reported by *Subtelforum* from 2015 to 2019. Results are robust to using alternative averaging periods. Sample of 109 developed and developing economies.

A.1.8 Import and export extensive margins

Table A.1.8.1: Import extensive margin and domestic export participation: direct effects and regional spillovers

	t	$t + 1$	$t + 2$	$t + 3$	$p_{\text{eff.}}$	$p_{\text{plac.}}$
<i>Panel A1: Direct effects – Import extensive margin</i> (Treatment: cable disruptions)						
Avg. foreign exporters per partner <i>Sample mean</i> = 309.2	−35.8** (15.4)	−64.6** (23.1)	−64.8 (39.9)	−84.6 (57.4)	0.066	0.162
N	305	240	157	123		
Sw.	22	16	11	9		
Total foreign exporters <i>Sample mean</i> = 10,599	−1,601* (885)	−3,550** (1,604)	−6,156** (2,919)	−15,642** (6,614)	0.036	0.302
N	305	240	157	123		
Sw.	22	16	11	9		
Distinct foreign exporter countries <i>Sample mean</i> = 29.6	0.54 (0.78)	−0.34 (0.98)	−1.80* (0.98)	−4.41*** (1.49)	0.022	
N	305	240	157	123		
Sw.	22	16	11	9		
<i>Panel A2: Direct effects – Domestic export participation</i> (Treatment: cable disruptions)						
Domestic exporter exit rate <i>Sample mean</i> = 0.383	0.041 (0.043)	0.152** (0.068)				
N	28	17				
Sw.	5	4				
Surviving entrants <i>Sample mean</i> = 2,398	−170** (76)					
N	18					
Sw.	4					
<i>Panel B1: Spillover effects – Domestic export participation</i> (Treatment: disrupted economies in region)						
Distinct foreign exporter countries <i>Sample mean</i> = 31.5	2.04*** (0.19)	1.36*** (0.18)	−0.10 (0.40)		<0.001	
N	112	110	78			
Sw.	75	73	43			
Total domestic exporters <i>Sample mean</i> = 17,620	−21 (216)	75 (201)	366* (175)		0.046	
N	24	22	20			
Sw.	13	11	9			
Incumbent domestic exporters <i>Sample mean</i> = 11,314	128* (75)	145* (85)	376** (177)		0.194	
N	23	21	18			
Sw.	13	11	9			
Domestic exiters <i>Sample mean</i> = 5,071	−179* (93)	−205* (107)	−502* (293)		0.113	
N	23	21	18			
Sw.	13	11	9			
<i>Panel B2: Spillover effects – Domestic export participation</i> (Treatment: disruption episodes in region)						
Distinct foreign exporter countries <i>Sample mean</i> = 31.7	4.89*** (0.46)	3.07*** (0.41)	−0.19 (0.80)		<0.001	
N	112	110	78			
Sw.	75	73	43			
Total domestic exporters <i>Sample mean</i> = 17,620	−42 (431)	149 (403)	733* (349)		0.046	
N	24	22	20			
Sw.	13	11	9			
Incumbent domestic exporters <i>Sample mean</i> = 11,314	255* (150)	289* (170)	752* (355)		0.194	
N	23	21	18			
Sw.	13	11	9			
Domestic exiters <i>Sample mean</i> = 5,071	−359* (186)	−410* (214)	−1,004* (585)		0.113	
N	23	21	18			
Sw.	13	11	9			

Notes: World Bank EDD data. Panel A: CDH estimator with country and year FEs, country-specific linear trends, and region×year FEs. Panels B: country and year FEs and country-specific linear trends; region nonparametric trends omitted to preserve cross-regional identifying variation. Panel B treatment: count of disrupted economies (*peer_break*); Panel C: count of disruption episodes (*peer_maxbreak*). SEs clustered by country. *Distinct exporter countries*: origin countries with ≥ 1 observed exporting firm at destination (sample max. 57). N , Sw., and means inline; sample contracts at longer horizons. $p_{\text{eff.}}/p_{\text{plac.}}$: joint-test p -values for effects/placebos; blank = not estimated. Panel A2 results rest on 4–5 switchers (directional). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.1.9 Economic cost of disruptions in low connectivity settings

Table A.1.9.1: Dynamic effects and relative costs (net of spillovers) of cable disruptions on GDP per capita (log), under low connectivity settings.

Effect	Cost (USD pc)	Estimate	SE	LB CI	UB CI	N	Switchers
Low connectedness: Cable disruptions							
Effect 1	-86.21	-0.0178554	0.0083093	-0.0341413	-0.0015695	227	41
Effect 2	-117.34	-0.0243052	0.0131106	-0.0500015	0.0013911	182	31
Effect 3	-229.60	-0.0475558	0.0182410	-0.0833074	-0.0118041	130	21
Effect 4	-314.08	-0.0650551	0.0235643	-0.1112402	-0.0188700	108	20
Effect 5	-505.17	-0.1046329	0.0432699	-0.1894404	-0.0198254	92	16
Effect 6	-498.76	-0.1033071	0.0446246	-0.1907697	-0.0158445	80	15
Effect 7	-513.48	-0.1063543	0.0497341	-0.2038314	-0.0088772	66	15
<i>Sample average GDP per capita: 4827.979 USD</i>							
Low connectedness: Repair days							
Effect 1	-7.10	-0.0014723	0.0007624	-0.0029666	0.0000221	219	39
Effect 2	-6.91	-0.0014322	0.0012359	-0.0038545	0.0009901	175	29
Effect 3	-19.62	-0.0040653	0.0021174	-0.0082153	0.0000847	125	20
Effect 4	-29.30	-0.0060727	0.0028406	-0.0116402	-0.0005053	106	19
Effect 5	-45.38	-0.0094042	0.0047874	-0.0187873	-0.0000211	91	15
Effect 6	-53.39	-0.0110628	0.0057208	-0.0222754	0.0001498	84	15
Effect 7	-50.70	-0.0105054	0.0056461	-0.0215715	0.0005608	70	15
<i>Sample average GDP per capita: 4825.691 USD</i>							
No IXP: Cable disruptions							
Effect 1	-108.50	-0.0189173	0.0091647	-0.0368797	-0.0009549	202	39
Effect 2	-109.63	-0.0191145	0.0155196	-0.0495324	0.0113034	153	28
Effect 3	-395.19	-0.0689046	0.0333393	-0.1342484	-0.0035607	105	16
Effect 4	-536.35	-0.0935163	0.0387908	-0.1695448	-0.0174877	93	16
Effect 5	-1052.01	-0.1834262	0.0789860	-0.3382359	-0.0286165	82	13
Effect 6	-854.71	-0.1490254	0.0661469	-0.2786709	-0.0193798	74	13
Effect 7	-891.52	-0.1554429	0.0791558	-0.3105854	-0.0003003	59	12
<i>Sample average GDP per capita: 5735.356 USD</i>							
No IXP: Repair days							
Effect 1	-11.18	-0.0016748	0.0008356	-0.0033125	-0.0000370	197	37
Effect 2	-6.60	-0.0009885	0.0014740	-0.0038776	0.0019005	148	26
Effect 3	-27.68	-0.0041464	0.0032831	-0.0105811	0.0022883	107	17
Effect 4	-38.22	-0.0057256	0.0038817	-0.0133337	0.0018824	95	17
Effect 5	-66.05	-0.0098958	0.0070332	-0.0236807	0.0038891	85	14
Effect 6	-73.76	-0.0110500	0.0082536	-0.0272268	0.0051269	77	14
Effect 7	-60.60	-0.0090782	0.0079234	-0.0246078	0.0064515	61	13
<i>Sample average GDP per capita: 6674.825 USD</i>							

Online Appendix

OA.1 Disruption data: coding protocol and maintenance areas

OA.1.1 Internet disruption database: coding protocol

Sources.

The database is built from systematic monitoring of connectivity disruption reports published by *SubTel Forum*, the submarine cable industry’s primary technical platform, which documents national-level disruptions caused by cable infrastructure failures in real time. The *SubTel Forum* cable faults and maintenance desk is complemented by two secondary sources: the *Internet Disruption Report* (internetdisruption.report), a periodic compilation of global connectivity incidents cross-referenced against operator announcements; and Akamai’s *State of the Internet Connectivity* reports for earlier years. Each event is cross-checked against targeted web searches combining the country name, the cable system name, and the approximate date. The database covers 2008–2020; the estimation sample is restricted to 2008–2020.

Unit of observation.

The raw database records connectivity disruption events at the country–event level. A single physical cable fault can impair connectivity in multiple countries simultaneously: each affected country is recorded as a separate observation, reflecting the principle that effects are measured from the perspective of what each country experiences, not from proximity to the physical break. For instance, the December 2008 failure of the SEA-ME-WE 4 and FLAG systems, caused by a vessel anchor in the Mediterranean, generated 14 country-level disruption events across Asia, the Middle East, and Southern Europe.

Attribution criterion.

A country is coded as disrupted only when operators or contemporary sources explicitly report a measurable loss of international connectivity—degraded throughput, partial or full outage, or confirmed rerouting with service impact—attributable to a cable infrastructure failure. Countries that are topologically connected to the faulted cable but for which traffic is successfully rerouted without reported degradation are coded as not treated. This *material-impact* criterion ensures that treatment captures experienced connectivity loss rather than physical proximity to the fault.

Event taxonomy.

Each raw event is assigned one of the following categories on the basis of source descriptions. *Cable breaks* are disruptions explicitly linked to a physical cut or break requiring dispatch of a repair vessel; these constitute the primary treatment events. *Cable outages* are service-affecting faults without confirmed physical damage—repeater failures, power-feed faults, branching-unit malfunctions—and are retained as secondary treatment events. *Network disruptions* cover connectivity losses attributable to routing issues, DNS problems, or other technical failures not involving cable infrastructure damage. *Other-cause* events include government-ordered shutdowns, sabotage of terrestrial or submarine fiber, power outages, natural disasters (cyclones, earthquakes, floods), cyber attacks, and scheduled maintenance operations.

Treatment variable construction.

The estimation sample retains only cable breaks and cable outages as treatment-generating events, and constructs two country-year variables: (i) *cable disruptions*: the annual count of SMC-induced connectivity losses; and (ii) *repair days*: the annual sum of connectivity-loss days associated with these events. Government-ordered shutdowns, sabotage, power failures-caused disruptions, routing issues, and events of unknown cause are coded separately and excluded from the treatment variable. As robustness checks, I drop events associated with suspected or confirmed sabotage and events associated with natural disasters that could independently affect economic outcomes (Section 5).

OA.1.2 SMC maintenance areas



Fig. OA.1.1: SMC maintenance areas.
Source: Global Marine Systems Ltd..

OA.2 First-stage estimation tables

This section reports the tables underlying the first-stage check of Section 3.3: the verified-onset episodes used in the daily event study (Table OA.2.1) and the country-month panel estimates (Table OA.2.2).

Table OA.2.1: Verified-onset disruption episodes used in the first-stage event study

Onset date	Cable system(s)	Countries (n)	Examples
23 Jan 2008	FALCON	2	Iran, India
30 Jan 2008	SEA-ME-WE 4, FLAG	9	Egypt, Pakistan, India, UAE
01 Feb 2008	FALCON	2	Oman, UAE
19 Dec 2008	SMW-3/SMW-4/FLAG/GO-1	15	Egypt, Saudi Arabia, India, Taiwan
10 Jul 2009	SAT-3	4	Nigeria, Benin, Togo, Niger
17 Aug 2009	APCN-2	6	China, Hong Kong, Taiwan, Philippines
14 Apr 2010	SEA-ME-WE 4	7	Egypt, India, UAE, France, Italy
22 Mar 2013	SEACOM	2	Mozambique, Tanzania
27 Mar 2013	SEA-ME-WE 4	5	Egypt, Saudi Arabia, India, Pakistan
23 Apr 2015	AAG	3	Vietnam, Hong Kong, Singapore
18 Jun 2017	MainOne	2	Nigeria, Cameroon
03 Dec 2017	SEA-ME-WE 3	3	Australia, Singapore, Vietnam
30 Mar 2018	ACE	10	Mauritania, Sierra Leone, Liberia, Senegal
09 Jan 2020	FALCON, FLAG-FEA	7	Yemen, Sudan, Tanzania, Saudi Arabia
16 Jan 2020	SAT-3, WACS	16	Nigeria, South Africa, Angola, Gabon
20 Feb 2020	ECFS	6	Antigua, Dominica, Grenada, St Kitts
14 Mar 2020	EASSy	13	Egypt, Kenya, Tanzania, Djibouti
Total (RIPE RIS coverage)		107 country-events	62 distinct countries

Notes: Documented multi-country submarine-cable disruptions whose onset date is web-verified from contemporaneous operator and press sources, and used in the daily event study of Section 3.3. A country enters once per episode that affects it. “Countries (n)” is the coded scope. A few country-events without RIPE RIS coverage (Ethiopia, Vietnam in 2015 and 2017, Anguilla, Montserrat) are dropped, leaving 107 country-events across 62 countries.

Table OA.2.2: First stage: routed connectivity on the SubTel treatment, country-month panel

Outcome (country-month)	Two-way FE coef. (SE)
Outage days	0.792*** (0.234)
Outage episodes	0.130*** (0.031)
Severity ($\sum \text{drawdown} $)	32.82*** (9.86)
Worst-month drawdown (pp)	−6.29*** (1.43)
Treatment	SubTel cable disruptions (count, per country-month)
Fixed effects	country, year-month
Economies / Obs.	172 / 26,288

Notes: All outcomes are within-country measures of routed-prefix degradation (RIPE RIS), where a day’s *drawdown* is the percentage shortfall of routed prefixes relative to their trailing 60-day baseline. *Outage days* = number of days in the month with drawdown below −10%. *Outage episodes* = number of distinct runs of such days ($\text{gaps} \leq 3$ days merged). *Severity* = sum of $|\text{drawdown}|$ over the month’s outage days (depth $\times$ duration). *Worst-month drawdown* = the month’s deepest daily drawdown (percentage points, negative). The column is a two-way fixed-effects regression on the SubTel cable-disruption count, SE clustered by country. The heterogeneity-robust dynamic effects of [de Chaise-martin and d’Haultfoeuille \(2024\)](#) on the same panel are plotted in Figure 4(c)–(d) and reported in full, with placebos, in Table OA.2.3. A complementary Poisson (PPML) specification for the episode count gives a coefficient of 1.04 on the disruption count ($z = 6.8$). The monthly mean routed level (`mean_v4`) shows no statistically significant change at any horizon, consistent with routed connectivity returning to trend after the acute withdrawal. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.2.3: First stage: full CDH dynamic estimates, country-month panel

Event-time	Worst-month drawdown (pp)	Outage episodes	Outage days	Severity ($\sum \text{drawdown} $)
Effect ₁	−10.03*** (2.73)	0.163*** (0.056)	0.85* (0.50)	53.09** (25.34)
Effect ₂	−3.46 (2.15)	0.039 (0.053)	1.87** (0.88)	52.86** (25.60)
Effect ₃	−1.80 (1.59)	0.057 (0.048)	0.44 (0.52)	8.42 (11.82)
Effect ₄	0.41 (1.44)	−0.01 (0.049)	0.16 (0.52)	0.25 (10.71)
Effect ₅	0.27 (1.14)	0.019 (0.041)	−0.13 (0.36)	−2.86 (6.80)
Effect ₆	—	−0.063 (0.042)	−0.44 (0.35)	−8.93 (8.00)
Placebo ₁	0.38 (1.75)	0.010 (0.063)	−0.12 (0.41)	−1.52 (9.49)
Placebo ₂	−0.76 (1.46)	−0.011 (0.038)	0.39 (0.57)	7.26 (12.26)
Placebo ₃	−0.58 (1.37)	0.013 (0.049)	0.55 (0.57)	11.12 (14.19)
Joint p (effects)	0.004	<0.001	0.002	0.028
Joint p (placebos)	0.894	0.944	0.454	0.806

Notes: Dynamic effects from [de Chaisemartin and d'Haultfoeulle \(2024\)](#) (`did_multiplegt_dyn`, `normalized`) on the 172-country country-month panel, with treatment the SubTel cable-disruption count and standard errors clustered by country in parentheses. Effect _{ℓ} is the normalized effect ℓ months into a disruption and Placebo _{ℓ} the corresponding pre-trend lead; the worst-month drawdown is estimated over five post-periods and the other outcomes over six. Outcomes are defined as in Table [OA.2.2](#). 77 switchers contribute to the effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA.3 Leave-one-out robustness

To assess whether the headline estimates are driven by any single influential switcher, I re-estimate each baseline event-study sequentially excluding each treated country from the estimation sample. The resulting distribution of point estimates at horizons $t+5$ and $t+6$ is reported below for four key outcomes: GDP per capita growth (baseline specification, Figure OA.3.1), log GDP per capita (baseline, Figure OA.3.2; and broader counterfactual without region-by-year fixed effects, Figure OA.3.3), and log goods-and-services imports per capita which displays the highest sensitivity (-65%) to cable breaks (mechanism-section specification, Figure OA.3.4). Across all four outcomes, point estimates remain negative across the entire leave-one-out (LOO) distribution; the headline magnitudes are robust in sign and order of magnitude to any single-country exclusion. The most influential switcher varies by outcome—Armenia, Liberia, or Sierra Leone for the GDP outcomes; Sudan for the import channel—but no single exclusion moves the point estimate outside the full-sample 95% confidence band.

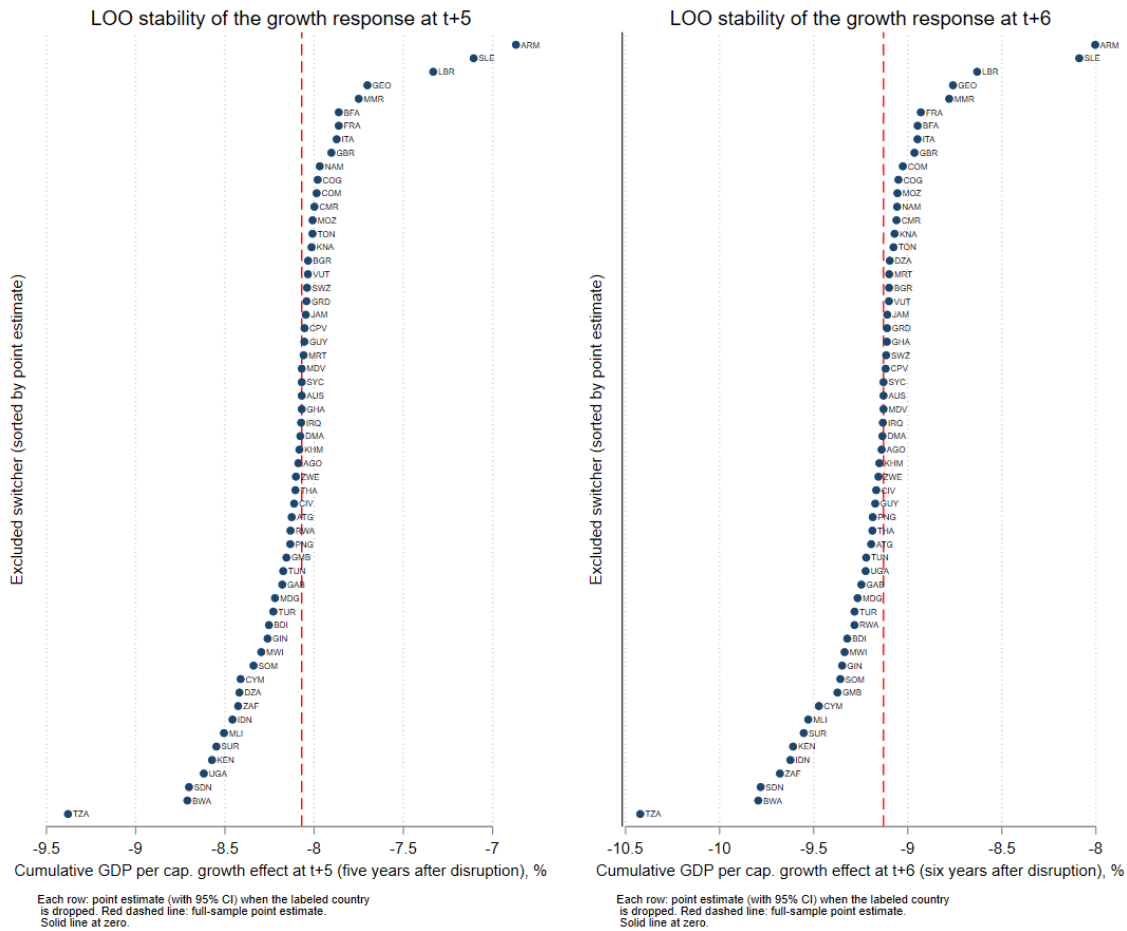


Fig. OA.3.1: Leave-one-out stability: GDP per capita growth response (baseline specification).

Notes: Each row reports the estimated effect at $t+5$ (left panel) or $t+6$ (right panel) from a leave-one-out re-estimation of the baseline event-study in which the labeled country is excluded from the sample. Red dashed line: full-sample point estimate; solid vertical line: zero. Specification follows Figure 5 (treatment: `cable_disrupt`; country, year, country-trend, region-by-year, and income-group-by-year fixed effects; standard errors clustered at the country level).

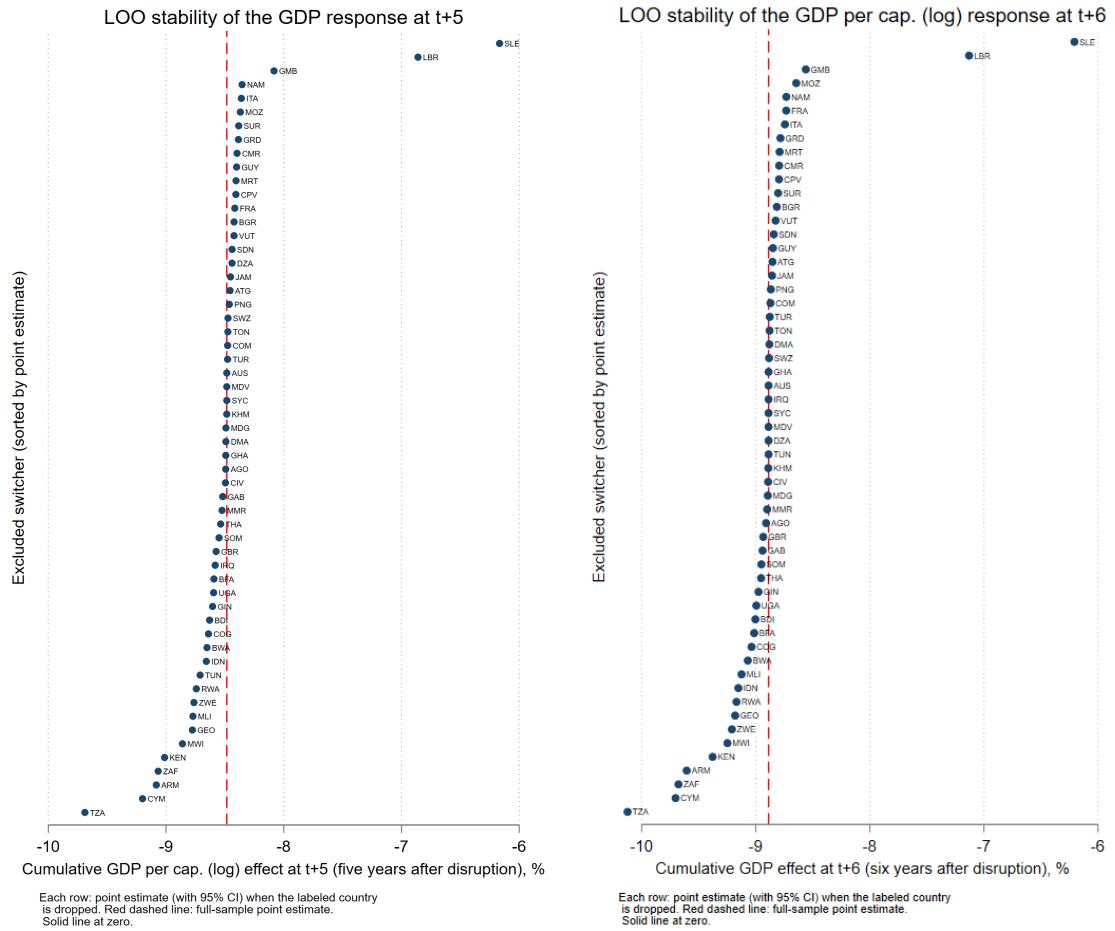


Fig. OA.3.2: Leave-one-out stability: log GDP per capita response (baseline specification).

Notes: Each row reports the estimated cumulative deviation in log GDP per capita at $t+5$ (left panel) or $t+6$ (right panel) from a leave-one-out re-estimation of the baseline event-study in which the labeled country is excluded from the sample. Red dashed line: full-sample point estimate; solid vertical line: zero. Specification follows Figure 5 (treatment: `cable_disrupt`).

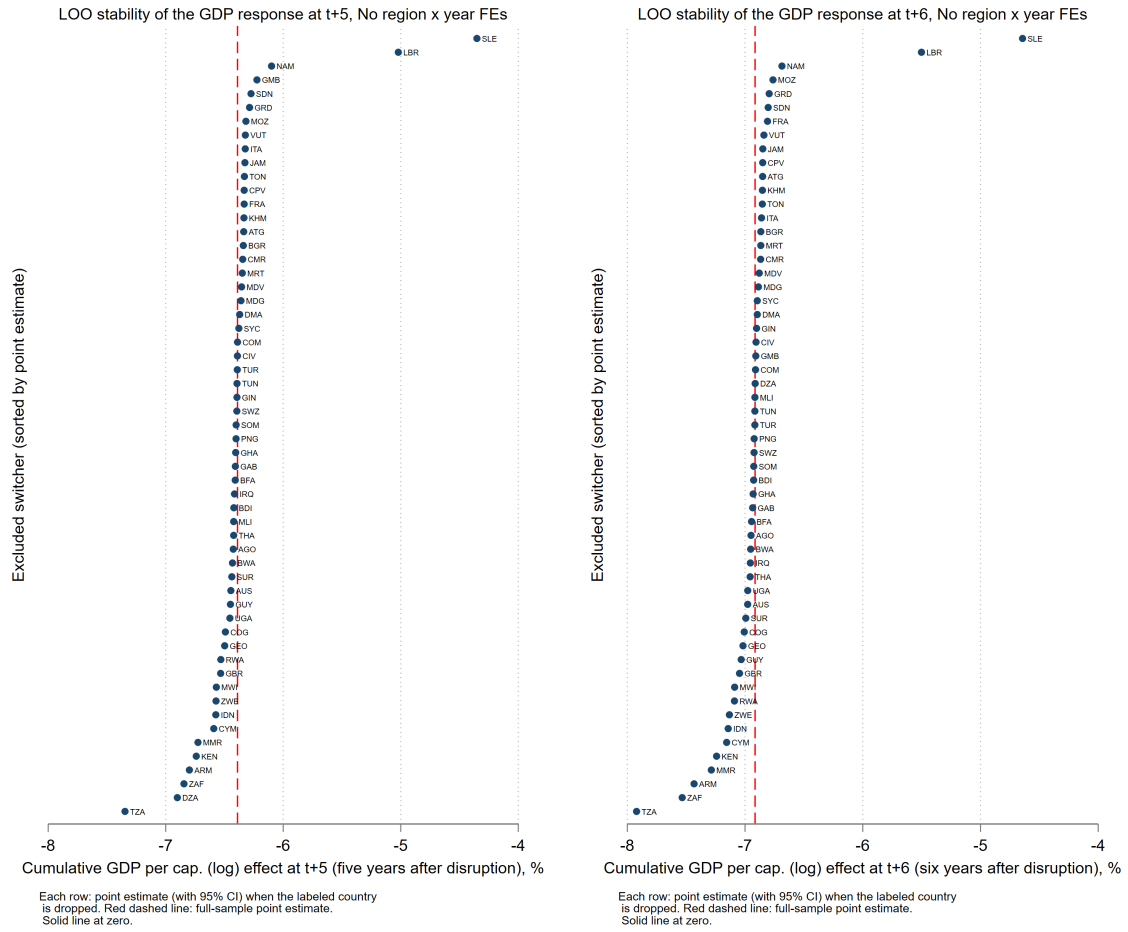


Fig. OA.3.3: Leave-one-out stability: log GDP per capita response, broader counterfactual (no region-by-year FE).

Notes: Each row reports the estimated cumulative deviation in log GDP per capita at $t+5$ (left panel) or $t+6$ (right panel) from a leave-one-out re-estimation of the broader-counterfactual event-study (region-by-year fixed effects dropped) in which the labeled country is excluded from the sample. Red dashed line: full-sample point estimate; solid vertical line: zero. Specification follows Figure 6 (treatment: `cable_disrupt`).

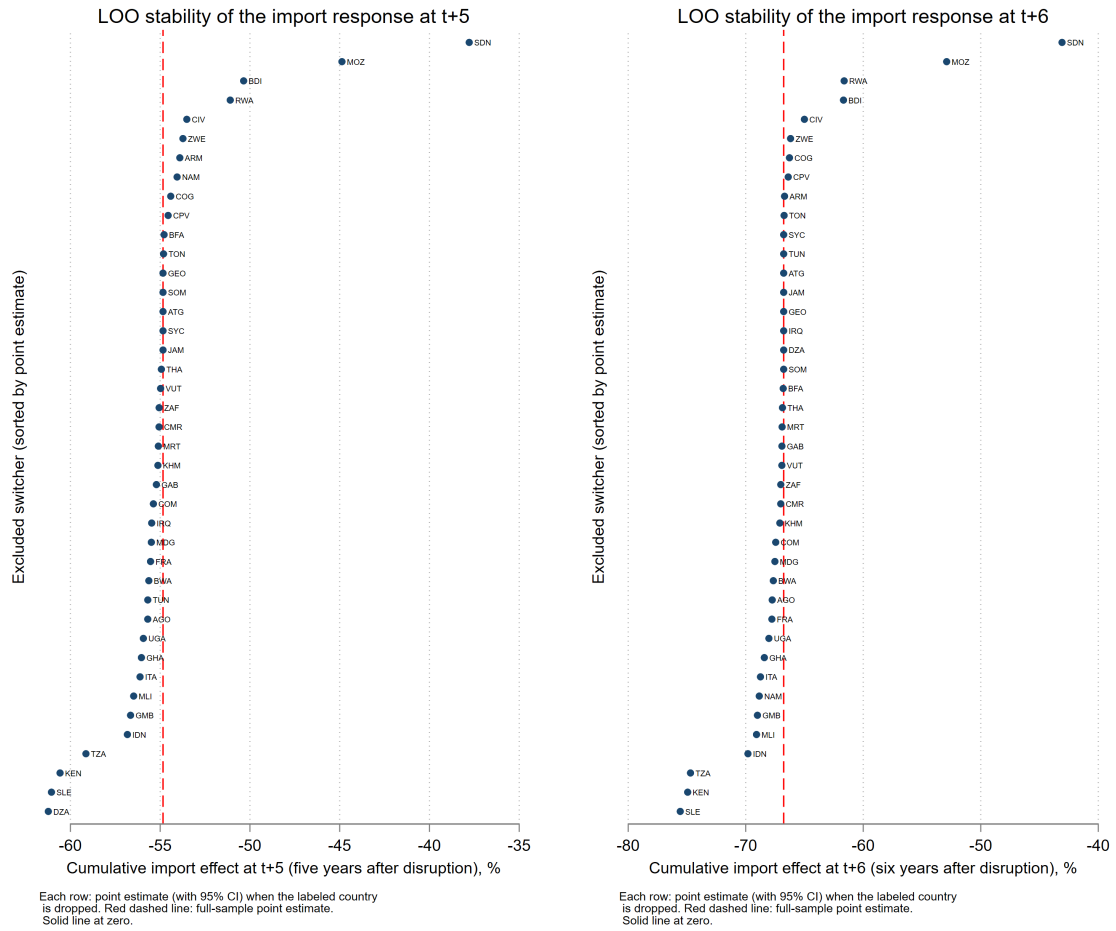


Fig. OA.3.4: Leave-one-out stability: goods-and-services import response.

Notes: Each row reports the estimated cumulative deviation in log goods-and-services imports per capita at $t+5$ (left panel) or $t+6$ (right panel) from a leave-one-out re-estimation in which the labeled country is excluded from the sample. Treatment: `cable_break` (mechanism-section convention). Red dashed line: full-sample point estimate; solid vertical line: zero.

OA.4 Formal framework for identification under regional spillovers

OA.4.1 Potential outcomes with partial (regional) interference

Let S_{rt} denote a measure of *regional exposure* in region r at time t (e.g., the number of disruptions affecting other countries in region r), and write $r(i)$ for country i 's region. Under partial interference at the regional level, potential outcomes may depend on both own exposure and regional exposure, $Y_{it}(D_{it}, S_{r(i)t})$, hypothesizing outcomes are not affected by shocks outside the region once I condition on controls and fixed effects. This structure corresponds to the 'Regional SUTVA' of (Huber and Steinmayr 2021), adapted to a cross-country setting where the aggregate units are countries and interference operates within World Bank regions.

This structure therefore maps naturally into three groups in a given year/event-time: (i) treated countries T with $D_{it} > 0$; (ii) *spillover-only* countries S with $D_{it} = 0$ but $S_{r(i)t} > 0$; and (iii) *pure controls* C with $D_{it} = 0$ and $S_{r(i)t} = 0$.

OA.4.2 Direct, spillover, and total (ITT) effects

With interference, the “direct effect” of a disruption is interpreted as the effect of increasing own exposure while holding fixed the regional exposure faced by both treated and untreated units. In this setting, this corresponds to the contrast

$$\text{Direct effect, net of spillovers : } T - S \equiv \mathbb{E}[Y_{it}(D_{it} = 1, S_{r(i)t}) - Y_{it}(D_{it} = 0, S_{r(i)t})],$$

i.e., treated vs. untreated *within the same exposed regional environment*. The spillover effect is defined as the effect of being untreated but located in an exposed region:

$$\text{Spillover on non-treated : } S - C \equiv \mathbb{E}[Y_{it}(0, S_{r(i)t} > 0) - Y_{it}(0, S_{r(i)t} = 0)].$$

Finally, the total effect of a disruption on treated countries relative to non-exposed controls, referred to as the *Intention-to-Treat* (ITT) under interference, is

$$\text{ITT / total effect : } T - C \equiv \mathbb{E}[Y_{it}(1, S_{r(i)t} > 0) - Y_{it}(0, S_{r(i)t} = 0)].$$

These satisfy the accounting identity:

$$T - C = (T - S) + (S - C). \quad (2)$$

In other words, the total effect on treated units relative to a pure control decomposes into a direct effect net of spillovers plus spillovers on non-treated units. In the empirical implementation, I construct an *implied ITT* by combining estimates of $(T - S)$ and $(S - C)$.

OA.4.3 Mapping to empirical specifications and identification assumptions

The baseline event-study with region-by-year fixed effects, $\delta_{r(i),t}$, identifies $(T - S)$ by comparing treated and untreated countries *within the same region-year*. Region-by-year fixed effects absorb all shocks common to the region in a given year—such as SMC systems deployment, or region-wide

diversion and reallocation forces—so identification comes from within-region-year differences in own exposure D_{it} .

To identify spillovers ($S - C$), I estimate a separate event-study on the non-treated sample (restricting to $D_{it} = 0$) in which treatment is a measure of regional exposure, that is, the number of disruption episodes in the region.⁴³ This design compares non-treated countries in exposed regions to non-treated countries in non-exposed regions, thereby capturing spillovers on untreated units.

Finally, I obtain the implied ITT effect ($T - C$) by combining the two components according to Equation (2). Because both components may be statistically dependent, I compute confidence intervals for the implied ITT using a joint country-cluster bootstrap: in each replication I resample countries, re-estimate the direct and spillover specifications under the main design, compute the average effect over $[t+3; t+6]$, and then form the total effect for that replication.

This decomposition is informative under two maintained assumptions. First, interference is primarily regional (partial interference), so that S_{rt} is a sufficient summary of exposure to others' disruptions once I condition on fixed effects and controls. Second, conditional parallel trends hold for the relevant comparisons: treated and untreated countries within region-years (for $(T - S)$), and non-treated countries in exposed vs. non-exposed regions (for $(S - C)$). Under these conditions, this framework makes it possible to interpret baseline estimates as direct effects net of spillovers and to recover an implied total effect relative to non-exposed controls.

Table OA.4.1 reports point estimates and 95% bootstrapped percentile intervals of the implied ITT obtained by combining $(T - S)$ and $(S - C)$ via a joint country-cluster bootstrap, averaged over horizons $k \in \{4, \dots, 7\}$. Two peer-exposure measures are used: a *breadth* measure (number of disrupted peers in the region-year) and a *frequency* measure (number of disruption episodes). The two measures paint a consistent picture but emphasize different features of regional interference. The breadth measure delivers more precise estimates of spillovers and implies a total effect of about -7 percentage points (pp) over $[t+3; t+6]$ (Panel A). The frequency measure yields substantially larger spillovers (Panel B), consistent with strong diversion when disruptions are recurrent, but the implied total effect is estimated less precisely and centers around -5 pp. Taken together, these estimates align in sign and order of magnitude with the direct $T - C$ estimate reported in Figure 6.

⁴³I also use the regional number of disrupted economies as complementary approach to estimating spillover effects. However, since multiple countries can be affected by only one disruption in a region, the resulting estimates cannot be used to measure the implied ITT effect.

Table OA.4.1: Average direct, spillover, and implied intention-to-treat effects at horizon $[t + 3; t + 6]$, with bootstrapped confidence intervals.

Effects	Valid reps	Point est.	Bootstrapped 95% CI
Direct effect (TS): $T - S$	415	-8.937	[-22.073; -2.436]
Panel A: Number of regional disrupted peers			
Spillovers (SC): $S - C$	415	1.664	[0.184; 3.319]
Implied ITT (TC): $TS + SC$	415	-7.273	[-20.008; -0.616]
Panel B: Number of regional disruption episodes			
Spillovers (SC): $S - C$	415	3.611	[0.3928; 7.196]
Implied ITT (TC): $TS + SC$	415	-5.326	[-18.606; 2.094]

Notes: All specifications include country fixed effects, year fixed effects, linear country trends; but exclude income-group-by-year fixed effects. TS corresponds to the direct effect estimated with region-by-year fixed effects (within-region identification). SC corresponds to the spillover effect estimated on the non-treated sample (comparing non-treated units in exposed vs. non-exposed regions). TC is constructed as $TC = TS + SC$. Following standard definitions under interference (e.g. [Özler \(2018\)](#)), the ITT (treated vs. pure control) decomposes as $(T - C) = (T - S) + (S - C)$. Reported statistics come from 415 valid country-cluster bootstrap replications in each panel; the 95% confidence interval is the percentile bootstrap interval (2.5th and 97.5th percentiles). TS, SC, and TC are computed as the average of event-study coefficients at horizons $[t+3; t+6]$.

OA.5 Exposure factors: detailed estimates and country lists

Table OA.5.1: Exposure factor: cable density - Dep. var: GDP per capita (cst USD, ln)

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects, countries with less than 2 SMCs at baseline						
Effect_1	-0.016151	0.0077128	-0.031732	-0.0014982	277	48
Effect_2	-0.015491	0.0127912	-0.0405615	0.0095792	213	34
Effect_3	-0.043123	0.0169108	-0.0762769	-0.0099877	159	23
Effect_4	-0.0609051	0.0215189	-0.1030814	-0.0187288	135	22
Effect_5	-0.0953038	0.0375909	-0.1689807	-0.0216269	118	18
Effect_6	-0.0913442	0.0376943	-0.1652237	-0.0174646	105	17
Effect_7	-0.0948740	0.0422522	-0.1776868	-0.0120611	89	17
Test of joint nullity of the effects: p -value = 0.09697087						
Placebo leads						
Placebo_1	-0.0043465	0.006149	-0.0163984	0.0077054	277	48
Placebo_2	0.0017591	0.0123483	-0.0224432	0.0259613	166	28
Placebo_3	0.0126587	0.0308894	-0.0732008	0.0478834	101	15
Placebo_4	-0.0326342	0.0507389	-0.1320806	0.0668122	43	9
Test of joint nullity of the placebos: p -value = 0.79889347						
Event-study effects, countries with more than 2 SMCs at baseline						
Effect_1	-0.0047134	0.0069626	-0.0183598	0.0089331	31	7
Effect_2	-0.0089523	0.0117188	-0.0319208	0.0140161	29	6
Effect_3	-0.0150795	0.0208495	-0.0559438	0.0257848	24	6
Effect_4	0.0357675	0.0307084	-0.0960205	0.0246676	21	5
Effect_5	-0.0511884	0.0394048	-0.1284204	0.0260436	21	5
Effect_6	-0.0508085	0.0366205	-0.1225834	0.0209663	21	5
Effect_7	-0.1056703	0.0985599	-0.2988441	0.0875036	17	3
Test of joint nullity of the effects: p -value = 0.55676114						
Placebo leads						
Placebo_1	-0.0180623	0.0129461	-0.0434362	0.0073116	14	4
Placebo_2	-0.0110334	0.0091271	-0.0289222	0.0068554	14	4
Placebo_3	-0.0238910	0.0204706	-0.0640127	0.0162307	7	3
Placebo_4	-0.0279690	0.0219853	-0.0710594	0.0151214	4	2
Test of joint nullity of the placebos: p -value = 0.55604552						
Panel B: Cable repair days						
Event-study effects, countries with less than 2 SMCs at baseline						
Effect_1	-0.001519	0.0007296	-0.0030218	-0.0001619	264	44
Effect_2	-0.001162	0.0011387	-0.0033939	0.0010698	201	30
Effect_3	-0.0036508	0.0019287	-0.0074310	0.0001293	153	22
Effect_4	-0.0056063	0.0025575	-0.0106189	-0.0005938	133	21
Effect_5	-0.0084040	0.0041259	-0.0164905	-0.0003175	117	17
Effect_6	-0.0098397	0.0049180	-0.0194788	-0.0002006	109	17
Effect_7	-0.0096833	0.0049906	-0.0194647	-0.0000980	93	17
Test of joint nullity of the effects: p -value = 0.00767882						
Placebo leads						
Placebo_1	-0.0002204	0.0004972	-0.0011949	0.0007541	264	44
Placebo_2	-0.0001944	0.0011585	-0.0024649	0.0020762	154	24
Placebo_3	0.0013185	0.0022234	-0.0056763	0.0030394	95	14
Placebo_4	-0.0035647	0.0046860	-0.0127490	0.0056196	40	8
Test of joint nullity of the placebos: p -value = 0.88315022						
Event-study effects, with more than 2 SMCs at baseline						
Effect_1	-0.0001899	0.0003995	-0.0009730	0.0005932	33	8
Effect_2	0.0002654	0.0011936	-0.0020740	0.0026049	33	8
Effect_3	0.0010651	0.0023863	-0.0036119	0.0057421	26	7
Effect_4	0.0021075	0.0056481	-0.0090525	0.0130875	23	6
Effect_5	0.0025431	0.0076081	-0.0123684	0.0174547	23	6
Effect_6	0.0030162	0.0086911	-0.0138722	0.0210961	23	6
Effect_7	0.0085037	0.0155296	-0.0219338	0.0389412	19	4
Test of joint nullity of the effects: p -value = 0.63489944						
Placebo leads						
Placebo_1	-0.0007526	0.0005394	-0.0018098	0.0003047	14	4
Placebo_2	-0.0006755	0.0005588	-0.0017707	0.0004197	14	4
Placebo_3	0.0012315	0.0010552	-0.0032996	0.0008366	7	3
Placebo_4	-0.0030327	0.0023768	-0.0076821	0.0016347	4	2
Test of joint nullity of the placebos: p -value = 0.5560458						

Table OA.5.2: Exposure factor: Connected partners - Dep. var: GDP per capita (cst USD, ln)

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects, countries with less than 10 connected partners at baseline						
Effect_1	-0.0178554	0.0083093	-0.0341413	-0.0015695	227	41
Effect_2	-0.0243052	0.0131066	-0.0500015	0.0013911	182	31
Effect_3	-0.0475558	0.0184210	-0.0833074	-0.0118041	130	21
Effect_4	-0.0650511	0.0236543	-0.1112402	-0.0180870	108	20
Effect_5	-0.1046329	0.0432699	-0.1894404	-0.0198254	92	16
Effect_6	-0.1303071	0.0446246	-0.2197697	-0.0158445	80	15
Effect_7	-0.1063543	0.0479341	-0.2038314	-0.0088772	66	15
<i>Test of joint nullity of the effects: p-value = 0.19718756</i>						
Placebo leads						
Placebo_1	-0.0037281	0.0057842	-0.0150649	0.0076088	227	41
Placebo_2	-0.0077203	0.0140648	-0.0358267	0.0198462	138	26
Placebo_3	-0.0318909	0.0337203	-0.0980716	0.0341097	76	14
Placebo_4	-0.0462155	0.0515078	-0.1471699	0.0540738	40	9
<i>Test of joint nullity of the placebos: p-value = 0.90050126</i>						
Event-study effects, countries with more than 10 connected partners at baseline						
Effect_1	-0.0129436	0.0105689	-0.0336582	0.0077111	49	13
Effect_2	0.0171878	0.0149755	-0.0121636	0.0465392	25	9
Effect_3	0.0364046	0.0211033	-0.0030501	0.0758593	22	8
Effect_4	0.0316143	0.0249748	-0.0173054	0.0805939	20	7
Effect_5	0.0261818	0.0293002	-0.0312456	0.0836092	19	7
Effect_6	0.0142449	0.0241369	-0.0330626	0.0615525	18	7
Effect_7	0.0421985	0.0406314	-0.0374377	0.1218346	13	5
<i>Test of joint nullity of the effects: p-value = 0.08365044</i>						
Placebo leads						
Placebo_1	-0.0052415	0.0108857	-0.0265209	0.0160379	40	10
Placebo_2	-0.0052151	0.0120235	-0.0183506	0.0278070	14	5
Placebo_3	0.0077719	0.0289499	-0.0489688	0.0645126	11	4
Placebo_4	-0.0276125	0.0271315	-0.0807893	0.0255643	7	2
<i>Test of joint nullity of the placebos: p-value = 0.69661468</i>						
Panel B: Cable repair days						
Event-study effects, countries with less than 10 connected partners at baseline						
Effect_1	-0.0014723	0.0077624	-0.0029666	0.0002221	219	39
Effect_2	-0.0014322	0.0123590	-0.0038545	0.0009901	175	29
Effect_3	-0.0040653	0.0211744	-0.0082153	0.0000847	125	20
Effect_4	-0.0060727	0.0284060	-0.0116402	-0.0000503	106	19
Effect_5	-0.0094042	0.0478740	-0.0187873	-0.0000211	91	15
Effect_6	-0.0110628	0.0570208	-0.0222754	-0.0001498	84	15
Effect_7	-0.0150554	0.0564610	-0.0215715	0.0005608	70	15
<i>Test of joint nullity of the effects: p-value = 0.03613592</i>						
Placebo leads						
Placebo_1	-0.000315	0.0005141	-0.0013226	0.0006926	219	39
Placebo_2	-0.0006576	0.0012866	-0.0031782	0.0018630	131	24
Placebo_3	-0.0026437	0.0025456	-0.0076330	0.0023456	71	13
Placebo_4	-0.0045599	0.0048876	-0.0114755	0.0049837	37	8
<i>Test of joint nullity of the placebos: p-value = 0.81649942</i>						
Event-study effects, countries with more than 10 connected partners at baseline						
Effect_1	-0.0009052	0.0010324	-0.0029287	0.0011183	48	13
Effect_2	0.0022539	0.0012231	-0.0001434	0.0046512	24	9
Effect_3	0.0039463	0.0021629	-0.0002929	0.0081555	24	9
Effect_4	0.0063482	0.0044160	-0.0023701	0.0150034	22	8
Effect_5	0.0072143	0.0058297	-0.0042117	0.0186403	22	8
Effect_6	0.0075111	0.0065138	-0.0052556	0.0202779	21	8
Effect_7	0.0131423	0.0130019	-0.0070490	0.0333337	15	6
<i>Test of joint nullity of the effects: p-value = 0.15801121</i>						
Placebo leads						
Placebo_1	0.0000621	0.0006754	-0.0012617	0.0013860	37	9
Placebo_2	-0.0002417	0.0007682	-0.0014774	0.0012639	11	4
Placebo_3	0.0003986	0.0014846	-0.0025112	0.0033083	11	4
Placebo_4	-0.0029851	0.0029331	-0.0087340	0.0027637	7	2
<i>Test of joint nullity of the placebos: p-value = 0.73482839</i>						

Table OA.5.3: Exposure factor: IXPs - Dep. var: GDP per capita (cst USD, ln)

	Estimate	SE	LB CI	UB CI	N	Switchers
Panel A: Cable disruptions						
Event-study effects, countries with no IXP						
Effect_1	-0.0189173	0.0091647	-0.0368797	-0.0009549	202	39
Effect_2	-0.0191145	0.0155196	-0.0495324	0.0113034	153	28
Effect_3	-0.0689046	0.0333930	-0.1342484	-0.0035607	105	16
Effect_4	-0.0931563	0.0387908	-0.1695448	-0.0174877	93	16
Effect_5	-0.1834622	0.0789860	-0.3382539	-0.0286165	82	13
Effect_6	-0.1490254	0.0661469	-0.2786709	-0.0193798	74	13
Effect_7	-0.1554429	0.0791558	-0.3105854	-0.0003003	59	12
<i>Test of joint nullity of the effects: p-value = 0.17107647</i>						
Placebo leads						
Placebo_1	-0.0025691	0.0074703	-0.0172107	0.0120726	202	39
Placebo_2	-0.0015334	0.0163308	-0.0354112	0.0304743	131	25
Placebo_3	-0.0171764	0.0400308	-0.0956354	0.0612826	74	12
Placebo_4	-0.0450509	0.0572891	-0.1573354	0.0672336	38	8
<i>Test of joint nullity of the placebos: p-value = 0.67173253</i>						
Event-study effects, countries with at least 1 IXP						
Effect_1	-0.0028077	0.0067164	-0.0159716	0.0103562	54	14
Effect_2	-0.0084263	0.0123375	-0.0326073	0.0157547	50	12
Effect_3	-0.0141184	0.0155629	-0.0456211	0.0163843	47	11
Effect_4	-0.0173247	0.0204336	-0.0573737	0.0227244	41	9
Effect_5	-0.0148378	0.0264481	-0.0666751	0.0369994	40	9
Effect_6	-0.0170459	0.0248257	-0.0657033	0.0316116	40	9
Effect_7	-0.0032057	0.0275441	-0.0571912	0.0507798	36	8
<i>Test of joint nullity of the effects: p-value = 0.13091225</i>						
Placebo leads						
Placebo_1	0.0091365	0.0169671	-0.0241185	0.0423914	28	11
Placebo_2	-0.0171635	0.0181874	-0.0528103	0.0184832	18	6
Placebo_3	-0.0506292	0.0373045	-0.1237446	0.0224862	12	4
Placebo_4	0.0530124	0.0626209	-0.0697223	0.1757474	4	1
<i>Test of joint nullity of the placebos: p-value = 0.19395334</i>						
Panel B: Cable repair days						
Event-study effects, countries with no IXP						
Effect_1	-0.0016748	0.0008356	-0.0033125	-0.0000370	197	37
Effect_2	-0.0009885	0.0011474	-0.0038776	0.0019005	148	26
Effect_3	-0.0041464	0.0032831	-0.0105811	0.0022883	107	17
Effect_4	-0.0057256	0.0038017	-0.0133337	0.0018824	95	17
Effect_5	-0.0098580	0.0073032	-0.0236807	0.0038891	85	14
Effect_6	-0.0110500	0.0082536	-0.0272268	0.0051269	77	14
Effect_7	-0.0090782	0.0079234	-0.0246078	0.0064515	61	13
<i>Test of joint nullity of the effects: p-value = 0.07557589</i>						
Placebo leads						
Placebo_1	-0.0000765	0.0006162	-0.0012843	0.0011313	195	36
Placebo_2	-0.0003498	0.0014791	-0.0032488	0.0025492	124	22
Placebo_3	-0.0019498	0.0028906	-0.0068154	0.0045156	74	12
Placebo_4	-0.0037348	0.0051408	-0.0138105	0.0063409	38	8
<i>Test of joint nullity of the placebos: p-value = 0.71868738</i>						
Event-study effects, countries with at least 1 IXP						
Effect_1	-0.0001739	0.0005928	-0.0013357	0.0009879	54	14
Effect_2	-0.0005918	0.0010377	-0.0025536	0.0015140	50	12
Effect_3	-0.0009557	0.0012898	-0.0034836	0.0015723	45	10
Effect_4	-0.0018085	0.0021442	-0.0060111	0.0023940	42	9
Effect_5	-0.0014426	0.0028982	-0.0071230	0.0042378	42	9
Effect_6	-0.0017528	0.0030337	-0.0076988	0.0041932	42	9
Effect_7	-0.0017801	0.0035891	-0.0088145	0.0052544	39	9
<i>Test of joint nullity of the effects: p-value = 0.08146742</i>						
Placebo leads						
Placebo_1	-0.0010987	0.0013780	-0.0016021	0.0037994	28	11
Placebo_2	-0.0003022	0.0009778	-0.0022187	0.0016143	18	6
Placebo_3	-0.0014389	0.0013947	-0.0041725	0.0012948	10	3
Placebo_4	-0.0017671	0.0020874	-0.0032341	0.0055882	4	1
<i>Test of joint nullity of the placebos: p-value = 0.12160442</i>						

OA.5.1 Low-connectivity country lists

Table OA.5.1: List of low-connectedness countries – criteria: # connected partners < 10.

ISO	Country	Income group (2008)	Region
AGO	Angola	Lower middle income	Sub-Saharan Africa
ARM	Armenia	Lower middle income	Europe & Central Asia
BDI	Burundi	Low income	Sub-Saharan Africa
BFA	Burkina Faso	Low income	Sub-Saharan Africa
BGR	Bulgaria	Upper middle income	Europe & Central Asia
BWA	Botswana	Upper middle income	Sub-Saharan Africa
CIV	Côte d'Ivoire	Lower middle income	Sub-Saharan Africa
CMR	Cameroon	Lower middle income	Sub-Saharan Africa
COG	Congo, Rep.	Lower middle income	Sub-Saharan Africa
COM	Comoros	Low income	Sub-Saharan Africa
CPV	Cabo Verde	Lower middle income	Sub-Saharan Africa
GAB	Gabon	Upper middle income	Sub-Saharan Africa
GEO	Georgia	Lower middle income	Europe & Central Asia
GHA	Ghana	Low income	Sub-Saharan Africa
GIN	Guinea	Low income	Sub-Saharan Africa
GMB	Gambia, The	Low income	Sub-Saharan Africa
GUY	Guyana	Lower middle income	Latin America & Caribbean
IRQ	Iraq	Upper middle income	Middle East & North Africa
KEN	Kenya	Low income	Sub-Saharan Africa
KHM	Cambodia	Low income	East Asia & Pacific
LBR	Liberia	Low income	Sub-Saharan Africa
MDG	Madagascar	Low income	Sub-Saharan Africa
MDV	Maldives	Lower middle income	South Asia
MLI	Mali	Low income	Sub-Saharan Africa
MOZ	Mozambique	Low income	Sub-Saharan Africa
MRT	Mauritania	Low income	Sub-Saharan Africa
MWI	Malawi	Low income	Sub-Saharan Africa
NAM	Namibia	Upper middle income	Sub-Saharan Africa
PNG	Papua New Guinea	Lower middle income	East Asia & Pacific
RWA	Rwanda	Low income	Sub-Saharan Africa
SDN	Sudan	Lower middle income	Sub-Saharan Africa
SLE	Sierra Leone	Low income	Sub-Saharan Africa
SOM	Somalia	Low income	Sub-Saharan Africa
SUR	Suriname	Upper middle income	Latin America & Caribbean
SWZ	Eswatini (Swaziland)	Lower middle income	Sub-Saharan Africa
SYC	Seychelles	High income	Sub-Saharan Africa
TON	Tonga	Upper middle income	East Asia & Pacific
TZA	Tanzania	Low income	Sub-Saharan Africa
UGA	Uganda	Low income	Sub-Saharan Africa
VUT	Vanuatu	Lower middle income	East Asia & Pacific
ZWE	Zimbabwe	Low income	Sub-Saharan Africa

Table OA.5.2: List of no-IXP countries – criteria: no IXP.

ISO	Country	Income group (2008)	Region
AGO	Angola	Lower middle income	Sub-Saharan Africa
ARM	Armenia	Lower middle income	Europe & Central Asia
BDI	Burundi	Low income	Sub-Saharan Africa
BFA	Burkina Faso	Low income	Sub-Saharan Africa
BGR	Bulgaria	Upper middle income	Europe & Central Asia
BWA	Botswana	Upper middle income	Sub-Saharan Africa
CIV	Côte d'Ivoire	Lower middle income	Sub-Saharan Africa
CMR	Cameroon	Lower middle income	Sub-Saharan Africa
COG	Congo, Rep.	Lower middle income	Sub-Saharan Africa
COM	Comoros	Low income	Sub-Saharan Africa
CPV	Cabo Verde	Lower middle income	Sub-Saharan Africa
GAB	Gabon	Upper middle income	Sub-Saharan Africa
GEO	Georgia	Lower middle income	Europe & Central Asia
GHA	Ghana	Low income	Sub-Saharan Africa
GIN	Guinea	Low income	Sub-Saharan Africa
GMB	Gambia, The	Low income	Sub-Saharan Africa
GUY	Guyana	Lower middle income	Latin America & Caribbean
IRQ	Iraq	Upper middle income	Middle East & North Africa
KEN	Kenya	Low income	Sub-Saharan Africa
KHM	Cambodia	Low income	East Asia & Pacific
LBR	Liberia	Low income	Sub-Saharan Africa
MDG	Madagascar	Low income	Sub-Saharan Africa
MDV	Maldives	Lower middle income	South Asia
MLI	Mali	Low income	Sub-Saharan Africa
MOZ	Mozambique	Low income	Sub-Saharan Africa
MRT	Mauritania	Low income	Sub-Saharan Africa
MWI	Malawi	Low income	Sub-Saharan Africa
NAM	Namibia	Upper middle income	Sub-Saharan Africa
PNG	Papua New Guinea	Lower middle income	East Asia & Pacific
RWA	Rwanda	Low income	Sub-Saharan Africa
SDN	Sudan	Lower middle income	Sub-Saharan Africa
SLE	Sierra Leone	Low income	Sub-Saharan Africa
SOM	Somalia	Low income	Sub-Saharan Africa
SUR	Suriname	Upper middle income	Latin America & Caribbean
SWZ	Eswatini (Swaziland)	Lower middle income	Sub-Saharan Africa
SYC	Seychelles	High income	Sub-Saharan Africa
TON	Tonga	Upper middle income	East Asia & Pacific
TZA	Tanzania	Low income	Sub-Saharan Africa
UGA	Uganda	Low income	Sub-Saharan Africa
VUT	Vanuatu	Lower middle income	East Asia & Pacific
ZWE	Zimbabwe	Low income	Sub-Saharan Africa

OA.6 Financial-channel exhibits

This section collects the event-study exhibits supporting the financial-channel results of Section 6 (domestic credit and cross-border banking).

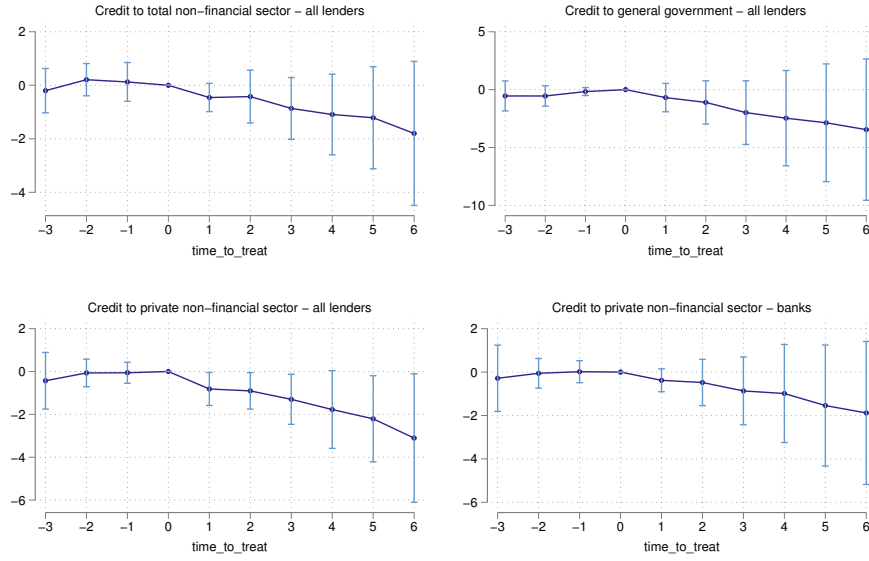


Fig. OA.6.1: Cable disruptions and credit to non-financial domestic agents (USD, Z-scores).

Data from the Bank of International Settlements (BIS). Variables are winsorized at 90% and standardized afterwards. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The estimation sample includes 27 high-income and emerging economies (271 observations), with six switchers (76 obs.): Australia, France, United Kingdom, Indonesia, Italy, and Turkey.

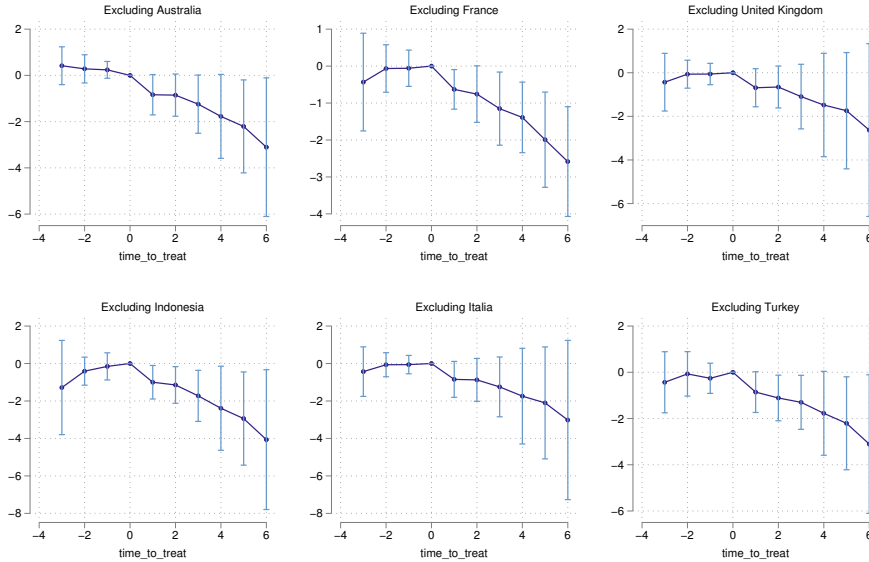


Fig. OA.6.2: Cable disruptions and credit to non-financial domestic agents (USD, Z-scores), sequentially excluding switchers.

Data drawn from the Bank of International Settlements (BIS). Variables are winsorized at 90% and standardized afterwards. Markers show point estimates; bars denote 95% confidence intervals. Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence.

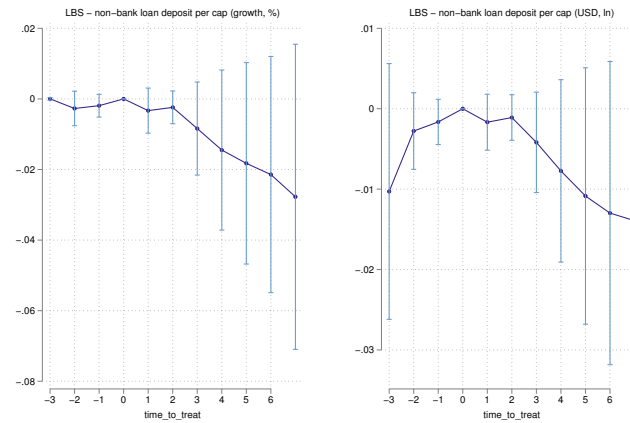


Fig. OA.6.3: Non-bank loan deposits per capita (USD, ln) - Location Banking Statistics.

Markers show point estimates; bars denote 90% confidence intervals. Data drawn from the Locational Banking Statistics (LBS) from the Bank of International Settlements (BIS). Specifications include country fixed effects, year fixed effects, linear country trends, and nonparametric region and income-group trends, with standard errors clustered at the country level. Year 6 corresponds to a period of five years after the cable failure occurrence. The estimation sample includes 137 countries (1,781 observations), including 50 switchers (650 obs.). High-income and middle-income economies respectively represent 33.6% and 54% of the estimation sample.

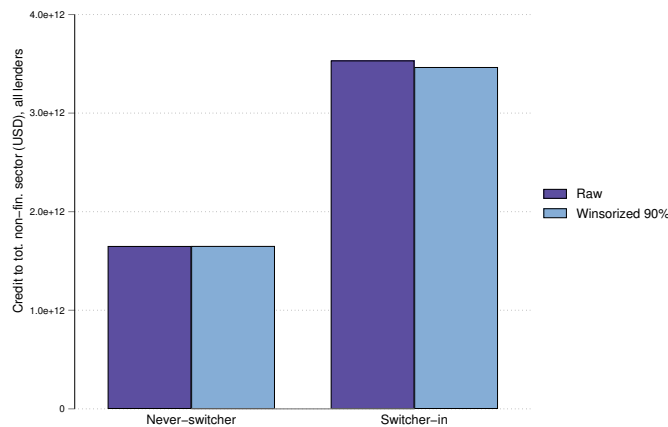


Fig. OA.6.4: Domestic credit to non-financial domestic actors (USD), switchers versus never switchers.

The sample includes 27 high-income and emerging economies (271 observations), with six switchers (76 obs.): Australia, France, United Kingdom, Indonesia, Italy, and Turkey. The credit variable is winsorized at 90% to ensure that results are not driven by outliers.

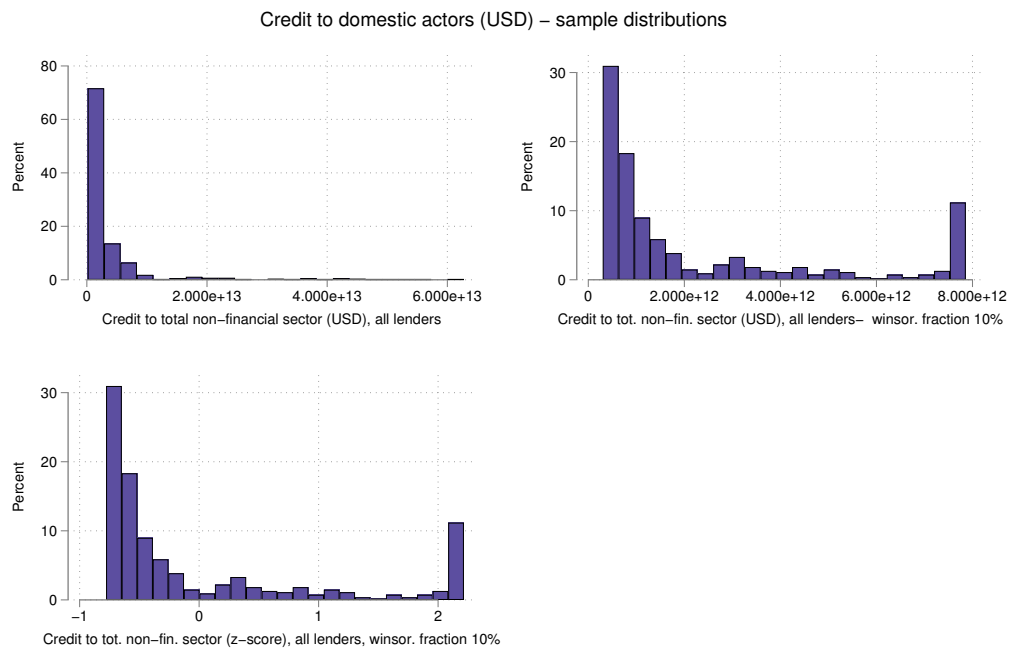


Fig. OA.6.5: Domestic credit to non-financial domestic actors (USD), variable distributions.